The extremal even unimodular lattices of rank 32

Their number, and the classification of those with automorphisms other than ± 1

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Version 2026-11-14.1. A versioned preprint; later versions may correct or extend it. The statements on which the proofs rest and their computational assumptions are listed in Tables 7 and 10.

Abstract

There are exactly 11,044,268 isometry classes of extremal even unimodular lattices of rank 32. Exactly 143,329 have automorphisms other than ± 1; we classify these lattices and determine the order of each automorphism group. The proof first classifies the lattices with additional automorphisms, using complete lists of pairs with an automorphism of odd prime order, an involution other than ± 1, or a Gaussian structure. The number of all classes is then obtained from King's rootless mass m by $N = 2m + \sum_{L \in S}(1 - 2/\,|\mathrm{Aut}\,L|)$. Representatives of the lattices with automorphism group $\{\pm 1\}$ are not constructed. The proof assumes the correctness of the finite computations specified in §7.

1. Introduction

What this section does. §1.1 recalls what was known about extremal lattices of rank 32. §1.2 states the two main theorems and a corollary. §1.3 explains the method in one page. §1.4 says what the proof assumes. §1.5 describes the organisation.

1.1 Extremal even unimodular lattices of rank 32

The minimum of an even unimodular lattice of rank n satisfies $\min L \le 2\lfloor n/24 \rfloor + 2$ [SPLAG, Ch. 7, Cor. 21], and L is *extremal* if equality holds; extremal lattices do not exist for large n [MOS75, Cor. to Thm 1]. In rank 32 an extremal lattice is one without vectors of norm 2. Rank 32 is the first rank in which the genus of even unimodular lattices is far too large to be listed: a mass-based estimate suggests about 1.2×10^9 classes [ChT26, §1]. It is also the first rank in which extremal lattices are numerous.

Several families were known. Koch and Venkov showed that there are exactly five extremal lattices of neighbour defect 0, the code lattices of the five doubly even self-dual codes of length 32 and minimum weight 8 [KV89, Satz 10] (the classification of the codes is quoted there from Conway and Pless), and continued the study of rank 32 in [KV91]. Koch and Nebe found that there are exactly ten extremal lattices of neighbour defect 8 [KN93, Thm 1]; the twelve codes of their Table 1 give these ten and two of the code lattices of neighbour defect 0 (the non-isometry is stated there with reference to an appendix by Nebe to [KV91]). Lattices with additional structure were classified by mass formulas: the eleven extremal lattices with an automorphism of order 3 without fixed vectors [Hen13, Thm 1, Table 4, Rem. 3(a)], using the mass of Eisenstein lattices [HKN09, Thm 1]; the structures over the Hurwitz order [BN97, Thm 5.1]; and the four extremal lattices from a genus of Hermitian lattices of rank 2 over $\mathbb{Z}[\zeta_{17}]$ [Jür15, Satz 3.6.9]. The mass of the even unimodular lattices of rank 32 with an automorphism g with $g^2 = -1$ is given in [HKN12, Thm 2]. Chenevier and Taïbi indicate that a classification of the whole

genus of rank 32 may now be within reach and announce a forthcoming study [ChT26, §1]. Appendix A locates the 27 previously known lattices with $|\mathrm{Aut}\ L| > 2$ in our lists.

Since every lattice has the automorphisms ± 1, King's mass m [Kin03] gives the lower bound $N \geq 2m \approx 1.0969 \times 10^7$ (§2.2). No exact count or classification had been published as of November 2026.

1.2 Main results

Let $\mathcal{L}$ be the set of isometry classes of extremal even unimodular lattices of rank 32 (§2.1). Every $L \in \mathcal{L}$ has the automorphisms ± 1, so $|\mathrm{Aut}\ L| \geq 2$.

Theorem A. $|\mathcal{L}| = 11{,}044{,}268$.

Theorem B. Let $S = \{L \in \mathcal{L} : |\mathrm{Aut}\ L| > 2\}$. Then $|S| = 143{,}329$, and $S = S_{\mathrm{gen}} \sqcup S_{C_2^2} \sqcup S_{\mathrm{CQ}}$, where

1. $S_{C_2^2}$ consists of the 67,284 lattices with $\mathrm{Aut}\ L = \{\pm 1, \pm\sigma\} \cong C_2 \times C_2$;
2. S_{CQ} consists of the 63,499 lattices for which $\mathrm{Aut}\ L$ is a 2-group with -1 as its only involution, that is, cyclic of order $4, 8, 16$ or generalized quaternion of order $8, 16, 32, 64$, with the numbers in Table 8;
3. S_{gen} consists of the remaining 12,546 lattices; each has an automorphism of odd prime order or an involution $\sigma \neq \pm 1$ with $|C(\sigma)| > 4$ or σ conjugate to $-\sigma$.

The lattices of S, with $\mathrm{Aut}\ L$, are listed in the database [D], version 2, under the numbering of §6.8.

Here $C(\sigma)$ is the centralizer of σ in $\mathrm{Aut}\ L$.

Corollary C. Exactly 10,900,939 extremal even unimodular lattices of rank 32 have automorphism group $\{\pm 1\}$. The distribution of $|\mathrm{Aut}\ L|$ over $\mathcal{L}$ is given in Table 9.

Table 1. The parts of S in Theorem B.

part	number	Aut L	proved in
S_{gen}	12,546	computed for each L [C1]	§6.3
$S_{C_2^2}$	67,284	$C_2 \times C_2$	§6.4, Proposition 6.4; the number by Lemma 6.6
S_{CQ}	63,499	C_{2^n} $(n = 2, 3, 4)$ or Q_{2^n} $(n = 3, \ldots, 6)$	§6.5, Table 8; the numbers by Lemma 6.6
S	143,329		
$\mathcal{L} \setminus S$	10,900,939	$\{\pm 1\}$	Corollary C

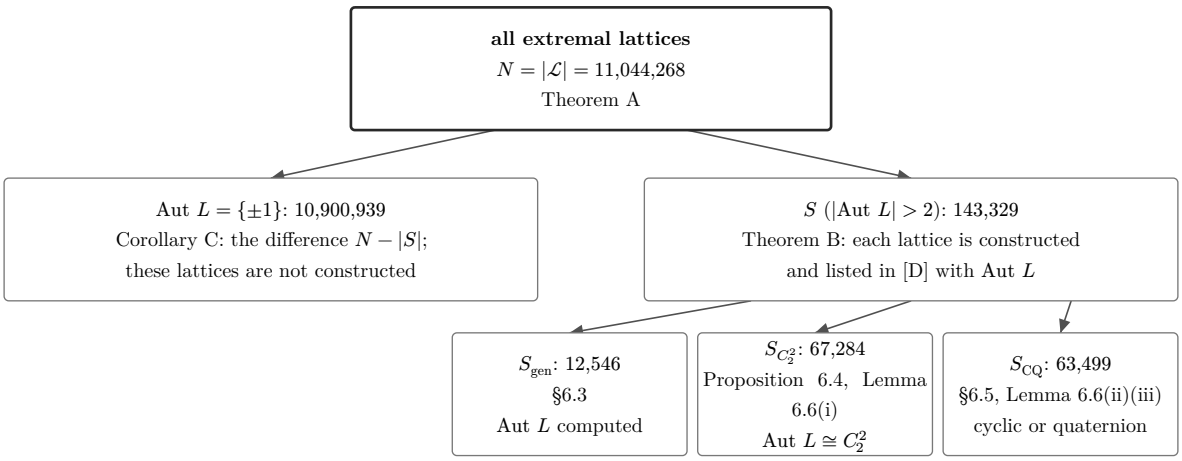


Figure 1. Where the number N comes from. The lattices of S are classified (Theorem B); the lattices with Aut $L = \{\pm 1\}$ are only counted. Identity (5.1) splits N in a different way: each lattice L contributes $2/\ |\mathrm{Aut}\ L|$ to $2m \approx 10{,}968{,}923.3$, and each lattice of S contributes the rest, $1 - 2/\ |\mathrm{Aut}\ L|$. These remainders add up to about 9,687.0 over S_{gen}, to exactly 33,642 over $S_{C_2^2}$ (each lattice gives 1/2) and to 512,251 /16 $\approx$ 32,015.7 over S_{CQ} (Table 8), in all about 75,344.7 (§6.7).

1.3 The method

Let G be a finite group containing the central involution -1. If $G \neq \{\pm 1\}$, choose x outside that subgroup. If its order has an odd prime divisor, a power of x has odd prime order. Otherwise its order is 2^a. For $a = 1$, x is an involution other than ± 1. For $a \geq 2$, $x^{2^{a-1}}$ is an involution; if it is -1, a further power has square -1. Thus these three kinds of structures detect every lattice of S.

We classify the pairs carrying these structures. For odd primes and involutions this means complete lists of components and of gluing classes (§4, Appendix B); Gaussian pairs are complete by their exact rootless mass (§4.3, Appendix C). Completeness requires valid candidates, pairwise non-isometry, correct group orders in the weights, and an independently justified mass. An equality of masses alone does not establish all four conditions.

We then identify the underlying integer lattices, compute their full automorphism groups where needed, and use finite-group arguments for the remaining cyclic, quaternionic and C_2^2 cases (§6). This establishes Theorem B without using the total N. Finally, summing $1 = 2/\ |G| + (1 - 2/\ |G|)$ over all lattices gives the formula $N = 2m + \sum_{L \in S}(1 - 2/\ |\mathrm{Aut}\ L|)$ of (5.1) and Theorem A (§5 and §6.7).

An earlier organization of the count by assigned group elements is kept in the auxiliary note [Aux]; it is not used in the proof.

1.4 What the proof assumes

The proof is computer-assisted. Its numerical inputs are King's rootless mass, complete lists of components and gluing pairs, the Gaussian rootless mass and list, exact identifications of the underlying integer lattices, and their full automorphism-group orders. It relies on complete short-vector enumeration, group and orbit computations, canonical forms and isometry decisions, and our implementations of the stated mass formulas. These dependencies are listed in §7. Returned automorphisms and positive isometries are checked in integers; completeness of groups and negative isometry decisions remain software assumptions.

The two exceptional full-group computations from Gram matrices alone, for BW_{32} and for the code lattice with $|\mathrm{Aut}\ L| = 2^{31}3^5 5^2 7$, are separated from the structural calculation of their conjugacy classes (§6.3). Group presentations or the order of a subgroup generated by construction matrices do not by

themselves provide an upper bound for the full lattice automorphism group. The aggregation (5.1) reads only King's mass and the classified lattices. This arithmetic check is conditional on the mathematical completeness and the group orders of the classification; it is not an independent proof of those inputs.

1.5 Organisation

§2 fixes conventions. §3 detects additional automorphisms and gives their possible types. §4 constructs complete lists of structured pairs. §5 reduces the total to the classification, proved in §6. §7 records algorithms and reproducibility. Appendices B, C and D supply the component classifications and mass formulas. The auxiliary note [Aux] keeps an earlier organization of the count. Statements and displayed equations are numbered within each section, so that (4.1) is an equation and Proposition 4.1 a statement; tables are numbered consecutively, with the letter of the appendix in the appendices. The figures are numbered in the same way.

How to read this paper. The paper can be read in several ways.

- *For the numbers only.* §1.2 states them, and Figure 1 shows how N splits. §6.8 and Table 9 give the distribution of $|\mathrm{Aut}\ L|$, and the database [D] lists the lattices of S.
- *To check the proof.* Figure 2 (§5.2) and Table 7 show which statement rests on which. Read §3 and §4 for the complete lists, then §6 for the classification and §5.1 for the count. The appendices prove the component lists (B) and the masses (C, D). Several long proofs begin with a paragraph headed "Strategy of the proof".
- *To use the data.* §6.8 explains the labels and §7.4 the files; the README files of the repository describe them.
- *For the groups themselves.* The second paper of this repository describes the 99 lattices with the largest groups.

The remarks headed "Points that are easy to misread" (§2 and §5) collect conventions on which misreadings are likely.

2. Preliminaries

What this section does. We fix conventions (§2.1), recall the mass and King's rootless mass (§2.2), discriminant forms, 2-elementary lattices and gluing (§2.3), and Hermitian structures given by automorphisms whose minimal polynomial is cyclotomic (§2.4). Everything here is standard; we give references and the precise forms we use.

Table 2. Notation.

symbol	meaning	§
$\mathcal{L}$, N	isometry classes of extremal even unimodular lattices of rank 32; $N = \lvert\mathcal{L}\rvert$	2.1
m	King's mass $\sum_{L\in\mathcal{L}} 1/\ \lvert\mathrm{Aut}\ L\rvert$	2.2
S	$\{L \in \mathcal{L} : \lvert\mathrm{Aut}\ L\rvert > 2\}$	1.2
$C(h)$	centralizer of h in Aut L	2.1
F, K, Z	fixed lattice L^h, and its orthogonal complement ($K = L^{-\sigma}$ for an involution, Z for odd prime order)	4.1.1
h, H, $\tilde{H}$	h is an element of Aut L, as in a pair (L, h); H is the Hermitian form with $(x, y) = \mathrm{Tr}\ H(x, y)$ (§2.4), and $\tilde{H}$ its rescaling in Lemma 4.4. Appendices C and D, where no element h occurs, write h for their Hermitian form (Table 3)	2.4

symbol	meaning	§
$\mathbb{K}$, ν	the cyclotomic field $\mathbb{Q}(\zeta)$ (§2.4) or $\mathbb{Q}(\zeta_p)$ (§4.2, Appendix D), not the lattice K; $\nu = \varpi\overline{\varpi}$ (Lemma 4.4, Appendix D)	2.4
σ, g	$\sigma \neq \pm 1$ an involution of Aut L; g an automorphism with $g^{2^{a-1}} = -1$, a Gaussian structure if $g^2 = -1$	2.4, 3.3
ω	the alternating form $H'(x,y) \bmod \varpi$ on $L/\varpi L$ (§2.4, and $h(x,y) \bmod \varpi$ in Appendix C); in §7.2 a structure of a Hermitian lattice; ζ_3 in Appendix D.4	2.4
s, δ	rank of the discriminant group $\mathbb{F}_p^s$; for $p = 2$ the parity invariant of a 2-elementary lattice	2.3
q, $O(q)$	discriminant quadratic form and its orthogonal group	2.3
$[\varphi]$, Stab(φ)	double coset of a gluing φ and its stabilizer	4.1.1
$p-(z,f)-s$	type of an element h of odd prime order p: $\operatorname{rank}_{\mathbb{Z}[\zeta_p]}(L^h)^\perp = z$, rank $L^h = f$, L^h has determinant p^s	3.3
$e(h)$	the number of elements of Aut L assigned to h in the auxiliary count; computed inside a full group as a check, not a counting input	[Aux, §3.2]
f	2 or 1 according as a Hermitian $\mathbb{Z}[i]$-lattice is not or is isometric to its complex conjugate (§§4.3, 6.1); in a type $p-(z,f)-s$ the rank of L^h	6.1
U, U	the unitary group $\mathrm{U}(L,H) = C(g)$ of a Hermitian lattice (L,H), also $\mathrm{U}(T)$, $\mathrm{U}(Z)$ for root lattices and components, and $\lvert\mathrm{U}\rvert$ its order (also the finite unitary groups $\mathrm{U}_b(\mathbb{F}_4)$ of Appendix D); the italic U is always a unimodular sublattice (Proposition 3.2, Lemma 4.4, Appendices B and D)	2.4
D	the isometry invariant of §7.2 of a pair $(L, \pm g)$; $D(M)$ is a discriminant group; [D] is the database	7.2
ϖ, π	ϖ is a prime element: $1-\zeta$ in §2.4, $1-\zeta_p$ in §4.2 and Appendix D, $1+i$ in §7.2 and Appendix C (where also $\varpi_i = 1-\zeta_{2^i}$); π is the circle constant	2.4
M_2	rootless mass of Hermitian $\mathbb{Z}[i]$-lattices of rank 16	4.3
(z,s), mass$^\emptyset$	type of a Hermitian $\mathbb{Z}[\zeta_p]$-lattice and the rootless mass of its genus	D

2.1 Lattices and automorphisms

A *lattice* is a free $\mathbb{Z}$-module L of finite rank n with a positive definite symmetric bilinear form (x,y), given by a Gram matrix in a basis; (x,x) is the *norm* of x, and $\min L$ the smallest norm of a non-zero vector. The dual is $L^\# = \{y \in L \otimes \mathbb{Q} : (y,L) \subset \mathbb{Z}\}$. L is *integral* if $L \subset L^\#$, *even* if all norms are even, and *unimodular* if $L = L^\#$. Even unimodular lattices exist exactly in ranks divisible by 8, and their minimum satisfies $\min L \leq 2\lfloor n/24 \rfloor + 2$ [SPLAG, Ch. 7, Cor. 21]; L is *extremal* if equality holds. In rank 32 this means $\min L = 4$, that is, L has no *roots* (vectors of norm 2).

Two lattices are isometric if a $\mathbb{Z}$-linear bijection preserves the forms. $\mathcal{L}$ is the set of isometry classes of extremal even unimodular lattices of rank 32, and $N = |\mathcal{L}|$. Aut L is the finite group of isometries of L onto itself; it contains ± 1, and -1 is central. For $h \in \mathrm{Aut}\ L$, $C(h)$ is its centralizer, $[h]$ its conjugacy class, and $L^h = \{x \in L : xh = x\}$ its fixed lattice (we let Aut L act on the right).

2.2 Masses

The *mass* of a set of lattices is $\sum 1/\ |\mathrm{Aut}\ L|$ over its isometry classes. The mass of all even unimodular lattices of rank n is given by the Smith–Minkowski–Siegel mass formula [SPLAG, Ch. 16]. King computed the mass of those without roots by inclusion–exclusion over the root systems they may contain [Kin03]; for $n = 32$ it is the mass m of $\mathcal{L}$,

$$m = \frac{1{,}310{,}037{,}331{,}282{,}023{,}326{,}658{,}917}{238{,}863{,}431{,}761{,}920{,}000} \approx 5{,}484{,}461.65 \tag{2.1}$$

2.3 Discriminant forms, 2-elementary lattices, gluing

For an even lattice M, the *discriminant group* $D(M) = M^\#/M$ carries the *discriminant form* $q_M(x + M) = (x, x) \bmod 2\mathbb{Z}$ and the bilinear form $b_M(x + M, y + M) = (x, y) \bmod \mathbb{Z}$, written b when M is clear [Nik79]; $O(q_M)$ is its orthogonal group, and $\mathrm{Aut}\ M \to O(q_M)$ has image $\overline{\mathrm{Aut}\ M}$. M is *p-elementary* if $pM^\# \subset M$, so that $D(M) \cong \mathbb{F}_p^s$. For $p = 2$ we put $\delta = 0$ if q_M takes values in $\mathbb{Z}/2\mathbb{Z}$ and $\delta = 1$ otherwise; the even 2-elementary lattices of given rank, s and δ form one genus when they exist [Nik79, Thm 3.6.2]. For odd p the genus of even p-elementary lattices is determined by the rank and s ([SPLAG, Ch. 15, Thm 13]; see reason 1 in the proof of Proposition 3.3).

Table 3. Normalizations of forms. q always denotes the discriminant form of an even $\mathbb{Z}$-lattice with values in $\mathbb{Q}/2\mathbb{Z}$.

object	form	where
even $\mathbb{Z}$-lattice M	$q_M(x + M) = (x, x) \bmod 2\mathbb{Z}$ on $D(M)$	§2.3
even $\mathbb{Z}$-lattice M	$b_M(x + M, y + M) = (x, y) \bmod \mathbb{Z}$ on $D(M)$	§2.3
p-elementary $D(M)$, p odd	$Q(a) = (p/2)q_M(a) \in \mathbb{F}_p$, i.e. $Q(a) = j$ when $q_M(a) = 2j/p$	§4.2, §B.1.1, β_3 in §D.4
the same, convention $3q$	$3q_M(a) \bmod 3 = 2Q(a)$; determinant characters differ by $(2/3)^{\dim} = (-1)^{\dim}$	§D.4.6
2-elementary $D(M)$	(ρ, δ, σ): rank, δ as above, Brown invariant	§2.3, §D.4.6
$\mathbb{Z}[\zeta_k]$-structure (§2.4)	$(x, y) = \mathrm{Tr}_{\mathbb{K}/\mathbb{Q}} H(x, y)$	§2.4
Gaussian lattices (Appendix C)	the form written h in Appendix C is $2H$ for the H of §2.4, so $(x, y) = \mathrm{Re}\ h(x, y)$; roots $h(x, x) = 2$; $q(x) = h(x, x) \bmod 2$; $\omega(x, y) = h(x, y) \bmod \varpi$ on $L/\varpi L$, $\varpi = 1 + i$	§2.4 ($H' = dH$, $d = 2$), §4.3, App. C
$\mathbb{Z}[\zeta_p]$, p odd (§4.2, App. D)	$\tilde{H} = \nu^{(p-3)/2} H$ with $(x, y) = \mathrm{Tr}_{\mathbb{K}/\mathbb{Q}} H(x, y)$ and $\nu = \varpi\overline{\varpi}$; $a(x, y) = \varpi \tilde{H}(x, y) \bmod \varpi$ on Z/Z^*; Appendix D writes h for $\tilde{H}$	Lemma 4.4, §D.1
Eisenstein lattices, $p = 3$ (§D.4)	$h = \tilde{H} = H$, roots $h(x, x) = 1$; $\beta_3(x, y) = 3h(x, y) \bmod \varpi$ on D_3; $\beta_2(x, y) = 2h(x, y) \bmod 2\mathcal{O}_2$ on D_2	§D.4.1

If F and Z are primitive sublattices of an even unimodular lattice L with $Z = F^\perp$, the projections identify $D(F)$ and $D(Z)$ by an isomorphism φ with $q_Z \circ \varphi = -q_F$, and $L = \{x + y : x \in F^\#, y \in Z^\#, \varphi(x + F) = y + Z\}$; conversely every such φ defines an even unimodular lattice [Nik79, §1.5, Prop. 1.6.1]. Even overlattices of an even lattice M correspond to the subgroups of $D(M)$ on which q_M vanishes [Nik79, Prop. 1.4.1]. §4.1.1 turns this into a correspondence for pairs (L, h) (Proposition 4.1).

Lemma 2.1 (orders of $O(q)$). Let F be an even positive definite lattice of rank d with discriminant form q on $V = D(F)$.

(i) If F is p-elementary for an odd prime p, with $V \cong \mathbb{F}_p^s$, then $O(q)$ is the orthogonal group of the non-degenerate quadratic form Q of Table 3, so $|O(q)| = |\mathrm{GO}_s(p)|$ for odd s and $|O(q)| = |\mathrm{GO}_s^\varepsilon(p)|$ for even s, where $\varepsilon = \chi\big((-1)^{s/2} \operatorname{disc} Q\big)$ and $\chi(\operatorname{disc} Q) = e^{2\pi i d/8}\varepsilon_p^{-s}$; here χ is the Legendre symbol modulo p, and $\varepsilon_p = 1$ or i according as $p \equiv 1$ or 3 modulo 4.

(ii) If F is 2-elementary with invariants s and δ, then $|O(q)|$ is $|\mathrm{GO}_s^\varepsilon(2)|$ if $\delta = 0$, with $\varepsilon = +$ if and only if $d \equiv 0 \pmod 8$; $|\mathrm{GO}_{s-1}^\varepsilon(2)|$ if $\delta = 1$ and s is odd, with $\varepsilon = +$ if and only if $d \equiv \pm 1 \pmod 8$; $2\,|\mathrm{Sp}_{s-2}(2)|$ if $\delta = 1$, s is even and $d \equiv 2 \pmod 4$; and $2^{s-2}\,|\mathrm{GO}_{s-2}^\varepsilon(2)|$ if $\delta = 1$, s is even and $d \equiv 0 \pmod 4$, with $\varepsilon = +$ if and only if $d \equiv 0 \pmod 8$.

Here GO is the full orthogonal group of a non-degenerate quadratic form and Sp the symplectic group: $|\mathrm{GO}_{2m}^\varepsilon(r)| = 2r^{m(m-1)}(r^m - \varepsilon)\prod_{i=1}^{m-1}(r^{2i}-1)$, $|\mathrm{GO}_{2m+1}(r)| = 2r^{m^2}\prod_{i=1}^{m}(r^{2i}-1)$ for odd r, and $|\mathrm{Sp}_{2m}(r)| = r^{m^2}\prod_{i=1}^{m}(r^{2i}-1)$; for $m = 0$, $|\mathrm{GO}_0^\varepsilon(r)| = |\mathrm{Sp}_0(r)| = 1$ (the zero space), and the formula for $\mathrm{GO}_{2m+1}(r)$ gives $|\mathrm{GO}_1(r)| = |\{\pm 1\}| = 2$.

Proof. Write $\gamma(q) = \sum_{x\in V} e^{\pi i q(x)}$; by Milgram's formula, $\gamma(q) = |V|^{1/2}\, e^{2\pi i d/8}$. For a non-degenerate quadratic form P of dimension $2m$ over $\mathbb{F}_2$ of type ε, $\sum_x (-1)^{P(x)} = \varepsilon 2^m$; and for a vector w, $P(x) + \beta(w,x) = P(x+w) + P(w)$, where β is the polar form, so $P + \beta(w,\cdot)$ has the type of P times $(-1)^{P(w)}$.

(i). As p is odd, q and Q determine each other, so $O(q) = O(Q)$. Diagonalizing $Q = \sum_j c_j x_j^2$ and using quadratic Gauss sums, $\gamma(q) = \chi(\operatorname{disc} Q)\varepsilon_p^s p^{s/2}$; compare with Milgram's formula. A form of even dimension s has type $+$ if and only if $(-1)^{s/2}$ disc Q is a square.

(ii). Here q takes values in $(1/2)\mathbb{Z}/2\mathbb{Z}$, and $\beta = 2b$ is a non-degenerate symmetric bilinear form over $\mathbb{F}_2$. Since $x \mapsto \beta(x,x)$ is linear, there is a unique c with $\beta(x,x) = \beta(x,c)$ for all x; it is fixed by $O(q)$, and $\delta = 0$ if and only if $c = 0$. Put $W = c^\perp = \{x : q(x) \in \mathbb{Z}/2\mathbb{Z}\}$; on W, $Q = q|_W$ is an $\mathbb{F}_2$-valued quadratic form with polar form β, and the radical of $\beta|_W$ is $W \cap \langle c\rangle$.

If $\delta = 0$, then $W = V$, $O(q) = O(Q)$ and $\gamma(q) = \varepsilon 2^{s/2}$.

If $\delta = 1$ and s is odd, then $c \notin W$, since otherwise $W/\langle c\rangle$ would carry a non-degenerate alternating form in the odd dimension $s-2$. So $V = W \perp \langle c\rangle$ with $q(c) = \pm 1/2$, Q is non-degenerate on W, and $O(q) = O(Q)$: isometries fix c, and an isometry of Q extended by $c \mapsto c$ preserves q. Moreover $\gamma(q) = \varepsilon 2^{(s-1)/2}\big(1 + e^{\pm \pi i/2}\big)$, which gives $d \equiv \pm 1 \pmod 8$ for $\varepsilon = +$ and $d \equiv 4 \pm 1 \pmod 8$ for $\varepsilon = -$.

If $\delta = 1$ and s is even, then $c \in W$, since $\beta|_W$ is alternating on a space of odd dimension; so $q(c) \in \{0, 1\}$, and every $x \notin W$ has $q(x) \in \{1/2, 3/2\}$. Restriction gives $\rho : O(q) \to O(Q)$. For $x \notin W$ put $P_x(w) = q(x+w) - q(x) = Q(w) + \beta(x,w)$ on W. Fix $e \notin W$ and $g \in O(Q)$. The vectors v with $\beta(v, gw) = \beta(e,w)$ for all $w \in W$ are v_0 and $v_0 + c$, both outside W; the map equal to g on W with $e \mapsto v$ lies in $O(q)$ if and only if $q(v) = q(e)$, and $q(v_0 + c) = q(v_0) + q(c) + 1$.

Case $q(c) = 0$. Exactly one of v_0, $v_0 + c$ has the value $q(e)$, so ρ is bijective. Q induces a non-degenerate form $\overline{Q}$ on $W/\langle c\rangle$; $O(Q) \to O\big(\overline{Q}\big)$ is onto, with kernel $\{w \mapsto w + \lambda(w)c\}$, $\lambda \in (W/\langle c\rangle)^*$, of order 2^{s-2}. Hence $|O(q)| = 2^{s-2}\,|\mathrm{GO}_{s-2}^\varepsilon(2)|$ with ε the type of $\overline{Q}$. As $P_e(c) = 1$, the sum over $e + W$ in $\gamma(q)$ vanishes, so $\gamma(q) = 2\sum_{W/\langle c\rangle}(-1)^{\overline{Q}} = \varepsilon 2^{s/2}$: $d \equiv 0 \pmod 4$, and $\varepsilon = +$ if and only if $d \equiv 0 \pmod 8$.

Case $q(c) = 1$. v_0 and $v_0 + c$ have the same value, so $\ker\rho = \{1, t\}$ with $t(x) = x + \beta(x,c)c$. For $x \notin W$, $P_x(c) = 0$, so P_x is a non-degenerate form on $W/\langle c\rangle$, and $P_{x+w} = P_x + \beta(w,\cdot)$. By the remark above, the type of P_{e+w} is that of P_e times $(-1)^{q(e+w)-q(e)}$; hence $q(x)$ is determined by the type of P_x, for all $x \notin W$. Now $P_{v_0} \circ g = P_e$, so P_{v_0} and P_e have the same type, $q(v_0) = q(e)$, and ρ is onto.

Since $Q(c) = 1$, restriction to $W/\langle c\rangle$ is an isomorphism $O(Q) \to \mathrm{Sp}(W/\langle c\rangle) \cong \mathrm{Sp}_{s-2}(2)$: a symplectic h lifts uniquely, by sending w to the lift of $h(\overline{w})$ with value $Q(w)$. Hence $|O(q)| = 2\ |\mathrm{Sp}_{s-2}(2)|$. Here the sum over W in $\gamma(q)$ vanishes (pair w with $w + c$), and the sum over $e + W$ is $\pm 2^{s/2} e^{\pi i q(e)}$, so $d \equiv 2 \pmod 4$. □

Example (Lemma 2.1(ii) in small cases). This example is not used in the proofs. For $F = A_1^2$ ($d = 2$), $D(F) \cong \mathbb{F}_2^2$, and the three non-zero classes have $q = 1/2, 1/2, 1$. So $\delta = 1$, $s = 2$ and $d \equiv 2 \pmod 4$, and the lemma gives $|O(q)| = 2\ |\mathrm{Sp}_0(2)| = 2$: the only non-trivial isometry exchanges the two classes with $q = 1/2$. For $F = D_4$ ($d = 4$), the three non-zero classes all have $q = 1$. So $\delta = 0$, $s = 2$, and since $d \not\equiv 0 \pmod 8$ the type is $\varepsilon = -$; the lemma gives $|\mathrm{GO}_2^-(2)| = 6$, all permutations of the three classes. For $F = \sqrt{2}E_8$, the component of type $(8, 24)$ (§4.1.3), $q(x) = (x, x)/2 \bmod 2$ on $E_8/2E_8$, so $s = 8$, $\delta = 0$, $d = 8$, and $|O(q)| = |\mathrm{GO}_8^+(2)| = 348{,}364{,}800 = |W(E_8)|\ /2$, in accordance with §4.1.3. (The first two values can be checked by listing the six bijections of the three non-zero classes.)

2.4 Hermitian structures

If $g \in \mathrm{Aut}\ L$ has minimal polynomial the cyclotomic polynomial Φ_k, then L is a torsion-free module over $\mathbb{Z}[\zeta_k]$ by $\zeta_k x = xg$, of rank $32/\varphi(k)$, and $(x, y) = \mathrm{Tr}\ H(x, y)$ for a unique Hermitian form H. An isometry $L \to L'$ conjugating g to g' is the same as an isometry of Hermitian lattices, so the pairs (L, g) up to conjugacy correspond to the isometry classes of Hermitian lattices, and $C(g)$ is the unitary group $\mathrm{U}(L, H)$. The pair (L, g^{-1}) corresponds to the complex conjugate lattice $\left(L, \overline{H}\right)$; it is isometric to (L, H) exactly when g is conjugate to g^{-1} in $\mathrm{Aut}\ L$. For $k = 2^a$, $g^{-1} = -g^{2^{a-1}-1}$, which is $-g$ for $a = 2$. For $k = 4$, the rootless mass M_2 is $\sum 1/\ |\mathrm{U}(L, H)|$ over the Gaussian classes whose trace form is extremal (§4.3).

Let $a \geq 2$, $\mathbb{K} = \mathbb{Q}(\zeta)$ with ζ a primitive 2^a-th root of unity, $\mathcal{O} = \mathbb{Z}[\zeta]$ its ring of integers, $\mathbb{K}^+ = \mathbb{Q}(\zeta + \zeta^{-1})$ and $\mathcal{O}^+ = \mathcal{O} \cap \mathbb{K}^+$. The element $\varpi = 1 - \zeta$ generates the unique prime of $\mathcal{O}$ above 2, and $\mathcal{O}/\varpi\mathcal{O} = \mathbb{F}_2$. Let $g \in \mathrm{Aut}\ L$ satisfy $g^{2^{a-1}} = -1$. Then L is an $\mathcal{O}$-module via $\zeta x = xg$, torsion free of rank $r = 32/2^{a-1}$, and since g is an isometry there is a unique form $H : L \times L \to \mathbb{K}$, $\mathcal{O}$-linear in the first variable and Hermitian, with $(x, y) = \mathrm{Tr}_{\mathbb{K}/\mathbb{Q}} H(x, y)$. We call r the rank of L over $\mathcal{O}$.

Lemma 2.2. If L is even unimodular, then r is even.

Proof. Let $\mathfrak{D}$, $\mathfrak{D}^+$ and $\mathfrak{D}_{\mathbb{K}/\mathbb{K}^+}$ be the differents of $\mathbb{K}/\mathbb{Q}$, $\mathbb{K}^+/\mathbb{Q}$ and $\mathbb{K}/\mathbb{K}^+$, so that $\mathfrak{D} = \mathfrak{D}_{\mathbb{K}/\mathbb{K}^+}\mathfrak{D}^+$. Since $\mathcal{O} = \mathcal{O}^+[\zeta]$ and ζ is a root of $X^2 - (\zeta + \zeta^{-1})X + 1$, we have $\mathfrak{D}_{\mathbb{K}/\mathbb{K}^+} = (\zeta - \zeta^{-1}) = (1 - \zeta^2) = (1 - \zeta)(1 + \zeta)\mathcal{O} = \varpi^2\mathcal{O}$, because $-\zeta$ is also a primitive 2^a-th root of unity. Put $\mathfrak{p}^+ = \varpi^2\mathcal{O} \cap \mathcal{O}^+$, the prime of $\mathcal{O}^+$ above 2.

Since $H(x, L)$ is an $\mathcal{O}$-module, $(x, L) \subset \mathbb{Z}$ if and only if $H(x, L) \subset \mathfrak{D}^{-1}$. Choose a generator d of $\mathfrak{D}$ and put $H' = dH$. Unimodularity gives $H'(L, L) \subset \mathcal{O}$, and if $H'(x, L) \subset \varpi\mathcal{O}$ then $H(\varpi^{-1}x, L) \subset \mathfrak{D}^{-1}$, so $\varpi^{-1}x \in L$. Hence $\omega(x, y) = H'(x, y) \bmod \varpi$ is a non-degenerate form on $L/\varpi L \cong \mathbb{F}_2^r$. It is bilinear and symmetric, since complex conjugation and every unit act trivially on $\mathcal{O}/\varpi = \mathbb{F}_2$.

It remains to show that ω is alternating. Fix $x \in L$ and put $c = H(x, x) \in \mathbb{K}^+$. For $\alpha \in \mathcal{O}$, evenness of L applied to αx and to $(1 + \alpha)x$ gives $2\mathrm{Tr}_{\mathbb{K}^+/\mathbb{Q}}(\alpha\overline{\alpha}c) \in 2\mathbb{Z}$ and $2\mathrm{Tr}_{\mathbb{K}^+/\mathbb{Q}}((\alpha + \overline{\alpha})c) \in 2\mathbb{Z}$. The elements $\alpha\overline{\alpha}$ and $\alpha + \overline{\alpha}$ ($\alpha \in \mathcal{O}$) span $\mathcal{O}^+$ over $\mathbb{Z}$: the traces $\alpha + \overline{\alpha}$ form an ideal containing 2 and $\zeta + \zeta^{-1}$, which is $\mathfrak{p}^+$ (for $a = 2$ it is $2\mathbb{Z}$, and for $a \geq 3$ the element $\zeta + \zeta^{-1} \equiv -\varpi\overline{\varpi} \pmod 2$ has valuation 1

at $\mathfrak{p}^+$), and $1 = 1 \cdot \overline{1}$ generates $\mathcal{O}^+/\mathfrak{p}^+ = \mathbb{F}_2$. Hence $\mathrm{Tr}_{\mathbb{K}^+/\mathbb{Q}}(tc) \in \mathbb{Z}$ for all $t \in \mathcal{O}^+$, that is $c \in (\mathfrak{D}^+)^{-1}$, and $H'(x,x) = dc \in \mathfrak{D}(\mathfrak{D}^+)^{-1} = \varpi^2\mathcal{O}$. So $\omega(x,x) = 0$. A non-degenerate alternating form exists only in even dimension. □

For $a = 6$ the rank is $r = 1$, so by Lemma 2.2 no even unimodular lattice of rank 32 has an automorphism g with $g^{32} = -1$. For $a \geq 7$ the degree 2^{a-1} of the minimal polynomial exceeds 32.

Three normalizations of a Hermitian form. Table 3 lists them; here is the reason for each, with the smallest examples. This remark is not used in the proofs.

- §2.4 uses the form H with $(x,y) = \mathrm{Tr}\ H(x,y)$, the trace to $\mathbb{Q}$; it compares directly with the integer lattice.
- For $\mathbb{Z}[i]$ (§4.3, Appendix C) the form $h = 2H$ is used, so that $(x,y) = \mathrm{Re}\ h(x,y)$ and a root has $h(x,x) = 2$, as in the integer lattice. For example, in $A_1[i] = A_1 \otimes \mathbb{Z}[i]$ (Proposition C.1, an A_1^2 over $\mathbb{Z}$) a root x has $(x,x) = 2$, $H(x,x) = 1$ and $h(x,x) = 2$.
- For an element of odd prime order p (Lemma 4.4, Appendix D) the rescaled form $\tilde{H} = \nu^{(p-3)/2}H$ is used, because the lattices U of Lemma 4.4 are unimodular for it, and the theory of unimodular Hermitian lattices then applies. For $p = 3$, $\tilde{H} = H$; for example $A_2 = \mathbb{Z}[\omega]$ with $(x,y) = \mathrm{Tr}(x\overline{y})$ has $H(1,1) = 1$.

Points that are easy to misread (conventions). These remarks add nothing to the proofs; they collect conventions of this paper on which misreadings are likely.

1. *Extremal means minimum 4.* In rank 32 an extremal lattice is exactly an even unimodular lattice without roots (§2.1). "Rootless" and "extremal" mean the same here.
2. $|\mathrm{Aut}\ L|$ *counts* ± 1. Every group order in this paper is the order of the full group $\mathrm{Aut}\ L$, which contains -1. So $|\mathrm{Aut}\ L| \geq 2$, and the "smallest possible group" is $\{\pm 1\}$, of order 2.
3. *A mass is not a number of lattices.* The mass m counts each lattice L with weight $1/\ |\mathrm{Aut}\ L|$. As $|\mathrm{Aut}\ L| \geq 2$, we have $2m \leq N$, with equality only if every group is $\{\pm 1\}$. Here $2m \approx 1.0969 \times 10^7$, while $N \approx 1.1044 \times 10^7$.
4. *Pairs are not lattices.* The lists of §4 list pairs (L,h) up to isometry, with weight $1/\ |C(h)|$. One lattice occurs once for each conjugacy class of its structures (§3.1), so a list of pairs may contain the same lattice several times. §6 groups the pairs lattice by lattice.
5. σ *and* $-\sigma$*;* g *and* $-g$. An involution σ and $-\sigma$ give the same lattice with F and K exchanged (§4.1.2). A Gaussian structure g and $-g$ give complex conjugate Hermitian lattices (§2.4). This is why the weight $f/\ |\mathrm{U}|$ of §4.3 has $f = 2$ or 1.
6. *Letters.* An element of $\mathrm{Aut}\ L$ is h and a Hermitian form is H; the lattice $L^{-\sigma}$ is K and a cyclotomic field is $\mathbb{K}$; the unitary group is U (upright) and a unimodular sublattice is U (italic): in this paper the upright U always denotes a unitary group and the italic U always a lattice; a prime element is ϖ and π is the circle constant. Two letters keep two meanings in the main text, because the certificates and the data use them: f is the rank of L^h in a type $p-(z,f)-s$ and also the factor 1 or 2 of a Gaussian class, and D is a discriminant group and also the invariant of §7.2 ([D] is the database). Table 2 lists the conventions of the main text, including the three uses of ω. Some letters have a further meaning inside one proof or one appendix, stated where it is used: for instance h is the Hermitian form of Appendices C and D, N a rank in Appendix D, and S, T the Gram matrices of Siegel's formula in Appendix C.

3. Detecting additional automorphisms

3.1 The three kinds of structure

A *pair* (L, h) consists of $L \in \mathcal{L}$ and $h \in \operatorname{Aut} L$. An isometry of pairs $(L, h) \to (L', h')$ is an isometry $L \to L'$ that conjugates h to h'. For fixed L, these classes are exactly the conjugacy classes in $\operatorname{Aut} L$.

Proposition 3.1. If $\operatorname{Aut} L \neq \{\pm 1\}$, then L has an automorphism of odd prime order, an involution other than ± 1, or an automorphism g with $g^2 = -1$.

Proof. Choose $x \in \operatorname{Aut} L \setminus \{\pm 1\}$. If its order has an odd prime divisor p, a power of x has order p. Otherwise its order is 2^a. For $a = 1$, x is the required involution. For $a \geq 2$, $x^{2^{a-1}}$ is an involution. If it is not -1, it is the required one; if it is -1, put $g = x^{2^{a-2}}$. □

3.2 From structured pairs to the classification

Take complete lists of the pairs in Proposition 3.1, identify their underlying integer lattices, and retain one representative per isometry class. Every lattice of S occurs, and every valid representative belongs to S. Thus proving that these representatives are distinct and determining their full group orders proves Theorem B. §6 gives the group-theoretic reductions that make this identification feasible. §5 then gives Theorem A from this classification and King's mass.

An earlier organization of the count by assigned group elements is given in the auxiliary note [Aux]; its numbers and higher cyclotomic mass formulas are not premises of this argument.

3.3 The types

Proposition 3.2. Let $\sigma \neq \pm 1$ be an involution of $L \in \mathcal{L}$, $F = L^\sigma = \{x \in L : x\sigma = x\}$, $K = L^{-\sigma}$ and $d = \operatorname{rank} F$. Then F and K are even 2-elementary lattices of minimum at least 4 with isomorphic discriminant groups $(\mathbb{Z}/2)^s$, and

$$d \in \{8, 12, 14, 15, 16, 17, 18, 20, 24\}.$$

Proof. F and K are primitive sublattices of L and orthogonal complements of each other, so the projections of the unimodular lattice L to $F \otimes \mathbb{Q}$ and $K \otimes \mathbb{Q}$ are $F^\#$ and $K^\#$, and $L/(F \perp K) \cong F^\#/F \cong K^\#/K$ [Nik79, §1.5, Prop. 1.6.1]. Since $2x = (x + x\sigma) + (x - x\sigma) \in F \perp K$ for $x \in L$, these groups are 2-elementary. As sublattices of L, F and K are even of minimum at least 4.

The form $\psi(x, y) = (x, y) \bmod 2$ on $F/2F^\#$ is well defined and alternating since F is even, and non-degenerate since $\{x \in F : (x, F) \subset 2\mathbb{Z}\} = F \cap 2F^\# = 2F^\#$. Let T with $2F^\# \subset T \subset F$ be the preimage of a maximal isotropic subspace; then $[T : 2F^\#] = 2^{(d-s)/2}$ and $\det T = 4^d 2^{-s} 2^{-(d-s)} = 2^d$. Hence $U = 2^{-1/2}T$ is an integral unimodular lattice of rank d with $\min U \geq \min F/2 \geq 2$, that is, without vectors of norm 1 (this is [Neb13b, Lemma 4.9]). The same holds for K with rank $32 - d$. Unimodular lattices without vectors of norm 1 exist in rank n only for $n = 0, 8, 12$ and $n \geq 14$ [SPLAG, Table 16.7]. Since $\sigma \neq \pm 1$, $0 < d < 32$, and both d and $32 - d$ must lie in that set. □

For an element h of odd prime order p, let $F = L^h$ of rank f and $Z = F^\perp \cap L$. Then Z is a module over $\mathbb{Z}[\zeta_p]$ via h, of rank z with $f + (p-1)z = 32$, and $F^\#/F \cong Z^\#/Z \cong (\mathbb{Z}/p)^s$ [Neb13b, Lemma 4.2]. We call $p - (z, f) - s$ the *type* of h [Neb13b, Def. 4.1].

Proposition 3.3. The type of an element of odd prime order of a lattice in $\mathcal{L}$ is one of the 19 types in Table 4.

Proof. Let h have type $p-(z,f)-s$. Then $p \le 31$ and $1 \le z \le 32/(p-1)$, $f = 32-(p-1)z$, $0 \le s \le \min(z,f)$ ($s \le f$ since $F^\#/F \cong \mathbb{F}_p^s$ is a quotient of $F^\#/pF^\# \cong \mathbb{F}_p^f$, and $s \le z$ by Proposition D.1), and $z-s$ is even by Lemma 4.4. There are 93 such quadruples with $p \in \{3,5,7,...,31\}$. Appendix B.1 (Table B1) excludes 74 of them, each for one of the following reasons; the remaining 19 are the types of Table 4.

1. *No genus* (7 quadruples): there is no genus of even p-elementary lattices of rank f (or $(p-1)z$) and determinant p^s; for odd p such a genus exists if and only if the rank n is even, $n+s(p-1) \equiv 0 \pmod 4$, and $8 \mid n$ when $s \in \{0,n\}$. This is [SPLAG, Ch. 15, Thm 13(a) and Table 15.4] for the genus $\mathrm{II}_{n,0}(p^{\pm s})$ in the notation there, where the signature is $(n,0)$ and $p^{\pm s}$ is the p-part of the discriminant group.
2. *Density* (59): F or Z has minimum at least 4 and determinant p^s, so its centre density is at least $p^{-s/2}$, which exceeds the Cohn–Elkies bound on the centre density in its dimension [CE03, Table 3]; the bounds of that table are proved by exact rational arithmetic [CE03, §7]. In each of the 59 cases $p^{-s/2}$ exceeds the printed bound by at least 3.7×10^{-3} (a factor of at least 1.034), far more than the rounding of the table to five decimals [C2].
3. *No rootless unimodular sublattice* (2): Z contains a Hermitian unimodular lattice U of the same rank (Lemma 4.4), which has no roots since L is extremal; but the mass of rootless Hermitian unimodular lattices of that rank is 0 [C2].
4. *Linear programming* (6): the theta series of F and Z are modular forms for $\Gamma_0(p)$ with character, exchanged with those of $\sqrt{p}F^\#$ and $\sqrt{p}Z^\#$ by the Fricke involution; with the constraints from $\min F, \min Z \ge 4$, Lemma 4.2 and Lemma 4.3, the exact linear program (over $\mathbb{Q}$) for their coefficients is infeasible; §B.1.1 gives a rational certificate of infeasibility for each of the six [C2, C3].

Remark. The proof uses only the linear programs for the six quadruples of reason 4. They are also excluded by complete lists of components (proved complete by their masses): for five of them the genus of F contains no lattice of minimum at least 4, and for $3-(8,16)-6$, whose certificate in §B.1.1 uses only the conditions on F (so that this genus of F has no lattice of minimum at least 4 either), no unimodular sublattice U of rank 8 admits a suitable overlattice Z [C2]. This is a cross-check and is not used.

Table 4. The possible detecting structures. For an involution, $d = \operatorname{rank} L^\sigma$; σ and $-\sigma$ have types $(d, 32-d)$ and $(32-d, d)$. For an element of odd prime order, the type is $p-(z,f)-s$ (Table 2).

kind	types	§
involution $\neq \pm 1$	$(8,24)$, $(12,20)$, $(14,18)$, $(15,17)$, $(16,16)$	4.1
order 3	$3-(6,20)-6$, $3-(8,16)-8$, $3-(9,14)-7$, $3-(9,14)-9$, $3-(10,12)-6$, $3-(10,12)-8$, $3-(10,12)-10$, $3-(12,8)-8$, $3-(13,6)-5$, $3-(16,0)-0$	4.2
order 5	$5-(4,16)-4$, $5-(5,12)-5$, $5-(6,8)-4$, $5-(6,8)-6$, $5-(7,4)-3$, $5-(8,0)-0$	4.2
order 7, 17, 31	$7-(4,8)-4$, $17-(2,0)-0$, $31-(1,2)-1$	4.2
$\mathbb{Z}[i]$	rank 16; $g^2=-1$	4.3

4. Complete lists of structured pairs

What this section does. We list components and gluings for involutions and odd prime order (§§4.1–4.2), and Gaussian pairs (§4.3). Completeness follows from exhaustive reductions or exact positive mass identities, provided each candidate is valid, listed classes are distinct and their weights are correct. §6 then identifies the underlying integer lattices.

4.1 Pairs of prime order and involutions

4.1.1 Gluings of prime order

Let h be an automorphism of prime order p of $L \in \mathcal{L}$, $h \neq -1$ if $p = 2$. Put $F = L^h$ and $Z = F^\perp \cap L$; for $p = 2$ we write $K = Z = L^{-h}$. Let $G_F = \mathrm{Aut}\ F$, let G_Z be the centralizer of $h|_Z$ in $\mathrm{Aut}\ Z$ (for $p = 2$, $h|_Z = -1$ and $G_Z = \mathrm{Aut}\ Z$), and let $\overline{G}_F$, $\overline{G}_Z$ be their images in the orthogonal groups of the discriminant forms q_F, q_Z. The lattice L is the gluing

$$L = L_\varphi = \{x + y : x \in F^\#, y \in Z^\#, \varphi(x + F) = y + Z\}$$

along the isomorphism $\varphi : F^\#/F \to Z^\#/Z$ induced by the projections, which satisfies $q_Z \circ \varphi = -q_F$.

Proposition 4.1. Fix F and $(Z, h|_Z)$ up to isometry. Let $\mathrm{Iso}(q_F, -q_Z)$ be the set of isomorphisms φ with $q_Z \circ \varphi = -q_F$.

1. For every $\varphi \in \mathrm{Iso}(q_F, -q_Z)$, L_φ is an even unimodular lattice and $\mathrm{id}_F \perp h|_Z$ extends to an automorphism h_φ of L_φ of order p with fixed lattice F.
2. $\varphi \mapsto \big(L_\varphi, h_\varphi\big)$ induces a bijection between the double cosets $\overline{G}_Z \setminus \mathrm{Iso}(q_F, -q_Z)/\overline{G}_F$ and the isometry classes of pairs (L, h) with these F and $(Z, h|_Z)$, where L runs over all even unimodular lattices of rank 32.
3. The stabilizer $\mathrm{Stab}(\varphi) = \{(u, v) \in G_F \times G_Z : \overline{v}\varphi\overline{u}^{-1} = \varphi\}$ is isomorphic to $C\big(h_\varphi\big)$.
4. Consequently

$$\sum_{[\varphi]} \frac{1}{|\mathrm{Stab}(\varphi)|} = \frac{|\mathrm{Iso}(q_F, -q_Z)|}{|G_F|\ |G_Z|}. \tag{4.1}$$

Proof. (i) L_φ is even unimodular since φ is an anti-isometry of discriminant forms [Nik79, Prop. 1.4.1, 1.6.1]. The element $h|_Z$ acts trivially on $Z^\#/Z$: for $p = 2$ because $Z^\#/Z$ is 2-elementary, and for odd p because $(1 - h)Z^\# \subset Z$ [Neb13b, proof of Lemma 4.2]. Hence $\mathrm{id}_F \perp h|_Z$ preserves L_φ. (ii) and (iii): an isometry $L_\varphi \to L_{\varphi'}$ commuting with h_φ, $h_{\varphi'}$ maps F to F and Z to Z, so it is a pair $(u, v) \in G_F \times G_Z$ with $\overline{v}\varphi\overline{u}^{-1} = \varphi'$, and conversely. Taking $\varphi = \varphi'$ gives (iii). (iv) is the orbit–stabilizer formula for the action of $G_F \times G_Z$ on $\mathrm{Iso}(q_F, -q_Z)$. □

When $\mathrm{Iso}(q_F, -q_Z)$ is non-empty, $|\mathrm{Iso}(q_F, -q_Z)| = |O(q_F)|$. Identity (4.1) is what makes the lists complete: the right side is computed from $|O(q_F)|$, $|G_F|$ and $|G_Z|$ alone, every term of the left side is positive, so a list of double cosets whose left side equals the right side contains every double coset. Every pair (L, h) with $L \in \mathcal{L}$ arises from components F and Z of minimum at least 4, so we list all components of minimum at least 4 in the relevant genera, list all double cosets for each pair of components, prove the lists complete by (4.1), and then keep the double cosets for which L_φ has no vector of norm 2. Each component list is in turn proved complete in Appendix B: for involutions by an enumeration that provably reaches every lattice in question (for type $(8, 24)$ together with identity (B.1), Theorem B.3), for elements of odd prime order by an exact equality with the mass of the genus or of its rootless part.

Example (a gluing in rank 16). This example is not used in the proofs; it shows Proposition 4.1 and identity (4.1) on the smallest familiar case. Let $L = E_8 \perp E_8$ and let σ exchange the two summands, $(x, y)\sigma = (y, x)$. Then $F = L^\sigma = \{(x, x)\}$ and $K = L^{-\sigma} = \{(x, -x)\}$ ($x \in E_8$) are both isometric to $\sqrt{2}E_8$. Their duals are $F^\# = \{(x, x)/2\}$ and $K^\# = \{(x, -x)/2\}$, so $D(F) \cong D(K) \cong E_8/2E_8$. The gluing φ sends $(x, x)/2 + F$ to $(x, -x)/2 + K$, and $(x, x)/2 + (x, -x)/2 = (x, 0)$, so L_φ is $E_8 \perp E_8$ again.

No other gluing of these two components exists. The argument of Proposition 4.1 works in any rank, and the right side of (4.1) is $|O(q_F)| \,/\, |\mathrm{Aut}\ F|^2$, where $|\mathrm{Aut}\ F| = |W(E_8)| = 696{,}729{,}600$ and $|O(q_F)| = |\mathrm{GO}_8^+(2)| = |W(E_8)|\ /2$ (the example after Lemma 2.1). So the right side is $1/(2\ |W(E_8)|)$. On the left, $C(\sigma)$ consists of the pairs (w, w) with $w \in W(E_8)$ and their products with σ, so $|C(\sigma)| = 2\ |W(E_8)|$. This single term already equals the right side, and every term is positive. Hence an even unimodular lattice of rank 16 with an involution whose fixed lattice and its complement are both $\sqrt{2}E_8$ is $E_8 \perp E_8$; for instance D_{16}^+ has no such involution. The same mechanism, that $W(E_8) \to O(q_F)$ is onto, gives exactly one double coset for each lattice K of type $(8, 24)$ (§4.1.3).

4.1.2 Orientation

The involutions σ and $-\sigma$ give the same integer lattice. If $F \nsim K$, the inverse gluing belongs to the list with the components interchanged, so one of the two orientations suffices. If $F \cong K$, inversion sends $[\varphi]$ to $[\varphi^{-1}]$ within the same list. These classes coincide exactly when σ is conjugate to $-\sigma$. §6.1 uses this relation to define involution entries. No element-assignment count is needed for this identification.

4.1.3 The five types

Let $\sigma \neq \pm 1$ be an involution of $L \in \mathcal{L}$ of type $(d, 32 - d)$, $d \leq 16$ (§4.1.2). The components $F = L^\sigma$ and $K = L^{-\sigma}$ are even, 2-elementary ($2F^\# \subset F$), of minimum at least 4, and their discriminant forms are anti-isometric. Hence they have the same invariants s ($F^\#/F \cong \mathbb{F}_2^s$) and δ ($\delta = 0$ if $q_F(x) = (x, x) \bmod 2\mathbb{Z}$ takes integral values, $\delta = 1$ otherwise). Appendix B.2 lists all such lattices in the ranks that occur (Table B2); for $d \leq 15$ there is exactly one F, of the form $\sqrt{2}U$ with U unimodular. Table 5 shows the components of each type.

Extremality. By the proof of Proposition 3.2, F contains $\sqrt{2}U$ with U unimodular and $\min U \geq 2$, so $F^\# \subset U/\sqrt{2}$ and every non-zero vector of $F^\#$ has norm at least 1; the same holds for K. A vector of L_φ outside $F \perp K$ is $x + y$ with $x \in F^\# \setminus F$, $y \in K^\# \setminus K$, and if it has norm 2 then $(x, x) = (y, y) = 1$. Vectors of $F \perp K$ have norm 0 or at least 4. Hence L_φ *is extremal if and only if there is no class* $a \in F^\#/F$ *such that both* a *and* $\varphi(a)$ *contain a vector of norm 1.* This depends only on the double coset of φ and is decided from the lists of the classes of norm 1 in $F^\#/F$ and $K^\#/K$.

Gluings. For each pair (F, K) of components, the double cosets of Proposition 4.1(ii) were listed by computing a canonical form of φ under $\overline{G}_K \times \overline{G}_F$ (§7.2) and generating new classes by composing φ with reflections $x \mapsto x + 2b(x, v)v$ of $F^\#/F$ in elements with $q_F(v) \equiv 1$, and by random walks, among them walks that keep a given element of a known stabilizer [C4].

How the classes were found plays no role: for each of the 83 pairs of components of types $(15, 17)$, $(14, 18)$, $(12, 20)$ and $(16, 16)$, the sum of $1/\ |\mathrm{Stab}(\varphi)|$ over the listed classes equals, as a fraction, the value of the right side of (4.1) computed below, where $|\mathrm{Iso}(q_F, -q_K)| = |O(q_F)|$ since the list is not empty. The value of $|O(q_F)|$ was computed as $R(q_F)$, where $R(q) = n(q)R(q|_{H^\perp})$, $n(q)$ is the number of ordered pairs (x, y) with $q(x) = q(y) = 0$ and $b(x, y) = 1/2$, and $H = \langle x, y\rangle$ for one such pair; for forms

on a group of rank at most 3 and for forms without such pairs, $R(q) = |O(q)|$ is counted exhaustively. The stabilizer of (x, y) in $O(q)$ is isomorphic, by restriction to $H^\perp$, to $O(q|_{H^\perp})$, and its orbit has at most $n(q)$ elements, so $|O(q)| \le n(q)\ |O(q|_{H^\perp})|$ and, by induction on the rank of the group, $R(q) \ge |O(q)|$, whether or not $O(q)$ is transitive on these pairs. It was computed once for each genus, since the lattices of a genus have isometric discriminant forms [Nik79, Cor. 1.9.4]. The listed classes are distinct, so their sum is at most the true right side of (4.1), which is at most the one computed with $R(q_F)$; since the two ends are equal, the list is complete and $R(q_F) = |O(q_F)|$. Independently of the program for R, every recorded value equals the closed formula of Lemma 2.1(ii), which the checks of §7.4 recompute from the rank of F and the invariants s, δ of its Gram matrix; so the right side of (4.1) is the true one. *In short*, $R(q)$ counts pairs (x, y) with $q(x) = q(y) = 0$ and $b(x, y) = 1/2$ and recurses on their orthogonal complement; this gives an upper bound for $|O(q)|$ and hence for the right side of (4.1), while the sum over the listed classes is a lower bound for the true right side; when the two agree, both the list and the value are certified. The closed formula of Lemma 2.1(ii) confirms the value independently. The numbers of classes and of extremal classes are in Tables 5 and 6.

Type (15, 17). $F = \sqrt{2}A_{15}^+$, where A_{15}^+ is the unimodular lattice of rank 15 without vectors of norm 1 [SPLAG, Table 16.7], and K is the unique lattice of rank 17 with $s = 15$, $\delta = 1$ and minimum at least 4. There are 82,086 classes, 20,502 of them extremal. The type $(17, 15)$ needs no list of its own: by §4.1.2 its pairs $(L, -\sigma)$ are the inverse gluings $[\varphi^{-1}]$ of the classes of this list, and Table 5 counts M for both orientations.

Type (14, 18). $F = \sqrt{2}(E_7^2)^+$, and K is one of five lattices of rank 18 with $s = 14$, $\delta = 1$. There are 54,649 classes, 16,590 of them extremal.

Type (12, 20). $F = \sqrt{2}D_{12}^+$, and K is one of the 66 lattices of rank 20 with $s = 12$, $\delta = 1$ and minimum at least 4 [C5]. There are 27,434 classes, 13,798 of them extremal.

Type (16, 16). There are ten components of rank 16, one for each (s, δ) with $s \in \{8, 10, 12, 14\}$ except $(8, 1)$, and three with $s = 16$: $\sqrt{2}E_8^2$ and $\sqrt{2}D_{16}^+$ ($\delta = 0$) and $\sqrt{2}(D_8^2)^+$ ($\delta = 1$). Components with equal (s, δ) give the eleven pairs of Table 6. In ten of them $F \cong K$, and (L, σ), $(L, -\sigma)$ are both in the list (§4.1.2); for the pair $\left(\sqrt{2}E_8^2, \sqrt{2}D_{16}^+\right)$ the mass with both orientations counts this list twice. The pair $s = 16$, $\delta = 1$ alone has 222,208 classes before retaining only the extremal pairs.

Type (8, 24). Here the gluing is trivial and the work is in the list of K. The only possible F is $\sqrt{2}E_8$ (Appendix B.2), with $\mathrm{Aut}\ F = W(E_8)$, and q_F is the quadratic form $x \mapsto (x, x)/2 \bmod 2$ on $E_8/2E_8$. It is well known that $W(E_8) \to O(q_F) = \mathrm{GO}_8^+(2)$ is onto with kernel $\{\pm 1\}$. By Proposition 4.1, $\overline{G}_F = O(q_F)$ acts simply transitively on $\mathrm{Iso}(q_F, -q_K)$, so for each K there is exactly one double coset; and for every $v \in \mathrm{Aut}\ K$ there are exactly two $u \in W(E_8)$ with $(u, v) \in \mathrm{Stab}(\varphi)$, so $|C(\sigma)| = 2\ |\mathrm{Aut}\ K|$. Thus the pairs (L, σ) of type $(8, 24)$ correspond to the lattices K, and their contribution to M, counted for σ and $-\sigma$, is $\sum_K 1/\ |\mathrm{Aut}\ K|$. By Proposition B.2, L is extremal if and only if $\min K \ge 4$ and $\min K^\# \ge 2$. There are exactly 14,482 such K (Theorem B.3 and §B.3), with

$$\sum_K \frac{1}{|\mathrm{Aut}\ K|} = \frac{4{,}770{,}461{,}726{,}538{,}527}{792{,}723{,}456{,}000} \approx 6{,}017.813$$

[C6].

Table 5. Involution pairs: components, gluing classes, extremal classes and mass $M = \sum 1/\ |\mathrm{Stab}(\varphi)|$ over the extremal classes, with both orientations. For type $(8, 24)$ only the lattices K that give extremal lattices are listed, so its classes are all extremal. Components of type $(16, 16)$ are detailed in Table 6 [C4, C5, C6, C7].

type	F	K	classes	extremal	M (approx.)
$(15,17)$	$\sqrt{2}A_{15}^{+}$; 1, $s=15$, $\delta=1$	1 class, $s=15$, $\delta=1$	82,086	20,502	9,664.170
$(14,18)$	$\sqrt{2}(E_7^2)^{+}$; 1, $s=14$, $\delta=1$	5 classes, $s=14$, $\delta=1$	54,649	16,590	6,876.012
$(12,20)$	$\sqrt{2}D_{12}^{+}$; 1, $s=12$, $\delta=1$	66 classes, $s=12$, $\delta=1$	27,434	13,798	5,507.962
$(8,24)$	$\sqrt{2}E_8$; 1, $s=8$, $\delta=0$	14,482 lattices, $s=8$	14,482	14,482	6,017.813
$(16,16)$	10 classes, $8 \le s \le 16$	same	233,544	46,805	9,883.635

Table 6. The eleven pairs of components of type $(16,16)$. In each pair $F \cong K$ except the one marked $F \nsim K$, whose M is counted for both orientations. "$C=4$" is the number of extremal classes with $|\mathrm{Stab}(\varphi)|=4$. A dash in the column of components: the genus of F contains exactly one lattice of minimum at least 4, and $F \cong K$ is that lattice.

s, δ	components	$\lvert\mathrm{Aut}\ F\rvert$	classes	extremal	$C=4$	M (exact, extremal classes)
8, 0	—	89,181,388,800	2	2	0	1/ 22,830,435,532,800
10, 0	—	21,139,292,160	9	4	0	255/ 3,758,096,384
10, 1	—	743,178,240	4	4	0	1/ 6,193,152
12, 0	—	195,689,447,424	122	33	0	7,139,633 / 7,247,757,312
12, 1	—	754,974,720	72	33	0	16,750,411 / 251,658,240
14, 0	—	53,271,016,243,200	1,461	135	0	151,662,361 / 469,762,048
14, 1	—	32,614,907,904	6,108	1,702	667	11,480,973,667 / 50,331,648
16, 0	$\sqrt{2}E_8^2$, $\sqrt{2}E_8^2$	970,864,271,032,320,000	1,337	81	0	5,800,534,177 / 13,212,057,600
16, 0	$\sqrt{2}E_8^2$, $\sqrt{2}D_{16}^{+}$ $(F \nsim K)$	as in the adjacent rows	1,080	51	0	$2 \cdot 3{,}198{,}263$ / 5,160,960
16, 0	$\sqrt{2}D_{16}^{+}$, $\sqrt{2}D_{16}^{+}$	685,597,979,049,984,000	1,141	74	0	925,264,055 / 1,056,964,608
16, 1	$\sqrt{2}(D_8^2)^{+}$, $\sqrt{2}(D_8^2)^{+}$	53,271,016,243,200	222,208	44,686	34,940	8,502,033,728,743 / 880,803,840

4.2 Elements of odd prime order

We use the notation of §3.3 and §4.1.1 with h of odd prime order p and type $p-(z,f)-s$.

Lemma 4.2. If $\min Z \ge 4$, then $\min Z^{\#} \ge 2(p-1)/p$.

Proof. Let $0 \ne y \in Z^{\#}$ and $\beta = (y,y)$. Since $(1-h^i)y \in Z$ is non-zero for $1 \le i < p$ [Neb13b, proof of Lemma 4.2], its norm is at least 4. As $\sum_{i=0}^{p-1} yh^i = 0$, we have $\sum_{i=1}^{p-1}(y,yh^i) = -\beta$, and $4(p-1) \le \sum_{i=1}^{p-1}(y-yh^i, y-yh^i) = 2(p-1)\beta + 2\beta = 2p\beta$. □

Lemma 4.3. Let $\min F \ge 4$ and $\min Z \ge 4$. Then L_φ has a vector of norm 2 if and only if there are $a \in F^{\#}/F$ containing a vector of norm $2/p$ and $b \in Z^{\#}/Z$ containing a vector of norm $2-2/p$ with $\varphi(a) = b$.

Proof. A vector of norm 2 is $x+y$ with $x \in F^{\#}$, $y \in Z^{\#}$ and $(x,x)+(y,y)=2$; neither is 0 by the assumption on the minima. Since $F^{\#}/F$ has exponent p and F is even, $(x,x) \in (2/p)\mathbb{Z}$, and $(y,y) \ge 2-2/p$ by Lemma 4.2. Hence $(x,x) = 2/p$ and $(y,y) = 2-2/p$. Conversely, if $x \in a$ has norm $2/p$ and $y \in b$ has norm $2-2/p$, then $x+y \in L_\varphi$ has norm 2. □

In particular, if $F^{\#}$ has no vector of norm $2/p$ or $Z^{\#}$ has none of norm $2-2/p$, every gluing is extremal.

Lemma 4.4. Let $\varpi = 1-\zeta_p$. There is a Hermitian form $\tilde{H}$ on Z over $\mathbb{Z}[\zeta_p]$, a rescaling of the one with trace $(\cdot,\cdot)$, such that $Z^* := \{x : \tilde{H}(x,Z) \subset \mathbb{Z}[\zeta_p]\}$ equals $\varpi Z^{\#}$, $Z^* \subset Z \subset \varpi^{-1}Z^*$, and Z/Z^* carries a non-degenerate alternating form over $\mathbb{F}_p$. Hence $z-s$ is even, and the lattices U with $Z^* \subset U \subset Z$ corresponding to maximal isotropic subspaces are unimodular for $\tilde{H}$, of rank z and index $p^{(z-s)/2}$ in Z.

Proof. Let $\mathbb{K} = \mathbb{Q}(\zeta_p)$, $\mathcal{O} = \mathbb{Z}[\zeta_p]$, on which h acts as ζ_p, and $\mathbb{K}^+$ its real subfield. The different of $\mathbb{K}/\mathbb{Q}$ is $\mathfrak{D} = \varpi^{p-2}\mathcal{O}$, and, since $\zeta_p \equiv 1 \pmod{\varpi}$, complex conjugation acts trivially on $\mathcal{O}/\varpi = \mathbb{F}_p$. Let H be the Hermitian form with $(x,y) = \mathrm{Tr}_{\mathbb{K}/\mathbb{Q}} H(x,y)$, so that $Z^{\#} = \{x : H(x,Z) \subset \mathfrak{D}^{-1}\}$. Put $\nu = \varpi\overline{\varpi}$, a totally positive element of $\mathbb{K}^+$ with $\nu\mathcal{O} = \varpi^2\mathcal{O}$, let $\beta = \nu^{(p-3)/2}$, so that $\beta\mathcal{O} = \varpi^{p-3}\mathcal{O}$, and $\tilde{H} = \beta H$, again Hermitian since β is real. Then $\tilde{H}(x,Z) \subset \mathcal{O}$ if and only if $H(x,Z) \subset \varpi\mathfrak{D}^{-1}$, so $Z^* = \varpi Z^{\#}$. Since ϖ acts as $1-h$ and $(1-h)Z^{\#} \subset Z$ [Neb13b, proof of Lemma 4.2], $Z^* \subset Z$; and $H(Z,Z) \subset \mathfrak{D}^{-1}$ gives $\tilde{H}(Z,Z) \subset \varpi^{-1}\mathcal{O}$, so $\varpi Z \subset Z^*$. The index is $[Z:Z^*] = [Z^{\#} : \varpi Z^{\#}]/[Z^{\#} : Z] = p^{z-s}$.

On Z/Z^* put $a(x,y) = \varpi\tilde{H}(x,y) \bmod \varpi \in \mathbb{F}_p$. It is well defined since $\tilde{H}(Z^*,Z) \subset \mathcal{O}$, and non-degenerate since $\varpi\tilde{H}(x,Z) \subset \varpi\mathcal{O}$ means $x \in Z^*$. As $\overline{\varpi} = -\zeta_p^{-1}\varpi \equiv -\varpi \pmod{\varpi^2}$, $a(y,x) = -a(x,y)$. Finally $\tilde{H}(x,x) \in \mathbb{K}^+ \cap \varpi^{-1}\mathcal{O}$, and elements of $\mathbb{K}^+$ have even ϖ-adic valuation because ϖ is ramified in $\mathbb{K}/\mathbb{K}^+$; so $\tilde{H}(x,x) \in \mathcal{O}$ and $a(x,x) = 0$. Thus a is a non-degenerate alternating form, and $z-s$ is even.

The dual of Z^* with respect to $\tilde{H}$ is Z (because $\tilde{H}(x, \varpi Z^{\#}) \subset \mathcal{O}$ if and only if $H(x, Z^{\#}) \subset \mathfrak{D}^{-1}$). Hence for $Z^* \subset U \subset Z$ the $\tilde{H}$-dual U^* satisfies $U^* \subset Z$, and U^*/Z^* is the orthogonal of U/Z^* for a. If U/Z^* is maximal isotropic, it equals its orthogonal, so $U^* = U$. □

Roots of unimodular sublattices. Let U be as in Lemma 4.4, with the form $\tilde{H}$ of Lemma 4.4. If $x \in U$ is a root, then $\mathcal{O}x$ is an orthogonal summand of U isometric to the root lattice A_{p-1} with h acting as a Coxeter element. Indeed, the roots xh^i span $\mathcal{O}x$, a root lattice of rank $p-1$ with an automorphism h of order p without fixed vectors; h cannot permute p components, so it preserves the component containing x, which therefore is all of $\mathcal{O}x$; and the only irreducible root lattice of rank $p-1$ with an automorphism of order p is A_{p-1}, on which all such automorphisms are conjugate (they are p-cycles in $W(A_{p-1}) \times \{\pm 1\}$). For $A_{p-1} = \varpi\mathcal{O}$ with $(a,b) = \mathrm{Tr}_{\mathbb{K}/\mathbb{Q}}(a\bar{b})/p$, the form of Lemma 4.4 is $\tilde{H}(a,b) = (\varpi\overline{\varpi})^{(p-3)/2} a\bar{b}/p$, so $\tilde{H}(\varpi,\varpi) = (\varpi\overline{\varpi})^{(p-1)/2}/p$ is a unit. So $\tilde{H}(x,x)$ is a unit, and since U is unimodular, $\mathcal{O}x$ splits off. A root y of U is $y = a+b$ with $a \in \mathcal{O}x$, $b \perp \mathcal{O}x$, and $(a,a)+(b,b) = 2$ with both terms in $2\mathbb{Z}$, so $y \in \mathcal{O}x$ or $y \perp x$. Hence

$$U = A_{p-1}^k \perp U', \quad |\mathrm{U}(U)| = (2p)^k k!\, |\mathrm{U}(U')|, \tag{4.2}$$

with U' rootless and determined by U. Summing (4.2) over a genus expresses its mass through the rootless masses of smaller rank; this is how the rootless masses used in Proposition 3.3 (reason 3) and below are obtained (Appendix D).

Components. For h of type $p-(z,f)-s$, F is an even p-elementary lattice of rank f and determinant p^s, and Z is a Hermitian $\mathcal{O}$-lattice of rank z on which h acts as ζ_p; both have minimum at least 4. For odd p the genus of F is determined by (f,s): a genus is determined by its rank, signature and discriminant form [Nik79, Cor. 1.9.4], and the discriminant form on $\mathbb{F}_p^s$ is fixed by the existence condition of [SPLAG, Ch. 15, Thm 13(a)]; and the genus of Z is determined by (z,s) (Proposition D.1). Appendix B.4 lists the components and proves the lists complete (Table B4): the lattices F by a

classification of their genus by Kneser neighbours, completed when $\sum 1/\,|\mathrm{Aut}\ F|$ reaches the mass of the genus, or, for two genera, by the rootless mass of the genus; the lattices Z through their unimodular sublattices U, whose lists are complete by the rootless Hermitian mass, and then $Z \supset U$ by Lemma 4.4, step by step in s, with a mass identity at each step.

Gluings. For each pair (F, Z), the double cosets of Proposition 4.1(ii) were computed with GAP as double cosets in $O(q_F)$, a group whose order $|O(q_F)|$ is given by Lemma 2.1(i) (the recorded orders agree with it, §7.4), realized as a permutation group on $\mathbb{F}_p^s$, or by a canonical form when GAP did not finish [C8, C9]; in every case identity (4.1) holds as an equality of fractions. Extremality is decided by Lemma 4.3. If $F^\#$ has no vector of norm $2/p$ or $Z^\#$ has none of norm $2-2/p$, every gluing is extremal; this holds for 8 of the 16 types with $F \neq 0$.

Order 3. The ten types have 6,617 extremal pairs; 5,383 of them have $|C(h)| = 6$. For $3-(12,8)-8$, the largest type, $F = \sqrt{3}E_8$ with $\overline{G}_F = W(E_8)$, Z runs over 51 classes, and the double cosets are the orbits of $\mathrm{U}(Z)$ on the 56,862 cosets of $W(E_8)$ in $\mathrm{GO}_8^+(3)$; there are 15,423 pairs, of which 5,872 are extremal. For $3-(6,20)-6$, Z is the Coxeter–Todd lattice K_{12} with its Eisenstein structure and $\overline{G}_Z$ has index 2 in $O(q_Z) \cong \mathrm{GO}_6^-(3)$, so each F gives one or two double cosets; GAP shows that two occur for exactly one of the 22 lattices F with $F^\#$ free of vectors of norm $2/3$, which gives 23 extremal pairs. For the other two lattices F, whose duals contain vectors of norm $2/3$, no gluing is extremal: a class $a \in F^\#/F$ containing such a vector has $q_F(a) = 2/3$, so $q_Z(\varphi(a)) = 4/3$ for every φ, and every class of $K_{12}^\#/K_{12}$ with $q = 4/3$ contains a vector of norm $4/3$, so Lemma 4.3 applies. Indeed, two distinct vectors of norm $4/3$ in one class differ by a non-zero vector of K_{12}, of norm at least 4, so their inner product is at most $-2/3$; the sum of k of them has norm at most $4k/3 - 2k(k-1)/3$, so a class contains at most three of them. Hence the 756 vectors of norm $4/3$ of $K_{12}^\#$ meet at least 252 classes, all with $q = 4/3$, while by Milgram's formula ($|K_{12}^\#/K_{12}| = 3^6$, signature 12) exactly 252 classes have $q = 4/3$ [C10]. For $3-(16,0)-0$, $F = 0$ and $L = Z$ is a rootless Hermitian lattice of rank 16 over $\mathbb{Z}[\zeta_3]$; there are exactly eleven such lattices [Hen13, Thm 1, Table 4, Rem. 3(a)]. We constructed eleven such lattices without the Gram matrices of [Hen13], by neighbour methods and from codes, computed $C(h) = \mathrm{U}(Z)$ ourselves, and checked that the orders $|\mathrm{U}(Z)|$ agree with [Hen13, Table 4] and that $\sum 1/\,|\mathrm{U}(Z)|$ equals the rootless mass of the genus (Appendix D).

Orders 5, 7, 17, 31. The nine types have 218 extremal pairs. For $5-(7,4)-3$, $F = \sqrt{5}A_4^\#$ and the classes of $F^\#/F$ containing a vector of norm $2/5$ are exactly those with $q_F = 2/5$, so L_φ is extremal if and only if $Z^\#$ has no vector of norm $8/5$, independently of φ; 13 of the 23 lattices Z qualify, and since $\mathrm{Aut}\ F \to O(q_F)$ is onto, each gives one pair with $C(h) \cong \mathrm{U}(Z)$. For $5-(6,8)-6$, $F = F_1 \perp F_1$ with $F_1 = \sqrt{5}A_4^\#$, the classes of φ correspond to unordered pairs $\{V_1, V_2\}$ of subspaces with $D(Z) = V_1 \perp V_2$, and counting the $\mathrm{U}(Z)$-orbits of the good pairs over the four lattices Z gives 155 pairs. For $5-(8,0)-0$, $F = 0$ and $L = Z$; the nine classes were found by ϖ-neighbours, which preserve the $\mathbb{Z}[\zeta_5]$-structure, and the list is complete because the sum of $1/\,|\mathrm{U}(Z)|$ equals the rootless mass of the genus (Appendix D). For $17-(2,0)-0$, $L = Z$ is a Hermitian $\mathbb{Z}[\zeta_{17}]$-lattice of rank 2; its genus consists of seven Hermitian lattices, six of which have extremal trace form, and these six give exactly four lattices up to isometry [Jür15, Satz 3.6.9 and its proof]; independently, the sum over these four lattices of the number of elements of order 17 divided by $|\mathrm{Aut}\ L|$ equals the rootless mass $1/12$ of the genus [C9]; there are six pairs. For $31-(1,2)-1$, F is the binary form $\left(\begin{smallmatrix} 4 & 1 \\ 1 & 8 \end{smallmatrix}\right)$ with $\mathrm{Aut}\ F = \{\pm 1\} = O(q_F)$, Z runs over the rank-1 lattices of its genus other than A_{30}; every lattice of rank 1 has unitary group $\langle -\zeta_{31} \rangle$ of order 62 and the genus has mass $9/62$ (§D.1), so it has 9 classes (the relative class number $h^-(\mathbb{Q}(\zeta_{31})) = 9$

[Sch98, p. 1226]); so the eight rootless ones have total weight 4/31 (Appendix D). The gluing is unique; this gives eight pairs with $|C(h)| = 62$.

Example (identity (4.1) for the type $3-(9,14)-7$). This is a case in which identity (4.1) can be followed by hand; the example is not used in the proofs. By Table B4, the genus $(14,7)$ of F and the genus $(9,7)$ of Z each contain exactly one lattice of minimum at least 4, so there is one pair of components. For it, $|\mathrm{Aut}\ F| = 8{,}491{,}392 = 2^7 3^6 \cdot 7 \cdot 13$ and $|\mathrm{U}(Z)| = 22{,}674{,}816 = 2^7 3^{11}$. The form Q of Table 3 on $F^{\#}/F \cong \mathbb{F}_3^7$ is non-degenerate of odd dimension 7. Hence, by Lemma 2.1(i), $|O(q_F)| = |\mathrm{GO}_7(3)| = 2 \cdot 3^9(3^2-1)(3^4-1)(3^6-1) = 18{,}341{,}406{,}720$. The right side of (4.1) is therefore

$$\frac{18{,}341{,}406{,}720}{8{,}491{,}392 \cdot 22{,}674{,}816} = \frac{5}{52{,}488}.$$

The computed list has two double cosets, with $|\mathrm{Stab}(\varphi)| = 11{,}664$ and $104{,}976$, and

$$\frac{1}{11{,}664} + \frac{1}{104{,}976} = \frac{10}{104{,}976} = \frac{5}{52{,}488}.$$

So the left side already equals the right side. As every term is positive, there is no third double coset, whatever method found the two. Both gluings are extremal [C8]. The two lattices are `32.23328.1` and `32.839808.1` of [D], with $|\mathrm{Aut}\ L| = 2^5 3^6$ and $2^7 3^8$; the second paper of this repository describes them (its Theorem 6.1).

4.3 Gaussian pairs and their rootless mass

Let $\mathcal{O} = \mathbb{Z}[i]$. By §2.4, a pair (L, g) with $g^2 = -1$ is the same as an $\mathcal{O}$-lattice L of rank $r = 16$ with a Hermitian form H whose trace form $\mathrm{Tr}\ H$ is even unimodular; isometries of pairs are the $\mathcal{O}$-linear isometries of Hermitian lattices, so $C(g) = \mathrm{U}(L, H)$, the unitary group. L is extremal exactly when (L, H) has no *root*, a vector with $\mathrm{Tr}\ H(x, x) = 2$. Hence

$$M_2 = \sum_{(L,H)} \frac{1}{|\mathrm{U}(L,H)|},$$

the sum over isometry classes of rootless Hermitian $\mathcal{O}$-lattices of rank r with even unimodular trace form: the *rootless mass*.

Theorem 4.5. $M_2 = 4{,}128{,}748{,}922{,}231{,}417 \ /\ 120{,}324{,}096{,}000$.

The proof is in Appendix C. It follows King's method for $\mathbb{Z}$-lattices [Kin03]: the rootless mass is a finite alternating combination (C.2) of the masses $A_r(T)$ of embeddings of root lattices T, with coefficients from the classification of root lattices over $\mathbb{Z}[i]$ and their root sublattices; each $A_r(T)$ is given by a Siegel-type formula (C.3) as a product of local densities; and the local densities at the ramified prime 2 reduce to counting on $L/2L$ (Proposition C.7, Theorem C.8). The computation is in exact rational arithmetic [C11]; the implementation is among the assumptions of Table 10.

Proposition 4.6. The list [C12] consists of 76,742 rootless Hermitian $\mathbb{Z}[i]$-lattices of rank 16, no two of which are isometric or complex conjugate to each other. With $f = 2$ for a lattice not isometric to its complex conjugate and $f = 1$ otherwise, the sum of $f/\ |\mathrm{U}|$ over the list equals the value of Theorem 4.5. Hence the list contains every rootless Hermitian $\mathbb{Z}[i]$-lattice of rank 16 up to isometry and complex conjugation.

A pair (L, g) with $g^2 = -1$ and its pair $(L, -g)$ are the Hermitian lattice (L, H) and its complex conjugate, so (§2.4) $M_2 = \sum f/\,|\mathrm{U}|$ over the lattices up to isometry and complex conjugation; these are the Hermitian entries of order 4 of §6.1.

Strategy of the proof. The search that produced the list does not prove it complete. Completeness follows from Lemma 7.1, which needs three facts: the classes of the list are pairwise distinct, each weight $f/\,|\mathrm{U}|$ is correct, and the weights add up to M_2, which Theorem 4.5 computes independently of the list. A missing class would add a positive weight, so none is missing. The proof checks the three facts in this order.

Proof. The lattices were collected from the Hermitian entries of order 4 of the lattices met in the lists of §4.1 and §4.2 and in S_{gen}, and from random walks by neighbours in the genus [C12]. Two lattices were kept apart only if they differ in a canonical form or in the invariant D of the pair $(L, \pm g)$, both invariant under isometry and complex conjugation (§7.2). This proves the first statement. The weights $f/\,|\mathrm{U}|$ were obtained by three methods: from the automorphism group of a canonical form, by a backtrack search for the isometries commuting with g up to sign (it returns their group $N_{\pm}$, and $f/\,|\mathrm{U}| = 2/\,|N_{\pm}|$), and from $\mathrm{Aut}\,L$ computed with bliss. Every lattice has a weight from at least one method. For 11,421 lattices two methods were applied (bliss and the search for 11,144, bliss and the canonical form for 277), and they agree. Of the others, 62,449 have their weight from the canonical form alone: there a canonically selected set of Gaussian lines with distinct colours generates L. Each line is preserved by U; connectedness and the phase-normalization argument of §7.2 force $\mathrm{U} = \{\pm 1, \pm i\}$. For the discrete classes in the final list, the two normalized Gram hashes differ, which excludes an isometry to the complex conjugate; 2,547 have their weight from the backtrack search alone and 325 from bliss alone [C12]. The sum of the weights equals the value of Theorem 4.5 as a fraction [C12]. Since Theorem 4.5 is proved in Appendix C, Lemma 7.1 shows that the list is complete. □

The list supplies a lower bound for the Gaussian rootless mass independently of the mass formula. Equality establishes completeness under the validity, distinctness and group-weight conditions just proved.

5. Proof of Theorem A

What this section does. We reduce the total to the classification (§5.1) and list the statements on which the proof rests, with their proofs, assumptions and computations (§5.2, Table 7).

Points that are easy to misread (the count). These remarks add nothing to the proofs.

1. *(5.1) is not an independent check of Theorem B.* It is an exact identity, true for any finite set of lattices whose automorphism groups contain ± 1. It turns the classification of S into the number N, and it is only as reliable as the list of S and the orders $|\mathrm{Aut}\,L|$ it reads (§1.4).
2. *The lattices with* $\mathrm{Aut}\,L = \{\pm 1\}$ *are counted, not constructed.* Their number 10,900,939 (Corollary C) is $N - |S|$. No representative of them is computed in this paper.
3. *The auxiliary count is a check.* The thirteen-term computation of [Aux] reaches the same N by another route, but it shares the component lists with the main proof and is not a premise (§5.2).
4. *A "trivial" entry is not a trivial element.* An involution entry is trivial when $C(\sigma) = \{\pm 1, \pm\sigma\}$ and $\sigma \nsim -\sigma$; a Hermitian entry of order 4 is trivial when $N_G(\langle g \rangle) = \langle g \rangle$ (§6.1, Lemma 6.1).

Such an entry still comes from an automorphism other than ± 1, and its lattice still belongs to S.

5. *Identifiers and keys.* An identifier names the class of an entry in one of the lists of §4, and some rows of the records have none (§6.3). A key is an isometry invariant used to separate and sort (§7.2, §6.8). Equal keys prove isomorphism only in the situation of Lemma 7.2.
6. *Labels.* In a label $32.|\mathrm{Aut}\ L|.i$ (§6.8), the index i follows a fixed rule for sorting and has no further meaning.

5.1 Reduction to the classification

For every $L \in \mathcal{L}$, write $G_L = \mathrm{Aut}\ L$. Since G_L contains ± 1,

$$N = 2m + \sum_{L \in S} (1 - 2/\ |G_L|). \tag{5.1}$$

Indeed, for each class split 1 as $2/\ |G_L| + (1 - 2/\ |G_L|)$ and sum. The second summand vanishes exactly for the classes outside S.

Theorem B, proved in §6 from the complete structured-pair lists, supplies a non-redundant list of S and the orders $|G_L|$. Substitution in (5.1), in exact rational arithmetic, gives $N = 11{,}044{,}268$. The computation reads only King's mass and the saved classified-lattice table [C13]. This proves Theorem A conditional on precisely the classification and mass inputs described above. The proof of Theorem B does not use this numerical value.

Example (the method in rank 24). This example is not used in the proofs. In rank 24 the bound of §2.1 is again $\min L \leq 4$, and the only extremal lattice is the Leech lattice Λ, with $\mathrm{Aut}\ \Lambda = \mathrm{Co}_0$ [SPLAG, Table 16.1]. The auxiliary note runs the tools of §4 and of Appendices C and D in rank 24 and compares every value with the conjugacy classes of Co_0 [Aux, §5.6, Table 7c]. King's rootless mass of rank 24 is $1/\ |\mathrm{Co}_0|$, with $|\mathrm{Co}_0| = 8{,}315{,}553{,}613{,}086{,}720{,}000$. For involutions of type $(16, 8)$ the list of §4.1 has one double coset; it is extremal, and it is Λ, with $|\mathrm{Stab}(\varphi)| = |C(\sigma)| = 178{,}362{,}777{,}600$. For type $(12, 12)$ the list has six double cosets, and only one is extremal: Λ again, with $\sigma \sim -\sigma$. Since Λ is the only extremal lattice, $S = \{\Lambda\}$ in rank 24, and (5.1) gives

$$2/\ |\mathrm{Co}_0| + (1 - 2/\ |\mathrm{Co}_0|) = 1.$$

In rank 24 the mass term is negligible, and the count comes from S. In rank 32 it is the other way round: $2m \approx 10{,}968{,}923.3$ is 99.3% of N, and the 143,329 lattices of S add only about 75,344.7. The comparison in rank 24 checks the programs; it does not check the values of rank 32.

5.2 Dependencies of the main proof

The classification-based proof is assembled from the statements of Table 7. Each row states its output, the equivalence relation and weight under which a list is complete, how it is proved, which software assumptions of Table 10 it uses, and the computations of Table 11 that support it. Long proofs and finite calculations are in the appendices; the main text cites the statements.

Figure 2 draws the same dependencies as a diagram.

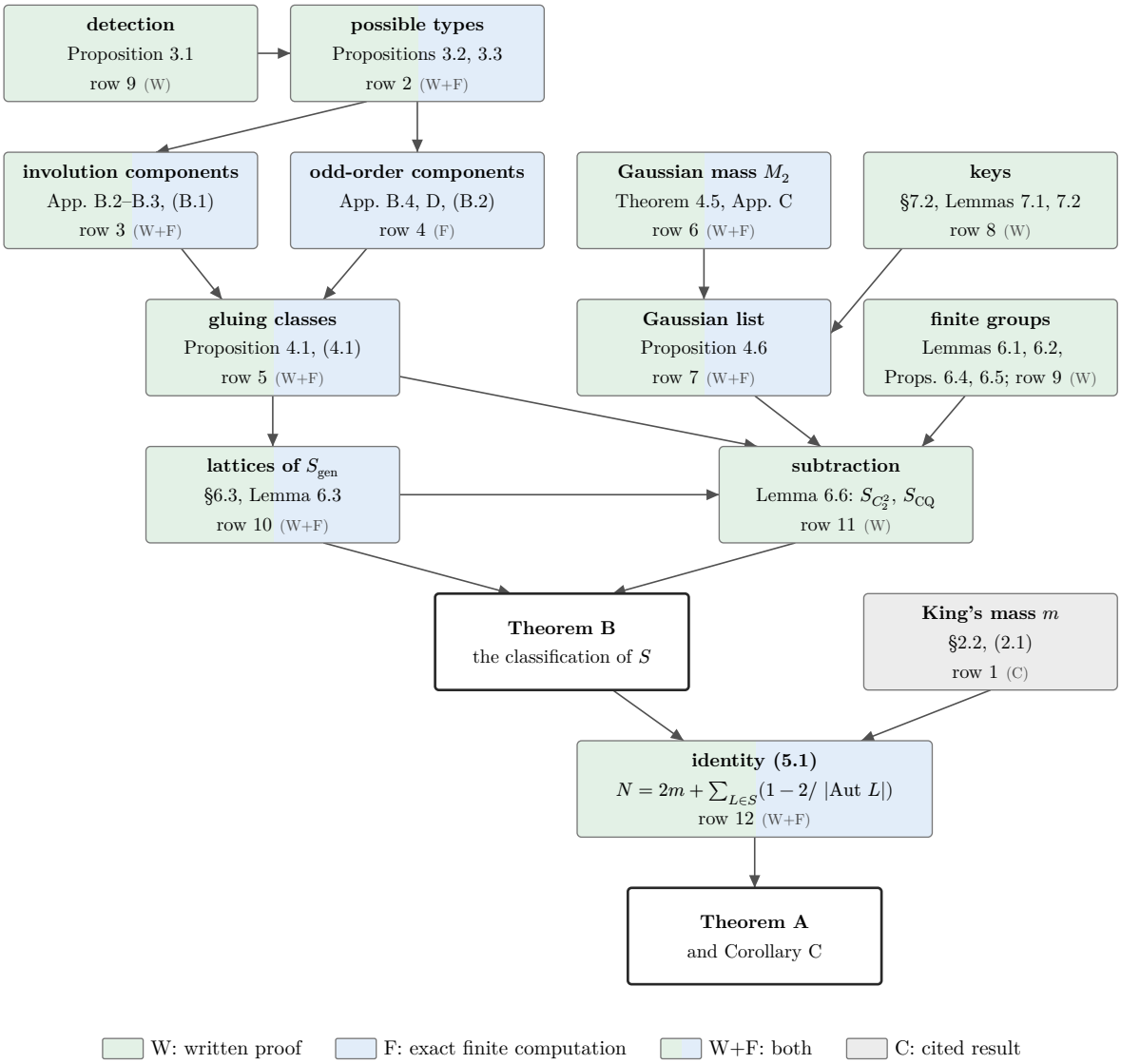


Figure 2. The dependencies of the proofs of Theorems A and B, drawn from Table 7 ("row k" is its row). An arrow means "is used by". Colours give the kind of proof, as in the column "proof" of Table 7; the computations behind each box are listed there. Proposition 3.1 also gives the three parts of S (§6.1), and the keys of §7.2 also separate the lattices of S_{gen}; these two uses are not drawn. Theorem B does not use King's mass or the value of N.

Table 7. The statements used by the proofs of Theorems A and B. "Hand" means a written proof in this paper; "cited" a published result; "finite" an exact finite computation. Theorem B uses rows 2–11, Theorem A in addition rows 1 and 12.

statement	classes, weights	proof	assumptions (Table 10)	computations
1. King's rootless mass m (§2.2)	$\mathbb{Z}$-isometry; $1/\lvert\text{Aut } L\rvert$	cited [Kin03, Table 1]; recomputation of the same inclusion–exclusion	King's table; our code	C14, C15
2. Possible types (Propositions 3.2, 3.3)	—	hand; cited [Nik79], [SPLAG], [CE03], [Neb13b]; six finite LP certificates; rootless Hermitian masses of Appendix D (reason 3)	dimensions of spaces of modular forms; theta coefficients; our implementation of the mass formulas of Appendix D	C2, C3
3. Involution components (Appendix B.2–B.3)	$\mathbb{Z}$-isometry; $1/\lvert\text{Aut}\rvert$	hand (Lemma B.1, Proposition B.2, Theorem B.3): complete by an enumeration that reaches	`qfminim`, `qfauto`, `qfisom`, bliss, GAP (the orbits of §B.3)	C5, C6, C7

statement	classes, weights	proof	assumptions (Table 10)	computations
		every lattice, except the lattices K of type $(8, 24)$ for A_2^{12}, which are complete by identity (B.1)		
4. Components of odd prime order (Appendix B.4, D)	F: $\mathbb{Z}$-isometry; $(Z, h\|_Z)$: isometries commuting with h, not identified under Galois; $1/\ \|\mathrm{Aut}\ F\|$, $1/\ \|\mathrm{U}(Z)\|$	finite: equality with genus and rootless masses; the 2-adic and 9-torsion counts of §D.4; cited [Hen13], [Jür15]	SageMath masses; our code for Appendix D; `qfauto`	C2, C8, C9, C16, C17, C18
5. Gluing classes (Proposition 4.1, §§4.1.3, 4.2)	pairs (L, h) up to isometry; orientation as in §4.1.2; $1/\ \|\mathrm{Stab}(\varphi)\|$	hand (Proposition 4.1, Lemmas 4.2–4.4); finite: identity (4.1) as an equality of fractions for each pair of components	GAP double cosets; canonical form of gluings; `qfminim` for extremality	C4, C6, C8, C9
6. Gaussian rootless mass M_2 (Theorem 4.5, Appendix C)	$\mathbb{Z}[i]$-linear isometry; $1/\ \|\mathrm{U}\|$; conjugates not identified	hand (C.1–C.10); finite: exact sum over 12,934 root types	our code for Appendix C; GAP for the Weyl group classes	C11, C19, C20
7. Gaussian list (Proposition 4.6)	isometry and complex conjugation; $f/\ \|\mathrm{U}\|$	Lemma 7.1 with row 6; keys of §7.2	bliss; backtrack search; canonical forms	C12
8. Keys (Lemmas 7.1, 7.2, §7.2)	one relation per key	hand: invariance of each key	bliss canonical labelling	—
9. Finite groups (Proposition 3.1, Lemmas 6.1, 6.2, Propositions 6.4, 6.5)	—	hand; cited [Gor80]	none	C21 (check)
10. The lattices of S_{gen} (§6.3, Lemma 6.3)	$\mathbb{Z}$-isometry; full $\|\mathrm{Aut}\ L\|$	shared exact identifiers and integer isometries for equality; Lemma 6.3 (exact types) and minimum-shell profiles for distinctness	bliss and GAP for full groups and classes; `qfminim`; exactness of identifiers (Table 10, our code)	C1, C22, C23, C24, C25
11. Subtraction (Lemma 6.6)	entries of each lattice	hand, from rows 3–10	the records of §6.3 list all involution and Hermitian entries of the full group	C25, C26, C27, C28, C29
12. Identity (5.1)	—	hand; finite exact sum	—	C13

The integer obtained in (5.1), the auxiliary thirteen-term sum and the distribution of group orders do not replace any row of Table 7.

The auxiliary thirteen-term computation of [Aux] uses rows 1–6 but neither row 10 nor row 11: it never computes a full automorphism group of rank 32. Its agreement with (5.1) is therefore a global check by a different arithmetic route of the full-group orders and the identifications of rows 10–11. It does not test the rows shared by both routes, and it cannot detect errors that cancel. It also relies on its own inputs $e(h)$ and M_3, M_4, M_5. It is not a premise of the proof.

6. Proof of Theorem B

What this section does. We group the pairs of §4 lattice by lattice. §6.1 defines the three parts. §6.2 proves Lemma 6.2: a lattice whose entries are all trivial has $|\mathrm{Aut}\ L| = 4$. §6.3 identifies the lattices of

S_{gen} and proves them distinct (Lemma 6.3). §6.4 and §6.5 prove Propositions 6.4 and 6.5 on the groups of $S_{C_2^2}$ and S_{CQ}. §6.6 removes the entries of S_{gen} (Lemma 6.6) and counts both parts. §6.7 obtains the total from the classified lattices and separates it from the auxiliary checks. §6.8 describes the numbering of the database.

6.1 Entries and the three parts

Fix $L \in S$ and put $G = \operatorname{Aut} L$. The detection argument of §1.3 gives the structures needed for the classification. We use the following *entries*:

- an *odd entry* is a conjugacy class $[h]$ with h of odd prime order;
- an *involution entry* is the set $\{[\sigma], [-\sigma]\}$ for $\sigma \neq \pm 1$, a single class if $\sigma \sim -\sigma$;
- a *Hermitian entry* of order 2^a is the set $\{[g], [-g]\}$ with $g^{2^{a-1}} = -1$.

For order 4 put $\mathrm{U} = C(g)$ and $f = 2$ or 1 according as g is not or is conjugate to $-g$. Its Gaussian mass weight is $f/\,|\mathrm{U}|$. No assignment count $e(h)$ is needed to define these entries or the three parts below. Auxiliary assignment statistics [Aux] may still be computed inside the full groups for invariant signatures and consistency checks.

In terms of the gluings of §4.1, an involution entry of type $(d, 32-d)$ with $d \neq 16$, or with $d = 16$ and $F \not\sim K$, is one gluing class $[\varphi]$ (seen from σ; from $-\sigma$ it is the class of φ^{-1} in the list of type $(32-d, d)$). For $d = 16$ and $F \cong K$ it is the pair $\{[\varphi], [\varphi^{-1}]\}$ of classes in the same list, which coincide exactly when $\sigma \sim -\sigma$.

An entry is *trivial* if it is an involution entry with $|C(\sigma)| = 4$ and $\sigma \not\sim -\sigma$, or a Hermitian entry of order 4 with weight $1/2$. A trivial involution entry has $C(\sigma) = \{\pm 1, \pm\sigma\}$. All other entries are *non-trivial*.

Lemma 6.1. A Hermitian entry of order 4 has weight $1/2$ if and only if $N_G(\langle g\rangle) = \langle g\rangle$.

Proof. We have $\langle g\rangle \subseteq \mathrm{U} \subseteq N_g := N_G(\langle g\rangle)$, and $N_g/\,\mathrm{U}$ embeds in $\operatorname{Aut}\langle g\rangle \cong C_2$, with $[N_g : \mathrm{U}] = 2$ exactly when g is conjugate to $g^{-1} = -g$, that is when $f = 1$. Hence $f/\,|\mathrm{U}| = 1/2$ if and only if $f = 2$ and $|\mathrm{U}| = 4$, if and only if $N_g = \mathrm{U} = \langle g\rangle$. □

We define the three parts of S; the subscripts name the automorphism groups that they turn out to have:

- S_{gen} ("general"): L has an odd entry or a non-trivial involution entry, that is, an element of odd prime order or an involution $\sigma \neq \pm 1$ with $|C(\sigma)| > 4$ or $\sigma \sim -\sigma$;
- $S_{C_2^2}$: $L \notin S_{\text{gen}}$ and L has an involution $\sigma \neq \pm 1$; by Proposition 6.4, $\operatorname{Aut} L \cong C_2 \times C_2$;
- S_{CQ}: L has neither an element of odd prime order nor an involution other than -1, so $\operatorname{Aut} L$ is a 2-group with -1 as its only involution, cyclic or generalized quaternion (Proposition 6.5).

These sets are disjoint by definition and exhaust S by Proposition 3.1. If there is no odd entry or involution entry, the group is a 2-group whose only involution is -1. In S_{CQ} the condition that $\operatorname{Aut} L$ is a 2-group matters: a group such as C_6 has -1 as its only involution but contains an element of order 3, so its lattices lie in S_{gen}. (In the database, 2,759 lattices of S_{gen} have -1 as their only involution, 2,308 of them with $\operatorname{Aut} L \cong C_6$; the others have groups such as C_{12}, Dic_3, $\mathrm{SL}(2,3)$, $\mathrm{SL}(2,5)$ and $\mathrm{SL}(2,17)$ [C30].)

6.2 Lattices with only trivial entries

Lemma 6.2. Let $L \in S$ and $G = \operatorname{Aut} L$.

1. $|G| = 4$ if and only if all entries of L are trivial. In that case L has exactly one entry.

2. If G is a 2-group of order at least 8, then L has no trivial entry.
3. L has at most one trivial entry, and if it has one, then $|G| = 4k$ with k odd.

Proof. We use that in a finite p-group every proper subgroup is properly contained in its normalizer.

Step 1. Let P be a 2-subgroup of G. (a) If $\sigma \in P$ is an involution $\neq \pm 1$, $-1 \in P$ and $H = \{\pm 1, \pm\sigma\} \neq P$, then the entry of σ is non-trivial. Indeed, take $y \in N_P(H) \setminus H$; then $y\sigma y^{-1}$ is an involution in H other than ± 1, so it is σ or $-\sigma$. In the first case $y \in C(\sigma) \setminus H$ and $|C(\sigma)| > 4$; in the second $\sigma \sim -\sigma$. (b) If $g \in P$ satisfies $g^2 = -1$ and $\langle g \rangle \neq P$, then $N_G(\langle g \rangle) \supseteq N_P(\langle g \rangle) \supsetneq \langle g \rangle$, so the entry of g is non-trivial by Lemma 6.1.

Proof of (ii). Apply Step 1 with $P = G$, which has order at least 8 and contains -1. Involution entries and Hermitian entries of order 4 are non-trivial, odd entries do not occur, and the remaining entries are non-trivial by definition.

Proof of (iii). If σ gives a trivial involution entry, let P be a Sylow 2-subgroup of G containing $H = \{\pm 1, \pm\sigma\}$; by Step 1(a), $P = H$. If g gives a trivial Hermitian entry, let P be a Sylow 2-subgroup containing $\langle g \rangle$; by Step 1(b), $P = \langle g \rangle$. In both cases $|P| = 4$, so $|G| = 4k$ with k odd. If L had two trivial entries, the corresponding Sylow 2-subgroups P_1, P_2 would be conjugate, $uP_1u^{-1} = P_2$. They cannot be $C_2 \times C_2$ and C_4. If $P_i = \{\pm 1, \pm\sigma_i\}$, then $u\sigma_1 u^{-1}$ is an involution of P_2 other than ± 1, hence $\pm\sigma_2$, and the two entries coincide. If $P_i = \langle g_i \rangle$, then $ug_1u^{-1} \in \{\pm g_2\}$, and again the entries coincide.

Proof of (i). If $|G| = 4$, then $G = \{\pm 1, \pm\sigma\} \cong C_2 \times C_2$ or $G = \langle g \rangle \cong C_4$ with $g^2 = -1$. In the first case the only entry is the involution entry of σ; $C(\sigma) = G$ and $\sigma \not\sim -\sigma$ since G is abelian, so it is trivial. In the second case the only entry is the Hermitian entry of g, and $N_G(\langle g \rangle) = G = \langle g \rangle$, so it is trivial by Lemma 6.1. Conversely, suppose all entries of L are trivial. By the detection argument of §1.3 and $L \in S$, L has an entry; by (iii) it has exactly one, and $|G| = 4k$ with k odd. If $k > 1$, G contains an element of odd prime order by Cauchy's theorem, whose odd entry is non-trivial. Hence $|G| = 4$. □

The lemma is a statement about finite groups containing a central involution -1; [C21] checks it on all such groups of order at most 190.

6.3 The lattices of S_{gen}

Terms of this section. They are defined in the text below; here they are collected in the order in which the argument uses them, and Figure 3 shows how the numbers arise.

- A *record* is the result of the full-group computation for one constructed lattice: |Aut L|, the conjugacy classes with their centralizers, and one *row* for each entry of L, with its *identifier* when one was computed. A record is *successful* if the computation finished.
- The identifier of a row is *exact* if it is the class of the entry in a list of §4. A list type (a pair of components of an involution type, or a type $p-(z,f)-s$) is *exact* if every row of that type in every successful record has an exact identifier.
- For an involution list with $F \cong K$, the *inverse identifier* of a class $[\varphi]$ is the identifier of $[\varphi^{-1}]$.
- A record is *unlabelled* if none of its rows received an exact identifier.
- Records are joined into *identification classes* when they share an exact identifier, when an inverse identifier links them, or when an explicit integer isometry links them.

- The *minimum-shell profile* and the *full-group signature* are isometry invariants; they separate the unlabelled lattices.

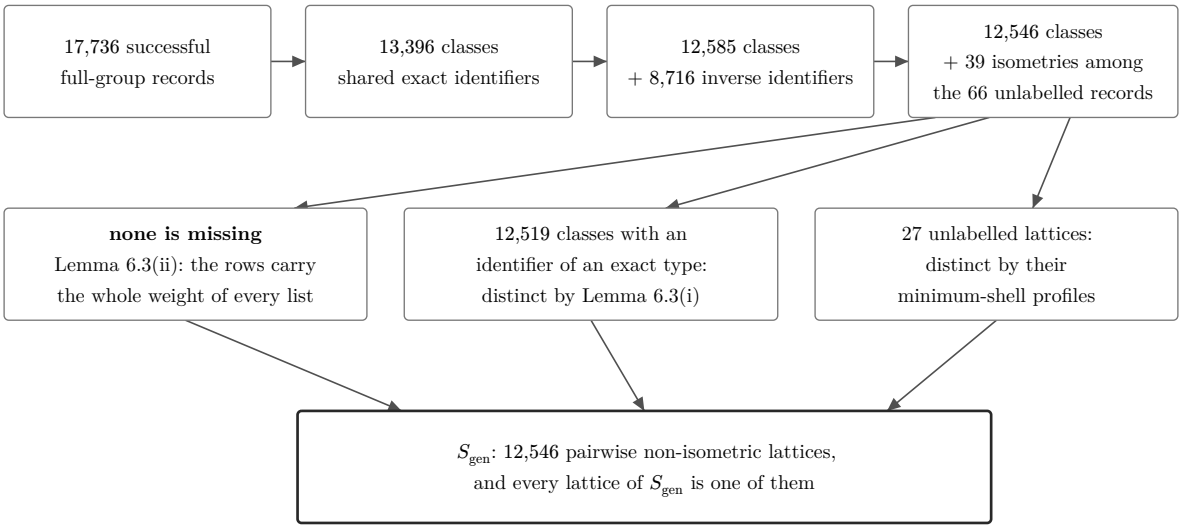


Figure 3. How the records of §6.3 become the lattices of S_{gen}. Top row: records are joined into identification classes by positive identifications only. Middle row: why different classes are different lattices (Lemma 6.3(i) and the profiles), and why no lattice is missing (Lemma 6.3(ii)).

Every lattice of S_{gen} has a non-trivial involution entry or an odd entry, and every such entry is a double coset in the lists of §4 with $|\mathrm{Stab}(\varphi)| > 4$, or with $|\mathrm{Stab}(\varphi)| = 4$ and $[\varphi] = [\varphi^{-1}]$, or of odd prime order. From entries of these lists (and, starting from their published descriptions (codes, Hermitian and quaternionic structures), from 26 of the 27 lattices of Appendix A; the remaining one, $[4.L_3(4).2^2]_{32,2}$ of [Neb98q] with $|\mathrm{Aut}\ L| = 2^{10}3^2 5 \cdot 7$, is reached from an entry of type $(12, 20)$) we construct lattices L_φ, compute $\mathrm{Aut}\ L$ with bliss and its conjugacy classes and centralizers with GAP, and list all entries of L with their weights [C1]. We call the result of this computation for one constructed lattice, that is $|\mathrm{Aut}\ L|$, the conjugacy classes and centralizers and the rows of the entries of L with their identifiers, a *record*, and the record *successful* if the computation finished and returned these data. Two constructed lattices are identified when they share an entry (the same double coset in the same list, including the partner class $[\varphi^{-1}]$ for $F \cong K$); the 66 *unlabelled* constructed records, none of whose computed entries received an exact identifier (their odd entries are of order 5, or of order 3 of the inexact types below) were grouped into 27 lattices (not related to the 27 of Appendix A) by explicit isometries, checked in integers. This gives 12,546 lattices. For each of them:

- as an auxiliary check, the weights computed from all group elements add up to $1 - 2/\ |\mathrm{Aut}\ L|$;
- every computed entry that can be matched with an entry of the lists of §4 has the same $|C|$ and the same auxiliary assignment count e computed inside the full group [Aux] (no mismatch).

For the ten lattices with $|\mathrm{Aut}\ L| > 10^7$, all identified with known lattices or with each other by explicit isometries, the conjugacy classes of two of them (BW_{32} and one code lattice, with $|\mathrm{Aut}\ L| = 2^{31}3^5 5^2 7 \cdot 17 \cdot 31$ and $2^{31}3^5 5^2 7$) were obtained from the structure of the group, not in a faithful permutation representation (the second paper of this repository computes them directly, on the minimal vectors, and finds the same classes of involutions and of elements of odd prime order, with the same centralizer orders): with $E = O_2(\mathrm{Aut}\ L)$, the classes of $\mathrm{Aut}\ L$ are obtained from the classes of $\mathrm{Aut}\ L/E$ and, for a representative tE of each, the orbits on the coset tE, under conjugation, of the preimage of the centralizer of tE in $\mathrm{Aut}\ L/E$ [C1].

For these two lattices the full group was also computed from their Gram matrices alone, with no supplied group order or generating set [C22]. The input Grams are exactly those used for the known representatives in this classification. The code lattice has order 91,321,742,131,200 and BW_{32} has order 48,126,558,103,142,400. Each is computed by the algorithm of §7.1 as twice the orbit size of a minimal line $\pm x$ times the order of the stabilizer of x; every candidate image of the line is tested, so this is the order of the full group. The groups in which the conjugacy classes were computed [C1] have these orders, so they are the full automorphism groups. These full-group computations are separate from the structural conjugacy-class calculations above. The rerun uses the same algorithm as the earlier full-group calculations, not an independent mathematical method.

The positive identifications among the unlabelled records are preserved as 39 unimodular integer isometries [C23]. Each satisfies $T^{\mathsf{T}} G_a T = G_b$ and $|\det T| = 1$. The 27 resulting unlabelled lattices are pairwise non-isometric by the following invariant [C24]. Let $\overline{X} \subset L/2L$ be the image of the lines $\pm x$ of minimal vectors; the 73,440 images are distinct, which the program checks. For $v \neq 0$ let $c(v)$ be the number of unordered pairs $\{a, b\} \subset \overline{X}$ of distinct elements with $a + b = v$, and attach to each $a \in \overline{X}$ the integers $s_j(a) = \sum_{b \in \overline{X}, b \neq a} c(a+b)^j$, $j = 1, 2, 3$. An isometry of lattices induces a linear isomorphism of $L/2L$ that maps $\overline{X}$ onto $\overline{X}$, so the multiset of the triples $(s_1, s_2, s_3)(a)$ is an isometry invariant, and so is the triple chosen from it by least frequency and then lexicographic order. We call this triple the *minimum-shell profile*. The profiles of the 27 lattices are pairwise different. Different profiles prove non-isometry; equal profiles are never used to infer an isometry.

As a second, independent reason, the unlabelled records are separated from all others by their *full-group signature*: the order of $\mathrm{Aut}\ L$ together with the multiset, over the entries of L, of their kind, their type and the order of their centralizer, and for Hermitian entries also $|\mathrm{U}|$ and f. These quantities are determined by $\mathrm{Aut}\ L$ and its conjugacy classes, so they are isometry invariants; no record with an exact identifier has the signature of an unlabelled record [C24]. Of the ten lattices with $|\mathrm{Aut}\ L| > 10^7$, nine are lattices of Appendix A, and 54 explicit isometries identify their records with them; the tenth, with $|\mathrm{Aut}\ L| = 2^{17}3^4 5$, is not among the lattices of Appendix A, and 13 explicit isometries identify its records with each other; all 67 are checked in integers [C24].

The identifications above are positive: two records are put together when they share an exact identifier (defined below), when their entries are the two classes $[\varphi]$, $[\varphi^{-1}]$ of one involution entry with $F \cong K$ (§4.1.2; all 8,716 such inverse identifiers (5,005 unordered pairs, 3,711 of them recorded in both directions; the certificate also lists the reverses of six of them) were recomputed [C24]), or by an explicit integer isometry [C23, C24]. Call the resulting classes of records *identification classes*. Step by step, the 17,736 successful full-group records fall into 13,396 classes by shared exact identifiers, into 12,585 when the inverse identifiers are added, and into 12,546 by the 39 isometries among the 66 unlabelled records; the 67 isometries for the ten lattices with $|\mathrm{Aut}\ L| > 10^7$ join no further classes, they confirm identifications already made by identifiers [C24, C25]. Distinctness of different identification classes needs a separate argument, because not every computed entry receives an identifier.

The identifier of an entry row is *exact* if it is the class of the entry in a list of §4. For involutions it is computed from the identified components and the canonical form of the gluing (§7.2); for four rows of type $(8, 24)$ in groups of order above 10^7 it was completed from the canonical key of type $(8, 24)$ (§7.2), whose saved prefix matches a unique lattice K, and from the centralizer order. For odd prime order the components are identified and an identifier is attached only if exactly one extremal class of the list has those components and that $|\mathrm{Stab}(\varphi)|$; otherwise the row has none. A single candidate is

the class of the entry by the completeness of the list, the identification of the components and the correctness of $|C(h)| = |\mathrm{Stab}(\varphi)|$. For each pair of components there is one ordered list, so the identifier of an involution row does not depend on whether σ or $-\sigma$ represents the entry. A list type (a pair of components of an involution type, or a type $p-(z,f)-s$) is *exact* if every row of that type in every successful full-group record has an exact identifier.

Lemma 6.3 (distinctness and exhaustion). Assume that each successful record lists all conjugacy classes of the full automorphism group of its lattice, with their types and centralizer orders, and that its identifiers are exact as defined.

(i) If some record of an identification class $\mathcal{J}_1$ has an identifier x on a row of an exact type T, then the lattice of $\mathcal{J}_1$ is not isometric to the lattice of any other identification class.

(ii) Let $L_1, ..., L_t$ be pairwise non-isometric lattices, one for each identification class. For a list ℓ of §4.1–§4.2 (for $p \geq 5$, the lists of the types with the same p and f taken together) give an extremal class X the weight $w(X) = 1/\ |\mathrm{Stab}(\varphi)|$, and give a row of type ℓ of some L_i, with element h, the weight $1/\ |C(h)|$, or $2/\ |C(h)|$ if ℓ is an involution list with $F \cong K$ and $h \nsim -h$. Let $T_0(\ell)$ be a set of trivial classes of ℓ (§6.1; empty for the lists of odd prime order, which have none) such that no class of $T_0(\ell)$, and for $F \cong K$ no partner $[\varphi^{-1}]$ of one, is a class of an entry of any L_i. If for every ℓ the rows of type ℓ of $L_1, ..., L_t$ have total weight $\sum w(X)$ over the extremal classes X of ℓ not in $T_0(\ell)$, then every lattice of S_{gen} is isometric to one of $L_1, ..., L_t$.

Strategy of the proof. Part (i) transports an entry: an isometry of lattices carries an entry with an exact identifier to an entry with the same identifier, and records that share an identifier were put into one identification class. Part (ii) reads identity (4.1) backwards. Each row of the records is an entry: one class of a list of §4, or, for an involution list with $F \cong K$, the classes $[\varphi]$ and $[\varphi^{-1}]$, which may be two. Distinct rows give disjoint sets of classes. If the rows already carry the whole weight of the list (less the trivial classes $T_0(\ell)$), no class is left over, so no lattice of S_{gen} can be missing.

Proof. (i). Let $\mathcal{J}_2$ be an identification class whose lattice is isometric to that of $\mathcal{J}_1$, and let u be an isometry. It maps the entry of x to an entry of the lattice of $\mathcal{J}_2$ of the same type T and the same class in the list of §4, since isometric pairs have the same components and the same double coset (Proposition 4.1). A record of $\mathcal{J}_2$ lists all entries of its full group, so it has a row for this entry; rows of odd prime order are indexed by conjugacy classes, so the transported class has its own row. As T is exact, that row carries the identifier of its class, which is x, or for an involution type with $F \cong K$ possibly x or the inverse class x'. Records sharing x lie in one identification class, and x and x' are joined by an inverse identifier. Hence $\mathcal{J}_2 = \mathcal{J}_1$.

(ii). A row of type ℓ of L_i with element h gives the pair (L_i, h), which is extremal; by Proposition 4.1 and the completeness of ℓ (§4), its isometry class is an extremal class X of ℓ with $|\mathrm{Stab}(\varphi)| = |C(h)|$. For an involution list with $F \cong K$ the row stands for the entry $\{[h], [-h]\}$, whose classes $[\varphi]$, $[\varphi^{-1}]$ in ℓ coincide exactly when $h \sim -h$ (§6.1); in every case the weight of the row is the total weight of the classes of its entry. Distinct rows give disjoint sets of classes: two rows of one L_i are distinct entries, and $(L_i, h) \cong (L_i, h')$ only if h' is conjugate to h in $\mathrm{Aut}\ L_i$; rows of different L_i have non-isometric lattices. Hence the rows of type ℓ have the total weight of the set $A(\ell)$ of classes of ℓ that are classes of entries of $L_1, ..., L_t$, and $A(\ell)$ is disjoint from $T_0(\ell)$. By hypothesis $A(\ell)$ and the set of extremal classes not in $T_0(\ell)$ have the same total weight; as all weights are positive, they are equal.

Now let $L \in S_{\text{gen}}$. It has an odd entry or a non-trivial involution entry; a class of this entry lies in one of the lists of §4.1–§4.2, which cover all types that occur (§3.3), and is extremal, and it is not trivial, so it is not in $T_0(\ell)$. Hence it is the class of an entry of some L_i: for elements h of Aut L and h_i of Aut L_i, $(L, h) \cong (L_i, h_i)$, so $L \cong L_i$. □

The exact types are all lists of involutions (including type $(8, 24)$ after the completion just described) and the types $3-(12,8)-8$, $3-(10,12)-6$, $3-(10,12)-8$, $3-(10,12)-10$ and $3-(9,14)-7$. The types $3-(9,14)-9$ and $3-(8,16)-8$ have 401 and 50 rows with two to twelve candidates; the types $3-(13,6)-5$, $3-(6,20)-6$, $3-(16,0)-0$ and all types with $p \geq 5$ have no exact identifiers. Of the 12,546 identification classes, 12,519 contain a record with an identifier on a row of an exact type [C25]; by Lemma 6.3 each of them is distinct from every other class. The remaining 27 classes are exactly the unlabelled lattices above, which are pairwise distinct by their minimum-shell profiles. Hence the 12,546 lattices of S_{gen} are pairwise non-isometric. By Lemma 6.3(ii) they are all of S_{gen}: for every list, the rows of the 12,546 lattices, one record for each, have the total weight of the extremal classes of the list less the trivial classes that §6.6 assigns to $S_{C_2^2}$, and none of these is the identifier of a row [C25]. This holds for each of the 83 involution lists, for type $(8, 24)$ and for the types of odd prime order, and is checked from the saved rows (§7.4). It covers also the types whose rows carry no identifier, and the entries that were never processed because their lattices had already been reached. The full-group signatures give a second reason why the unlabelled lattices differ from the others. The ambiguous rows cannot simply be ignored: for 297 of the 451 ambiguous order-3 rows some candidate lies in another identification class [C25].

6.4 The lattices of $S_{C_2^2}$

Proposition 6.4. A lattice $L \in S$ lies in $S_{C_2^2}$ if and only if it has a trivial involution entry and is not in S_{gen}. Then Aut $L = \{\pm 1, \pm\sigma\}$, and L has exactly one entry.

Proof. If $L \in S_{C_2^2}$, it has an involution $\sigma \neq \pm 1$, and since $L \notin S_{\text{gen}}$ its entry is trivial. Conversely a trivial involution entry gives an involution $\neq \pm 1$. By Lemma 6.2(iii), $|\text{Aut } L| = 4k$ with k odd; since $L \notin S_{\text{gen}}$, Aut L has no element of odd prime order, so $k = 1$ by Cauchy's theorem, and Lemma 6.2(i) applies. □

6.5 The lattices of S_{CQ}

A finite 2-group with exactly one involution is cyclic or generalized quaternion $Q_{2^n} = \langle x, y : x^{2^{n-1}} = 1, y^2 = x^{2^{n-2}}, yxy^{-1} = x^{-1} \rangle$, $n \geq 3$. Indeed, such a group has no noncyclic abelian subgroup, since a noncyclic abelian 2-group contains $C_2 \times C_2$ and hence three involutions; and a p-group without noncyclic abelian subgroups is cyclic or, for $p = 2$, generalized quaternion of order 2^n, $n \geq 3$ [Gor80, Thm 5.4.10 (ii)]. For $L \in S_{\text{CQ}}$ every element $g \neq \pm 1$ has 2-power order 2^a with $g^{2^{a-1}} = -1$, so all entries of L are Hermitian entries.

Proposition 6.5. Let $L \in S$ have a Hermitian entry of order 4, $g^2 = -1$, which is not an entry of a lattice of S_{gen}.

1. If $g \nsim -g$, then $L \in S_{\text{CQ}}$, Aut L is cyclic, and $|\text{Aut } L| = |\text{U}|$. This entry is the only Hermitian entry of order 4 of L.
2. If $g \sim -g$, then $L \in S_{\text{CQ}}$, Aut $L \cong Q_{2^n}$, and L has exactly three Hermitian entries of order 4, all with $g \sim -g$, with $|\text{U}| = 4, 4, 2^{n-1}$.

Proof. $L \notin S_{\mathrm{gen}}$ by assumption, and $L \notin S_{C_2^2}$ since $C_2 \times C_2$ has no element with $g^2 = -1$. So $L \in S_{\mathrm{CQ}}$ and $G = \mathrm{Aut}\ L$ is cyclic or generalized quaternion. If $G = C_{2^n}$, its elements of order 4 are $\pm g$; G is abelian, so $g \nsim -g$ and $\mathrm{U} = G$. If $G = Q_{2^n}$, its elements of order 4 are $\pm x^{2^{n-3}}$, conjugate under y, with centralizer $\langle x \rangle$ of order 2^{n-1}, and the elements yx^i, which form two conjugacy classes (i even, i odd), each closed under $yx^i \mapsto -yx^i = yx^{i+2^{n-2}}$, with centralizer $\langle yx^i \rangle$ of order 4. In particular every element of order 4 of Q_{2^n} is conjugate to its negative, so $g \nsim -g$ forces G to be cyclic. In both cases the claims follow. □

6.6 Subtraction of S_{gen} and the parts $S_{C_2^2}$, S_{CQ}

Lemma 6.6 (subtraction). Let $\mathcal{T}$ be the set formed from the extremal classes $[\varphi]$ of the lists of §4.1 with $|\mathrm{Stab}(\varphi)| = 4$ and $[\varphi] \neq [\varphi^{-1}]$, taking the class $[\varphi]$ if $F \nsim K$ (where $[\varphi] \neq [\varphi^{-1}]$ holds automatically, the inverse lying in the list with the components interchanged) and the pair $\{[\varphi], [\varphi^{-1}]\}$ if $F \cong K$; by Proposition 4.1(iii) and §4.1.2 these are exactly the trivial involution entries, and let $\mathcal{G}$ be the list of Proposition 4.6. Assume:

(1) the lists of §4.1–§4.2 and $\mathcal{G}$ are complete, the classes of $\mathcal{G}$ are pairwise distinct up to isometry and complex conjugation, and for each class $X \in \mathcal{G}$ the order $|\mathrm{U}(X)|$ and the value $f(X)$ are correct;

(2) $L_1, ..., L_t$ are pairwise non-isometric lattices of S_{gen} and every lattice of S_{gen} is isometric to one of them (§6.3, Lemma 6.3(i) and (ii));

(3) for each i, the trivial involution entry of L_i, if any, is identified with its element of $\mathcal{T}$, and every Hermitian entry of order 4 of L_i is identified with its class in $\mathcal{G}$; here the entries are those of the full group $\mathrm{Aut}\ L_i$.

Let $\mathcal{T}_{\mathrm{gen}} \subset \mathcal{T}$ and $\mathcal{G}_{\mathrm{gen}} \subset \mathcal{G}$ be the sets so identified. Then:

(i) $L \mapsto$ (its unique entry) is a bijection from $S_{C_2^2}$ onto $\mathcal{T} \setminus \mathcal{T}_{\mathrm{gen}}$, and $|\mathrm{Aut}\ L| = 4$ for these lattices;

(ii) $L \mapsto$ (its Hermitian entry of order 4) is a bijection from the lattices of S_{CQ} with cyclic group onto $\{X \in \mathcal{G} \setminus \mathcal{G}_{\mathrm{gen}} : f(X) = 2\}$, and $|\mathrm{Aut}\ L| = |\mathrm{U}(X)|$;

(iii) the relation "same underlying lattice" partitions $\{X \in \mathcal{G} \setminus \mathcal{G}_{\mathrm{gen}} : f(X) = 1\}$ into sets of three; these sets correspond bijectively to the lattices of S_{CQ} with quaternion group, and for a set with largest $|\mathrm{U}| = 2^{n-1}$ the group is Q_{2^n}.

In particular S is the disjoint union of $\{L_1, ..., L_t\}$ and the lattices described in (i)–(iii).

Strategy of the proof. The lists of §4 contain every trivial involution entry and every Hermitian entry of order 4, of every lattice. The entries that belong to lattices of S_{gen} are known from the records of §6.3. The remaining entries belong to lattices outside S_{gen}, and the finite-group results say how many entries each of these lattices has: one for $C_2 \times C_2$ (Proposition 6.4) and for a cyclic group, three for a quaternion group (Proposition 6.5). So counting the remaining entries counts these lattices.

Proof. (i). Let $\tau \in \mathcal{T}$ with underlying lattice L. If $L \in S_{\mathrm{gen}}$, then $L \cong L_i$ for some i by (2), and τ is a trivial involution entry of L_i. By Lemma 6.2(iii), L_i has at most one, so $\tau \in \mathcal{T}_{\mathrm{gen}}$ by (3). Conversely, an element of $\mathcal{T}_{\mathrm{gen}}$ has underlying lattice $L_i \in S_{\mathrm{gen}}$. Hence $\mathcal{T} \setminus \mathcal{T}_{\mathrm{gen}}$ consists of the trivial involution entries of lattices outside S_{gen}, and these lattices lie in $S_{C_2^2}$ by Proposition 6.4. Every $L \in S_{C_2^2}$ has a trivial involution entry by Proposition 6.4, and it occurs in $\mathcal{T}$ by the completeness in (1). Distinct elements of $\mathcal{T}$ are distinct entries by Proposition 4.1(ii) and §4.1.2. Since L has exactly one entry by Proposition 6.4, two distinct elements of $\mathcal{T} \setminus \mathcal{T}_{\mathrm{gen}}$ have non-isometric lattices.

(ii) and (iii). Let $X \in \mathcal{G}$ be the class of $(L, \pm g)$. As in (i), $X \in \mathcal{G}_{\text{gen}}$ if and only if $L \in S_{\text{gen}}$; here the identification in (3) gives X itself because the classes of $\mathcal{G}$ are distinct by (1). If $L \notin S_{\text{gen}}$, then $L \in S_{\text{CQ}}$ by Proposition 6.5. By §2.4, $f(X) = 1$ exactly when g is conjugate to $-g$ in Aut L. Every $L \in S_{\text{CQ}}$ occurs: Aut L is a 2-group of order greater than 2 (as $L \in S$) whose only involution is -1, so it contains g with $g^2 = -1$, whose class lies in $\mathcal{G}$ by (1). If $f(X) = 2$, Proposition 6.5(i) shows that X is the only Hermitian entry of order 4 of L and that $|\text{Aut } L| = |\text{U}(X)|$; this gives (ii). If $f(X) = 1$, Proposition 6.5(ii) shows that L has exactly three such entries, all with $f = 1$, and that their unitary groups have orders $4, 4, 2^{n-1}$ with Aut $L \cong Q_{2^n}$. Distinct lattices have disjoint sets of entries, which gives (iii). The last statement follows from the definitions of the three parts and Proposition 3.1. □

The hypotheses (2) and (3) are where the computations of §6.3 enter. For Hermitian entries, (3) is met through the invariance of the keys (Lemma 7.2) and the common key function of §7.2; for an involution entry with $F \cong K$, through the identifier of one of $[\varphi]$, $[\varphi^{-1}]$ and the recomputed table of 8,716 inverse identifiers [C28, C24]. Hypothesis (3) requires the entries of the *full* group: an entry missing from the computed list of some L_i would leave its class in $\mathcal{T} \setminus \mathcal{T}_{\text{gen}}$ or $\mathcal{G} \setminus \mathcal{G}_{\text{gen}}$, and L_i would be counted twice; a wrong identification would remove the class of a different lattice. By Lemma 6.2(iii), only representatives with $|\text{Aut } L_i| \equiv 4 \pmod 8$ can have a trivial involution entry. No odd entry and no identifier of an odd entry is used here. In the saved full-group records, every involution type is exact in the sense of §6.3 [C25], so every trivial involution entry has an exact pair identifier; no record has two trivial involution entries, and no record with a trivial involution entry has a group order not congruent to 4 modulo 8; every Hermitian entry of order 4 has a key, and all 11,545 distinct keys of these 16,163 entries belong to the list of Proposition 4.6, with the same |U| and f [C29]. The four involution entries of type $(8, 24)$ whose identifiers were completed from the canonical key of type $(8, 24)$ (§§6.3, 7.2) belong to lattices with $|\text{Aut } L| > 10^7$; they are non-trivial and do not enter $\mathcal{T}_{\text{gen}}$ in any case. These record checks confirm the bookkeeping of (3); that each record lists all conjugacy classes of the full group is the software assumption of Table 10 on bliss and GAP.

Counting the two parts. A lattice of S_{gen} can also have a trivial involution entry; then $|\text{Aut } L| = 4k$ with $k > 1$ odd by Lemma 6.2(iii), so it has an odd entry, and the computation of §6.3 lists that trivial entry among its entries. We say that the double cosets so listed are *met by* S_{gen}. Hence, by Lemma 6.6(i), $S_{C_2^2}$ is in bijection with the double cosets not met by S_{gen} with $|\text{Stab}(\varphi)| = 4$ and $[\varphi] \neq [\varphi^{-1}]$, except that for $F \cong K$ the two classes $[\varphi]$, $[\varphi^{-1}]$ of one entry give one lattice. Counting them gives 67,284 [C26, C28]; for the pair with $s = 16$, $\delta = 1$, for instance, 33,006 classes not met by S_{gen} give 16,503 lattices.

The Hermitian entries of order 4 of all lattices in $\mathcal{L}$ are the 76,742 classes of Proposition 4.6. Removing those that are entries of lattices of S_{gen} leaves 65,197 classes, which Lemma 6.6(ii)–(iii) describes. By Proposition 6.5(i), the 62,650 classes with $g \not\sim -g$ are the lattices of S_{CQ} with cyclic group, with $|\text{Aut } L| = |\text{U}| = 4, 8, 16$ for 62,468, 175, 7 of them. The values of $|\text{U}| = |C(g)|$ are those of the enumeration of Proposition 4.6, where for the 62,468 classes with $|\text{U}| = 4$ the canonical form is discrete, which forces $\text{U} = \langle g \rangle$, and for the 182 classes with $|\text{U}| \geq 8$ they come from a backtrack search and were all confirmed with bliss. The remaining 2,547 classes, with $g \sim -g$, fall into 849 triples with equal isometry invariants, and each triple was shown to be a single lattice by two explicit isometries checked in integers [C27, C28]. By Proposition 6.5(ii), the patterns of |U| in the triples, $(4, 4, 4)$, $(4, 4, 8)$, $(4, 4, 16)$, $(4, 4, 32)$, occurring 793, 48, 6, 2 times, give the quaternion groups of orders 8, 16, 32, 64. The saved triples form an exact partition of these 2,547 Gaussian classes. Different triples cannot represent

the same integer lattice, since Proposition 6.5 gives exactly three Gaussian entries for each quaternion lattice and Proposition 4.6 makes the entries distinct. This proves Table 8.

Table 8. The lattices of S_{CQ} (Theorem B).

Aut L	lattices	$1-2/\lvert\mathrm{Aut}\ L\rvert$	total	entries of order 4
C_4	62,468	1/2	31,234	1
C_8	175	3/4	525/4	1
C_{16}	7	7/8	49/8	1
Q_8	793	3/4	2,379 /4	3
Q_{16}	48	7/8	42	3
Q_{32}	6	15/16	45/8	3
Q_{64}	2	31/32	31/16	3
total	**63,499**		**512,251 /16**	

Example (one lattice of each kind). This example is not used in the proofs.

- `32.4.112978` lies in $S_{C_2^2}$. Its only entry is an involution entry for the pair of components $\sqrt{2}(D_8^2)^+$, $\sqrt{2}(D_8^2)^+$ ($s=16$, $\delta=1$, the last row of Table 6). The entry consists of two classes $[\varphi]\neq[\varphi^{-1}]$ of that list, both with $|\mathrm{Stab}(\varphi)|=4$, and neither is met by S_{gen}. By Lemma 6.6(i) the two classes give one lattice, with Aut $L=\{\pm1,\pm\sigma\}$; they are the pairs (L,σ) and $(L,-\sigma)$. The lattice contributes $1-2/4=1/2$ to (5.1).
- `32.4.1` lies in S_{CQ}, with Aut $L\cong C_4$. Its only entry is one Gaussian class of Proposition 4.6, with $f=2$ and $|\mathrm{U}|=4$. By Lemma 6.6(ii), Aut $L=\langle g\rangle$.
- `32.8.4736` lies in S_{CQ}, with Aut $L\cong Q_8$. It has three Gaussian classes, each with $f=1$ and $|\mathrm{U}|=4$: the pattern $(4,4,4)$ of §6.6. By Lemma 6.6(iii) they are one lattice with group Q_8, and the three classes are its three subgroups of order 4.

6.7 The total from the classified lattices

The three parts proved complete in §§6.3–6.5 are disjoint and exhaust S. Consequently the correction in (5.1) is obtained by adding $1-2/\lvert\mathrm{Aut}\ L\rvert$ over their representatives. Each lattice of $S_{C_2^2}$ contributes 1/2 (its group has order 4); the cyclic and quaternionic contributions use Table 8, and the general part uses the full-group orders of §6.3.

The saved database aggregation [C13] computes this correction directly from the classified rows, with no expected total as its pass condition. This is the arithmetic used in §5.1. The element-assignment computation of [Aux] shares the component lists, so agreement with it is an auxiliary consistency check, not an independent proof of completeness. Because that computation uses no full automorphism group of rank 32, it is a global check of rows 10 and 11 of Table 7 by a different arithmetic route (§5.2); it does not test the shared component and gluing lists.

For the general-part calculation, every classified general-part row is linked to an original successful full-group record with the same group order [C13]. That the entries which were not processed, because their lattices had already been reached from other entries, are entries of these lattices is Lemma 6.3(ii), for all types, including those without identifiers. These provenance checks do not establish that the original lists of pairs are exhaustive. The completeness arguments of §4 and Lemma 6.3 remain necessary.

6.8 The database and the distribution of $|\mathrm{Aut}\ L|$

Numbering. In [D] each $L \in S$ has the label `32.`$|\mathrm{Aut}\ L|$`.`i, where $i = 1, 2, \ldots$ numbers the lattices with the same $|\mathrm{Aut}\ L|$ in the following order.

1. The isomorphism type of $\mathrm{Aut}\ L$: groups with an identifier $[n, k]$ in the Small Groups Library by k, then groups identified by a structure description by that description, then the remaining 19 groups (nine identified by the orders of a chief series and the ten of order above 10^7) by the orders of a chief series; these 19 lattices have 18 distinct orders, and the two of order 196,608 have chief series of different orders (recorded in `data/classified-lattices.jsonl`).
2. The part of Theorem B: S_{gen}, then $S_{C_2^2}$, then S_{CQ}. The part is recorded in the field `part` of `data/classified-lattices.jsonl`; the order of the parts is the one just given.
3. A key, which is an isometry invariant of L (the first two kinds below are computed from the lists of §4): the smallest identifier of a class of the lists of §4.1 and §4.2 that is an entry of L (all lattices of $S_{C_2^2}$ and 12,519 of S_{gen}); otherwise the smallest canonical form or invariant D (§7.2) of a pair $(L, \pm g)$ with $g^2 = -1$ (all lattices of S_{CQ} and 5 of S_{gen}); otherwise the hash of the configuration modulo 2 of the lines spanned by vectors of norm 4 (22 lattices of S_{gen}). Within each part and group these keys are distinct.

This order is a convention without mathematical meaning. The labels were fixed with version 1 of [D] and are kept in later versions. The Gram matrices and generators of [D] are those of the computations of §§6.3–6.5; in version 1 the generators of the 182 lattices with $\mathrm{Aut}\ L = C_8$ or C_{16} generated only the subgroup of order 4, and version 2 adds a generator of $\mathrm{Aut}\ L$, found by the search of Proposition 4.6. The last label is that of BW_{32}, `32.48126558103142400.1`.

Table 9 gives the distribution of $|\mathrm{Aut}\ L|$ over $\mathcal{L}$. For the 12,546 lattices of S_{gen} it lists the orders computed with bliss (§6.3, Table 10); the 67,284 lattices of $S_{C_2^2}$ have $|\mathrm{Aut}\ L| = 4$, those of S_{CQ} are in Table 8, and the remaining 10,900,939 lattices have $\mathrm{Aut}\ L = \{\pm 1\}$ (Corollary C). The column of numbers sums to $|S| = 143{,}329$ without the first row and to N with it, and the orders satisfy (5.1) as an equality of fractions [C31, C30]. In S_{gen}, 104 distinct orders occur. The most frequent are 8 (7,055 lattices), 6 (2,308) and 16 (1,233); the largest is $|\mathrm{Aut}\ \mathrm{BW}_{32}|$.

The group of each lattice of S_{gen} was also identified [C30]. The generators are checked in integers to be automorphisms, and the conjugacy classes, with their sizes summing to $|\mathrm{Aut}\ L|$, were computed with GAP. For $|\mathrm{Aut}\ L| \le 10^7$, that is, for all but ten lattices of S_{gen}, the group was identified as follows. For 12,447 lattices it was identified by its number in the Small Groups Library of GAP, with 365 distinct groups; the most frequent are $C_4 \times C_2$ (3,737 lattices), C_2^3 (2,495), C_6 (2,308) and D_8 (823). For 80 lattices it was described by the structure description of GAP (a description, not a proof of isomorphism), and for the remaining nine only by the orders of a chief series. Each of the 12,536 groups was also given by a presentation. Its relations were checked on the generators, and the order of the finitely presented group was computed and equals $|\mathrm{Aut}\ L|$. For nine groups of order above 10^5 the identification run did not count this order; it was counted afterwards, by coset enumeration along a chain of point stabilizers in a permutation representation of small degree, with Reidemeister–Schreier presentations of the stabilizers [C30]. None of these identifications is used in Theorems A and B.

Table 9. The number of $L \in \mathcal{L}$ with a given $|\mathrm{Aut}\ L|$, read down the columns. Only the orders $4, 8, 16, 32, 64$ receive lattices of $S_{C_2^2}$ or S_{CQ} (Table 8); all others except 2 (the lattices outside S) are orders of lattices of S_{gen} [C30].

$\lvert\mathrm{Aut}\ L\rvert$	lattices	$\lvert\mathrm{Aut}\ L\rvert$	lattices	$\lvert\mathrm{Aut}\ L\rvert$	lattices	$\lvert\mathrm{Aut}\ L\rvert$	lattices
2	10,900,939	240	3	3,072	5	62,208	1
4	129,752	256	27	4,032	1	65,536	1
6	2,308	288	19	4,096	1	196,608	2
8	8,023	320	2	4,608	2	207,360	1
10	6	324	1	4,896	1	245,760	1
12	674	384	21	5,184	1	262,144	1
16	1,288	432	2	5,760	2	294,912	1
18	4	480	5	6,144	2	322,560	1
20	13	512	12	7,680	2	368,640	1
24	272	576	3	8,192	1	393,216	1
32	290	672	1	9,216	3	589,824	1
36	17	768	10	10,240	1	839,808	1
40	23	800	1	10,368	1	1,179,648	1
48	163	864	1	11,520	1	1,548,288	1
54	2	896	1	12,288	1	2,073,600	1
64	112	960	2	15,360	1	26,542,080	1
72	17	1,024	7	16,128	1	44,040,192	1
80	5	1,152	5	16,384	1	53,084,160	1
96	39	1,440	2	17,280	1	74,317,824	1
108	2	1,536	6	18,432	1	975,175,680	1
120	8	1,728	2	19,584	1	1,244,160,000	1
128	61	1,920	1	20,736	1	1,509,949,440	1
144	11	2,016	1	23,328	1	16,911,433,728	1
160	3	2,048	5	24,000	1	91,321,742,131,200	1
192	23	2,304	2	27,648	2	48,126,558,103,142,400	1
216	4	2,592	1	51,840	1		
224	1	2,880	1	58,752	1		

7. Computations and assumptions

> **What this section does.** §7.1 lists the programs. §7.2 states the two lemmas through which our lists use keys (Lemmas 7.1 and 7.2) and proves that each key is an isometry invariant. §7.3 lists everything the proofs assume (Table 10). §7.4 lists the computations with their certificates and running times (Table 11) and describes the files that accompany this paper and how they are checked. The research record [Rec] keeps the programs and runs behind the lists.

7.1 Programs

All computations used one environment: PARI/GP 2.17.3 [PARI] (through cypari2 2.2.4) for `qfminim`, `qfauto` and `qfisom`; GAP 4.15.1 [GAP] for orders of permutation groups, double cosets, conjugacy classes and centralizers; bliss 0.77 [JK07, JK11], called through its C interface, for canonical labellings and automorphism groups of graphs; SageMath 10.9 [Sage] for genus symbols and masses of genera; and our own code in Python 3.12 and C. The exact linear programs of Proposition 3.3 were solved with the

Parma Polyhedra Library (PPL) through SageMath; the proof uses only their finite rational certificates (§B.1.1), which are checked without it. Calls of `qfauto` and `qfisom` go through a wrapper that retries with random changes of basis and checks every returned matrix in integers.

Full automorphism groups. The orders $|\mathrm{Aut}\ L|$ of the lattices of S_{gen} and of the two exceptional known lattices are computed by one algorithm [C1, C22]. Let X be the set of vectors of norm 4 of L; the program checks that it has the 146,880 elements of an extremal lattice of rank 32 (§B.3). Fix $x \in X$. The *rooted local Gram graph* of x has as vertices x and the vectors $y \in X$ with $(x, y) = 2$, and every pair of vertices is joined by an arc coloured by their inner product (no arc is omitted); x receives its own vertex colour, so that the automorphisms of the coloured graph are exactly those fixing x. These vectors span $L \otimes \mathbb{Q}$, which is checked. An automorphism of L fixing x permutes these vertices and preserves the colours, and it is determined by that permutation. Conversely every colour-preserving automorphism of the graph that fixes x extends to a unique rational isometry, and the stabilizer $\mathrm{Aut}(L)_x$ consists of those extensions that preserve L. They are found exactly: with M the sublattice spanned by the vertices, they are the elements stabilizing the subgroup L/M of the discriminant group of M (a set stabilizer in GAP; when that group is not 2-elementary, a Schreier orbit of L under the rational group). Next, every line $\pm y \subset X$ is a candidate image of $\pm x$ unless an isometry invariant (the class profile modulo 2) excludes it. For each candidate not already in the orbit, an isometry of L taking x to y exists if and only if some isomorphism of the rooted local Gram graphs of x and y extends to an isometry preserving L, and this is decided exactly: the isomorphisms from the graph of x to that of y form one coset τG_x of the automorphism group G_x of the graph of x, where τ is one of them, extended to a rational isometry taking M_x to M_y. Then $\tau^{-1}(L)$ is an overlattice of M_x in $M_x^{\#}$, and an isometry of L taking x to y exists if and only if some element of G_x maps the subgroup L/M_x of $M_x^{\#}/M_x$ to $\tau^{-1}(L)/M_x$. This transporter problem is decided by GAP in the same action of G_x on $M_x^{\#}/M_x$ that gives the stabilizer. Since every candidate is tested, the orbit of $\pm x$ is determined, not only bounded below, and $|\mathrm{Aut}\ L| = 2\ |\mathrm{Aut}(L) \cdot (\pm x)| \cdot |\mathrm{Aut}(L)_x|$. All returned generators are checked in integers. The computation assumes the completeness of the short-vector enumeration, that bliss returns all automorphisms of the graph and canonical labellings, the correctness of GAP's stabilizer and orbit computations, and that our code implements these steps (Table 10).

7.2 Isomorphism keys

Our lists identify objects (lattices, pairs (L, h), double cosets) by *keys*: functions κ computed from a representative. A key is *invariant* if isometric objects have equal keys. The lists are used through the following two lemmas, which need only invariance, never the converse.

Lemma 7.1. Let $\mathcal{S}$ be a finite set of isomorphism classes with weights $w(X) > 0$ and $\sum_{X \in \mathcal{S}} w(X) = \mu$, and let κ be an invariant key. If $X_1, \ldots, X_n \in \mathcal{S}$ have pairwise distinct keys and $\sum_i w(X_i) = \mu$, then $\mathcal{S} = \{X_1, \ldots, X_n\}$.

Proof. Distinct keys imply distinct classes, so $\sum_i w(X_i) \le \mu$, with equality only if every class of $\mathcal{S}$ occurs. □

Lemma 7.2. In the situation of Lemma 7.1, two objects of $\mathcal{S}$ with equal keys are isomorphic.

Proof. Let $X \in \mathcal{S}$. By Lemma 7.1, X is isomorphic to some X_i, so $\kappa(X) = \kappa(X_i)$, and the $\kappa(X_i)$ are pairwise distinct. So κ separates the classes of $\mathcal{S}$. □

Lemma 7.2 is what allows the subtraction of Lemma 6.6: the Hermitian entries of order 4 of the lattices of S_{gen}, wherever they come from, are classes of the set of Proposition 4.6, so each is the unique class of that list with its key, and no explicit isometry is needed.

The keys are of the following kinds. For each we say how it is computed and why it is invariant.

1. *Canonical form of the generating shell.* Let $t(L)$ be the least even $t \geq 2$ such that the vectors of norm at most t generate L, and $X_t = \{x \in L : 0 < (x,x) \leq t(L)\}$ (with both signs). The graph $\Gamma(L)$ has vertex set X_t and an arc $x \to y$ coloured by a fixed function of (x,y) (and of $(x,\omega y)$ for a lattice with a structure ω); arcs with $|(x,y)|$ below a threshold depending only on $|X_t|$ are omitted. It is turned into a vertex-coloured graph by one layer per occurring colour, bliss computes a canonical labelling, and the key is $(t(L), |X_t|)$ and a hash of the relabelled arc set [C5]. *Invariance:* $t(L)$ and X_t are defined by norms and generation only, so an isometry $u : L_1 \to L_2$ (commuting with ω) maps $X_t(L_1)$ onto $X_t(L_2)$; it preserves inner products, hence colours and omitted arcs, so it is an isomorphism of coloured graphs; the layer of a colour is its rank among the occurring colours, which are the same on both sides, so u extends to an isomorphism of the layered graphs; a canonical labelling maps isomorphic graphs to the same graph, and the hash is a function of that graph. The choice of basis, the order of the vectors and the representatives of $\pm x$ only renumber vertices. A non-injective colour function or hash only makes the key coarser. With a structure ω, the key is an invariant of the Hermitian lattice up to isometries commuting with ω; it is used only for lists of Hermitian lattices up to such isometries (the lists of Appendix B.4), where a lattice and its complex conjugate are not identified.
2. *Configuration of the vectors of norm at most 4*: the same construction with $t(L)$ replaced by 4. It is invariant by the same argument; since these vectors need not generate L, it is used only to show non-isometry.
3. *The invariant D of $(L, \pm g)$, $g^2 = -1$.* With $V = L/\varpi L \cong \mathbb{F}_2^{16}$, let $n(c)$ and $r(c)$ be the numbers of lines $\langle \pm x, \pm xg \rangle$ of norm 4 and of norm 2 in the class $c \in V$ ($r = 0$ for the rootless lattices of Proposition 4.6; the same function is used for the lattices with roots met on the way), form the convolutions $n * n$, $n * n * n$ and $r * n$ over the group V, colour each c by a fixed function of the five values at c, and let D be a hash of the multiset of colours. *Invariance:* an isometry $\varphi : L \to L'$ with $\varphi g = \pm g' \varphi$ maps lines to lines and ϖL to $\varpi L'$, so it induces a group isomorphism $V \to V'$ carrying n, r to n', r'; convolutions commute with group isomorphisms; and x, $-x$, xg, $-xg$ lie in the same class modulo ϖ, so the counts do not depend on representatives. Replacing g by $-g$ does not change ϖL or the lines.
4. *Canonical Gram matrix of a pair $(L, \pm g)$, $g^2 = -1$* (used together with D for Proposition 4.6). Select the classes of V whose colour occurs once and which contain exactly one Gaussian line of norm 4, ordered by colour. Try the first $16, 20, 24, \ldots$ lines, up to 64, and choose the first such prefix that generates L as a $\mathbb{Z}[i]$-module. Generation means that the vectors $x_j, x_j g$ generate the integer lattice; it is checked by rank and Hermite normal form. If such a prefix exists, the pair is *discrete*. Whether a pair is discrete depends only on the colours of the classes of V, the numbers of lines in them and generation, which an isometry or a complex conjugation preserves as for D; so it is an invariant, and the key of a pair, its canonical Gram matrix if it is discrete and D otherwise, is an invariant as well. Starting with line 0, repeatedly select the least indexed unprocessed line adjacent to a processed line under the relation $H(x_j, x_k) \neq 0$. Its parent is the first adjacent line in the order in which the processed lines were reached. Rotate its representative by the unique unit for which the inner product with the parent has positive real part and non-negative imaginary part. Apply the

same procedure to the conjugate Hermitian form. The key is the smaller hash of the two resulting Gram matrices. *Connectedness:* if the non-orthogonality graph were disconnected, its components would generate a non-trivial orthogonal decomposition of L. Each summand would be an even unimodular integer lattice without roots. Its rank is a multiple of 8, and a non-trivial partition of 32 has a part of rank 8 or 16, where an even unimodular lattice has a root. Thus the graph is connected. *Invariance and the group bound:* changing representatives does not change the traversal or its parents. Inductively, the quadrant condition makes all normalized representatives agree up to the same unit on the first line, so the Gram matrix is unchanged. A linear isometry preserves this construction, and an anti-linear one exchanges the two constructions. Each element of U preserves every selected line. Its units on two adjacent lines coincide, and connectedness and generation give $\mathrm{U} = \{\pm 1, \pm i\}$. If the hashes of the two Gram matrices (of the form and of its conjugate) differ, the pair has no anti-linear automorphism, since one would exchange the two constructions. Equality of the Gram matrices themselves would establish one by mapping the generating vectors; equality of hashes alone is not used for that conclusion. For all 62,726 discrete classes in the final list these two hashes differ.

5. *Canonical form of a gluing*: the lexicographically smallest table of φ in its double coset $\overline{G}_K \varphi \overline{G}_F$, computed by a search along stabilizer chains of $\overline{G}_K$ and $\overline{G}_F$ [C4]. It is a function of the double coset by definition; that the program returns this minimum is a property of the program, which we assume (Table 10); it was checked on samples, and the canonical forms of all representatives were recomputed from scratch with the same result.
6. *Canonical form for the type* $(8, 24)$: the construction of the canonical form of the generating shell, applied to the 720 vectors of norm 2 of $K^{\#}$, with a separate call of bliss and a truncated hash [C6]; it is invariant by the same argument for all 14,482 lattices, whether or not the 720 vectors generate $K^{\#}$.

All keys are compared only between values produced by the same program with the same version of bliss. In particular, the keys of the Hermitian entries of order 4 of the lattices of S_{gen} [C1] and the keys of the classes of Proposition 4.6 [C12] are computed by the same function: the rule is the canonical Gram key for a discrete class and the invariant D otherwise. The same version of the program was used for both. For each of the 76,742 classes of Proposition 4.6, rerunning the enumeration used for its key (the same function on the saved Gram matrix of the underlying lattice of rank 32) returns exactly 73,440 vectors, each of norm 4 and no two equal up to sign. That lattice is rootless (by PARI `qfminim`, Table 10), hence extremal, and so has $146{,}880 = 2 \cdot 73{,}440$ vectors of norm 4 (§B.3); every pair $\pm v$ of norm 4 is therefore represented, the enumeration is complete for these classes, and D and the canonical Gram keys recomputed from it are the saved ones [C12]. For all 16,163 recorded Hermitian entries of order 4 of S_{gen}, the class of Proposition 4.6 with the same key has the same $|\mathrm{U}|$ and the same value of f [C29]. The lists of §4, Appendix B and Proposition 4.6 use keys only through Lemmas 7.1 and 7.2, and every identification "equal keys, hence isometric" made while building a list concerns objects of the set $\mathcal{S}$ to which Lemma 7.2 applies.

7.3 What the proofs assume

Table 10. What Theorems A and B assume. Outputs that are checked in integers (an element is an automorphism, a matrix is an isometry, a vector has the stated norm) are not assumptions; what is assumed is that the programs do not miss anything (a group is all of Aut, a list of vectors or of double cosets is complete, no further isometry exists) and that our own code implements the stated formulas.

assumption	used for	theorem
King's mass m [Kin03], recomputed with our implementation of the same inclusion–exclusion [C14]	term $2m$	A
PARI `qfminim`: the enumeration of short vectors is complete	minima, roots and norm-4 vectors of components and of glued lattices; extremality; the numbers $g(W_3)$ of §B.3 and $g(Z_+)$ of (B.2); the absence of roots in the 76,742 classes of Proposition 4.6 (on which the count of §7.2 rests); the theta coefficients of the bases of the six linear programs (§B.1.1)	A, B
PARI `qfauto`: the group generated by the returned automorphisms is all of Aut	|Aut| and |U| of components; orbit reduction in Lemma B.1 (a group that is too small only adds work)	A, B
PARI `qfisom`: the answer "not isometric" is correct	merging in the enumeration of Lemma B.1 for ranks at most 18 and in the classifications of genera by neighbours; for rank 20 every "not isometric" was confirmed by different keys (§7.2)	A, B
bliss: the canonical labelling is canonical, and the automorphism group returned is all of Aut	the canonical forms of §7.2 of the generating shell, of the configuration of vectors of norm at most 4 and for the type $(8, 24)$; |Aut K| for 14,462 lattices of type $(8, 24)$; the automorphisms of the rooted local Gram graphs of §7.1, hence |Aut L| for the 12,546 lattices of S_{gen} and the two exceptional known lattices	A, B
our code: the full-group algorithm of §7.1 (extension of graph automorphisms, the stabilizer of L/M, the test of every candidate line)	|Aut L| for S_{gen} and the exceptional known lattices	A, B
GAP: orders of permutation groups, double cosets, conjugacy classes, centralizers, set stabilizers	$\lvert O(q)\rvert$, $\lvert\mathrm{Stab}(\varphi)\rvert$, double cosets, conjugacy classes and invariant group signatures; the Aut P-orbits of good W and W_3 of §B.3 (complete) and their stabilizers, hence the left side of (B.1); the classes with $g^2 = -1$ and their centralizers in $W(F_4)$ and $W(E_8)$ (Proposition C.1); the stabilizers of §7.1	A, B
our canonical form of gluings returns the minimum over the double coset	the lists of double cosets for involutions, and for odd prime order where GAP did not finish (§4.2) (the canonical form of a gluing, §7.2)	A, B
SageMath: the mass of a genus of $\mathbb{Z}$-lattices (Smith–Minkowski–Siegel); dimensions of spaces of modular forms	completeness of the classifications of genera of components F (Appendix B.4) and the check in rank 16 (Appendix B.2); the bases of the six linear programs (§B.1.1)	A, B
published classifications [Hen13, Thm 1, Table 4, Rem. 3(a)], [Jür15, Satz 3.6.9]	types $3-(16,0)-0$ and $17-(2,0)-0$; both are also confirmed by rootless masses (§4.2, Appendix D)	A, B

assumption	used for	theorem
our implementations of the mass formulas of Appendices C and D, in exact arithmetic	M_2; the rootless component masses of Table D1 and §D.4; the vanishing rootless Hermitian masses of reason 3 of Proposition 3.3	A, B
the weights $f/\lvert\mathrm{U}\rvert$ of Proposition 4.6: canonical forms, backtrack search, bliss	M_2 confirmed by enumeration; $\lvert\mathrm{Aut}\ L\rvert$ for the lattices of S_{CQ} with cyclic group, and the 2,547 classes of the quaternion lattices (Lemma 6.6(iii))	A, B
$\lvert\mathrm{Aut}\ K\rvert$ for 20 lattices of type $(8,24)$ whose 720 short vectors of $K^{\#}$ do not generate it: the stabilizer of N/N' in the automorphism group of the graph of the 720 vectors (16, §B.3), `qfauto` (4)	the term of type $(8,24)$	A, B
the short-vector enumeration for the Gaussian keys (floating-point Fincke–Pohst) is complete	the keys D and canonical Gram matrices of §7.2 for the entries of S_{gen} (for the classes of Proposition 4.6 the enumeration is complete by the count of §7.2)	A, B
the canonical key of type $(8,24)$ (§7.2) is invariant and its saved prefix is unique among the 14,482 lattices K	completion of four identifiers of type $(8,24)$ in groups of order above 10^7 (§6.3)	A, B
our code: exact identifiers of entries and exact types (§6.3), minimum-shell profiles, orientation and exact sums	classification, Gaussian weights and (5.1)	A, B

7.4 Computations and reproduction

Table 11. Computations and code. References such as [C1] in the text point to the rows of this table.

ID	computation and code	CPU time

C1	The lattices of S_{gen} (Theorem B): the lattices reached by the entries of the lists of §4, their automorphism groups (bliss), conjugacy classes (GAP), all their entries, and the identification of equal lattices
	`certificates/general-part-identification.json`
	`research record: C1/`
		264 h

C2	Types of elements of odd prime order: existence of genera, rootless Hermitian masses, exact linear programs on theta series (PPL), and complete genus lists used to exclude all but the 19 types of Table 4
	`certificates/odd-prime-masses.json`
	`research record: C2/`
		—

C3	Six finite rational infeasibility certificates for the remaining prime-order types, with the bases of the theta series of the components fixed, every row checked exactly, and the bases and the signs of the discriminants checked
	`certificates/type-exclusion-lp.json`
	`research record: C3/`
		—

C4	Involution types $(d, 32-d)$ with $d \in \{12,14,15,16\}$: the gluing classes $[\varphi]$, their stabilizers, the extremality flag, and the mass identity $\sum 1/\lvert\mathrm{Stab}\rvert = \lvert O(q)\rvert\ /(\lvert\mathrm{Aut}\ F\rvert\ \lvert\mathrm{Aut}\ K\rvert)$ for every pair of components
	`certificates/involution-gluing-masses.json`
	`research record: C4/`
		—

ID	computation and code	CPU time

C5	Components of type $(12, 20)$: the 66 classes of rank-20 2-elementary lattices with $s = 12$, $\delta = 1$ and minimum ≥ 4, by the mass formula and independently by chains of index-2 overlattices; isometry decisions by bliss canonical forms and norm-4 configurations
	`certificates/involution-gluing-masses.json`
	`research record: C5/`
		≥ 37 h

C6	Involution type $(8, 24)$: the 14,482 rank-24 lattices K with $\min K \geq 4$ and $\min K^{\#} \geq 2$, via the reduction to Niemeier lattices (Theorem B.3), the orbits of good subspaces for the Leech lattice and A_1^{24}, and identity (B.1) for A_2^{12}
	`certificates/type-8-24-masses.json`
	`research record: C6/`
		—

C7	Components of involution types of ranks 8 to 20: chains of index-2 overlattices from $\sqrt{2}U$ (Lemma B.1); in rank 16 also the classification of the seventeen genera of even 2-elementary lattices by Kneser 3-neighbours, completed by the mass of the genus (a check, not used in the proofs)
	`certificates/involution-gluing-masses.json`
	`research record: C7/`
		—

C8	Elements of order 3: the ten types $3 - (z, f) - s$, their components, gluing classes and the numbers $e(h)$
	`certificates/odd-prime-masses.json`
	`research record: C8/`
		—

C9	Elements of order $p = 5, 7, 17, 31$: the nine types, their components and the numbers $e(h)$; the rootless masses (D.4)–(D.5) and their checks; the existence conditions of Proposition D.1 compared with the oddity formula; the terms $g(Z_+)$ of (B.2) for $p = 5$, recomputed for every class with the functions of the list program
	`certificates/odd-prime-masses.json`
	`research record: C9/`
		—

C10	The classes of $K_{12}^{\#}/K_{12}$: the 756 vectors of norm $4/3$ lie three in each of the 252 classes with $q = 4/3$ (a check of the argument of §4.2) `research record: C10/`	—

C11	The rootless mass M_2 of rank-16 Hermitian $\mathbb{Z}[i]$-lattices whose trace form is even unimodular (exact rational arithmetic)
	`certificates/gaussian-rootless-mass.json`
	`research record: C11/`
		64 min

C12	The 76,742 classes of rootless Hermitian $\mathbb{Z}[i]$-lattices of rank 16; the sum of their weights equals the value of [C11] as a fraction (computed by two programs); for every class, rerunning the enumeration used for its key returns exactly 73,440 vectors of norm 4, no two equal up to sign, and D and the canonical keys recomputed from it equal the saved ones (the time given is that of this check alone; the time of the lists was not recorded)
	`certificates/gaussian-list-weights.json`
	`research record: C12/`
		≥ 15.9 h

C13	Direct classification-based aggregation of the total from King's mass and the saved lattice table; the sources of the group orders of S_{gen}, and a check that no planned input was left out. No thirteen-term values, assignment counts or higher cyclotomic masses are read `research record: C13/`	—
C14	The mass $m = m_{32}(\emptyset)$ of rootless even unimodular lattices of rank 32 by King's inclusion–exclusion over root systems, reimplemented; agrees with the value tabulated in [Kin03] as a fraction	23 min

ID	computation and code	CPU time
	certificates/king-rootless-mass.json research record: C14/	
C15	King's exact value for the empty root system, and the value used in the paper; the primary mass normalization uses full automorphism orders certificates/king-rootless-mass.json research record: C15/	—
C16	Rootless masses of the Eisenstein genera (N, s, b) by (D.6), (D.8), (D.9) (Appendix D.4); the 105 classes of type $(13, 5)$; the checks of Table D4 certificates/odd-prime-masses.json research record: C16/	—
C17	Rootless masses of the p-elementary $\mathbb{Z}$-genera $(20, 6)$, $p = 3$, and $(16, 4)$, $p = 5$, by (D.10)–(D.12); the 24 lattices F; the checks of Table D5 certificates/odd-prime-masses.json research record: C17/	—
C18	Finite proof of the elementary quadratic 2-form element orbits by actual isometries on support patterns, for all 230 diagonal forms of dimensions 1 to 20 certificates/quadratic-2-form-orbits.json research record: C18/	—
C19	Full rank-16 root-type index and inclusion-exclusion sum, with regenerated exceptional coefficients and the local DP statistics of [C11] as inputs certificates/gaussian-rootless-mass.json research record: C19/	—
C20	Closed formulas for the root-sublattice coefficients checked against all 12,934 terms of [C19], and the classes with $g^2 = -1$ in $W(F_4)$, $W(E_8)$ and $2.\mathrm{Co}_1$ from the character tables of CTblLib (a method different from [C11]) certificates/root-coefficients.json research record: C20/	—
C21	Lemma 6.2 checked on all groups of order 4 to 190 with a central involution (supplementary; the lemma is proved in §6.2) research record: C21/	12 min
C22	Full automorphism groups of the two exceptional known lattices, recomputed from Gram matrices alone by the short-vector local-graph and integral-stabilizer algorithm; separate from the structural conjugacy-class calculation certificates/exceptional-full-groups.json research record: C22/	—
C23	39 positive identifications of unlabelled records by explicit unimodular integer isometries; separate from non-isometry decisions and signature invariance certificates/general-part-identification.json research record: C23/	—
C24	Full-group signatures of the unlabelled records compared with every successful record of the general part; the minimum-shell profiles of the 27 unlabelled lattices; the 67 integer isometries identifying the lattices with $\lvert\mathrm{Aut}\ L\rvert > 10^7$ (the signature report comes from the first version of the comparison program; the current version reproduces the same signatures and adds diagnostics) certificates/general-part-distinctness.json certificates/general-part-identification.json certificates/general-part-witnesses.json research record: C24/	—

ID	computation and code	CPU time

C25	Read-only count of exact entry identifiers by type in all successful general-part records, and the identification classes containing an identifier of an exact type (12,519 of 12,546), with one witness row for each class extracted by a separate program and replayed from the saved rows; conditional on full groups, complete class rows and correct lookups
	`certificates/general-part-distinctness.json`
	`certificates/general-part-witnesses.json`
	`research record: C25/`
		—

C26	The lattices of $S_{C_2^2}$ (Theorem B): the trivial involution entries not met by S_{gen} in [C1]
	`certificates/c2xc2-part.json`
	`research record: C26/`
		3 min

C27	The lattices of S_{CQ} (Theorem B): cyclic and generalized quaternion groups; the 849 triples of anti-linear classes, each shown to be one lattice by two explicit isometries checked in integers
	`certificates/quaternion-triples.json`
	`research record: C27/`
		≥ 22.5 h

C28	Inverse-gluing partition for $S_{C_2^2}$, matching the entries to the lattices by their identifiers, and all 1,698 integer isometries connecting the 849 quaternion triples, using the Gram matrices of the Gaussian classes
	`certificates/c2xc2-part.json`
	`certificates/quaternion-triples.json`
	`research record: C28/`
		—

C29	Read-only consistency of the saved full-group records of the general part with the subtraction lemma: trivial involution entries and their pair identifiers, Gaussian keys in the final list with equal unitary orders and anti-linear flags, and the provenance of the common Gaussian key function; the subtraction recomputed as set differences of individual entries and keys from the saved rows
	`certificates/general-part-witnesses.json`
	`certificates/subtraction-records.json`
	`research record: C29/`
		—

C30	Identification of Aut L for the 12,536 lattices of S_{gen} with $\lvert\mathrm{Aut}\ L\rvert \le 10^7$ (Small Groups Library, structure description, chief series; presentations and conjugacy classes, checked against the generators; the order of every presentation, for nine groups by coset enumeration along a chain of point stabilizers); Table 9 and its checks against $\lvert S\rvert$, (5.1) and Corollary C (supplementary) `research record: C30/`	about 112 h
C31	Identity (5.1) from the classified lattices and its split by part (two implementations) `research record: C31/`	1 s
C32	The 26 previously known extremal lattices of rank 32 with $\lvert\mathrm{Aut}\ L\rvert > 2$ found before the comparison with the catalogue, and their location in the lists of §4 (supplementary) `research record: C32/`	1.35 h
C33	The extremal lattices of rank 32 of the catalogue [NS] and the lattice of [BB92] for $m = 51$: an isometry onto a lattice of Table A1 found with bliss, with $T^{\mathsf{T}}GT = G'$ and $\det T = \pm 1$ checked in integers (supplementary) `research record: C33/`	—
C34	The numbers $e(\sigma)$ for involution entries (lifting to $C(\sigma)$ and counting elements assigned to σ), and their recount inside Aut L for a subset of entries `research record: C34/`	42 h

ID	computation and code	CPU time
C35	Rootless masses M_a over $\mathbb{Z}[\zeta_{2^a}]$ for $a = 3, 4, 5$ (auxiliary [Aux]); for $a = 4$ the list of 514 classes and its agreement with the formula; for $a = 5$ the direct count of the types of rank 2; the class of $g^4 = -1$ in $2.\mathrm{Co}_1$ `research record: C35/`	84 min
C36	The thirteen terms of the auxiliary count [Aux] read from the output files, and their sum (two implementations; auxiliary) `research record: C36/`	4 s
C37	Checks of the tools on known cases (auxiliary checks): masses, King's table, the unimodular lattices of rank at most 24, existence of genera, random gluings, the type $(8, 24)$, published classifications, automorphism groups `research record: C37/`	14 min
C38	The tools in rank 24 against the conjugacy classes of Co_0 (auxiliary checks), including the programs for e of [C34], [C8] and [C9] on the three involution types and nine types of odd prime order, and the range of the implementation of (C.2) in odd rank `research record: C38/`	—

This paper is accompanied by the repository `github.com/roiban1344/rank-32-extremal-even-unimodular-lattices`, version `v2026-11-14.1`. The directory `certificates/` holds the certificates read by the checks; Table 11 names, for each computation, its certificate and its place in the research record [Rec]. The directory `data/` holds the classified lattices of S with the orders and structure of their automorphism groups. The program `verify/verify.py` needs only Python and the files of the repository. It checks the six rational certificates of §B.1.1; identity (5.1) from King's mass and the classified lattices, with the totals of Theorems A and B and Table 8; every term of (C.2) from its coefficient, unitary group order and embedding average, with every embedding average $A_{16}(T)$ recomputed from the saved local statistics of §C.6, and the coefficients of A_n, D_n and G_m from the closed formulas of §C.3 (the four exceptional coefficients are inputs); in integers, the generators of the two largest groups and the 106 isometries of §6.3; the counts of §6 against the classified lattices, and from the saved rows of the full-group records the classes of Lemma 6.3(i) with a witness row for each, the identity of Lemma 6.3(ii) for every list with the distinctness of the rows that carry identifiers, and the subtraction of Lemma 6.6 as differences of sets of entries; the completeness identities from their terms, that is (4.1) for the pairs of components of §4.1 and §4.2, with $|O(q_F)|$ from the closed formulas of Lemma 2.1 (term by term for the 83 pairs of involution components and for 258 pairs of odd prime order, from the numbers of decompositions for the four pairs of $5 - (6, 8) - 6$ and from the recorded masses for the two pairs of $3 - (6, 20) - 6$ without extremal gluings, §4.2), the mass identities of Table B4, the steps of (B.2) used for Table B4 and their chains back to lists (z, z) that reach the rootless mass, identity (B.1) (for A_2^{12} from the terms of its 106,624 orbits of good W_3) and the sum of Proposition 4.6 with the number of weights obtained by each method; and the finite checks of the 9-torsion reduction and of the orbits of elementary quadratic 2-forms (§D.4.6). It does not recompute the terms of these identities (automorphism groups and stabilizers, numbers of decompositions, masses of genera), the rows of the full-group records, or isometry decisions; these rest on the statements of Table 7 and the assumptions of Table 10.

The programs that produced the lists, their runs and our verification notes are kept unchanged in the research record [Rec], a release asset described by its own index. They are a record of the work, not a library; documented modules will replace them in later versions. Table 12 lists the release assets with their SHA-256 digests. Files taken from other sources (King's table [Kin03], the pages of [NS]) are not included.

The CPU times of Table 11 add up to at least 498 hours. The 23 computations whose time was not recorded (C2, C3, C4, C6, C7, C8, C9, C10, C13, C15, C16, C17, C18, C19, C20, C22, C23, C24, C25, C28, C29, C33, C38) are not included.

Table 12. Files attached to the release, with their SHA-256 digests.

file	size	content
`research-record.tar.gz`	10.3 MB	the research record [Rec]: programs by computation, verification notes and an index `5d8e2f282e1e3569d9cc45bac7fae1637c734d1907c6e221ce5d832bab09d981`
`research-runs.tar.gz`	347.7 MB	the recorded runs of the programs of the research record (local paths masked) `f08b1cf95cae0210f1133d67b4280be0434e7a85df66f65911ed2b20b756c055`
`gaussian-walk-grams-1.jsonl.gz`	81.3 MB	Gram matrices of the Hermitian $\mathbb{Z}[i]$-lattices met by the first neighbour walk `7d03e622def823827fb26c3df58a688ffaacadc99f05dc1941ad9ffe16342ace`
`gaussian-walk-grams-2.jsonl.gz`	81.3 MB	Gram matrices met by the second neighbour walk `98ce247e53eb3000e5bdb6388cd7d7f4f12dbf8ecb424fb1bf494b5b9d12c36d`
`gaussian-classes.jsonl.gz`	69.9 MB	the 76,742 rootless Hermitian $\mathbb{Z}[i]$-lattices of Proposition 4.6: key, D, $\lvert U\rvert$, f, weight, Gram matrix G and W with $WGW^{\mathsf{T}} = G$, $W^2 = -1$ `7e8e1b92a2936de1fc7af40d7bb4fbbb1a789d810eff87495d1b676a916b8263`
`general-part-groups.tar`	29.7 MB	generators, presentations, conjugacy classes and group identifications for the 12,536 lattices of S_{gen} with $\lvert\mathrm{Aut}\ L\rvert \le 10^7$, by label, with a README `f75a42f14de3578580dd555823ca6373bc4fcde8b0451806e88b99d132429036`
`type-8-24-classes.tar`	119.1 MB	the 376,306 orbits of good W for $P = A_1^{24}$ of §B.3, each with a Gram matrix of K_W, $\lvert\mathrm{Aut}\ K_W\rvert$ and $\lvert\mathrm{Stab}(W)\rvert$ `d25b90009a6617708fe194b78d655bb448126efe6cec025d17d4637589068ce3`
`general-part-records.tar.gz`	1.3 MB	the 17,736 full-group records of §6.3 (Lemma 6.3): for each, the label, the group order, the conjugacy classes and the rows of entries, with a README `ef1187d1c734cff8a7862e1190d2989c9ef292f6748ee9abbbf16d422e733a08`
`database-v2.tar.gz`	90.2 MB	the database [D], version 2: for each lattice of S its Gram matrix and generators of Aut L, with a README, a manifest and an index by label `a0c9d3211bf71d3785bb74a827840a2726bdcd2140e5704c767dffedf4f3f023`

Appendix A. The previously known lattices

What this appendix does. It locates in S the 27 lattices of S that appear in the literature. The identifications with the literature are a comparison and are not used in the proofs of Theorems A and B; the published descriptions of 26 of these lattices (codes, Hermitian and quaternionic structures; the catalogue [NS] was used only for the comparison below) are among the starting points of the records of §6.3, which the proofs use through Lemma 6.3. We state on what each identification rests, using four grades of decreasing strength.

The grades are: (1) an explicit isometry to a lattice given by the source, checked in integers ($T^{\mathsf{T}} G_1 T = G_2$, $\det T = \pm 1$); (2) the same construction: the lattice is built from a code that is shown to be equivalent

to the code of the source by canonical forms, and equivalent codes give isometric lattices; (3) forced by two complete classifications: the source classifies a family completely, our lists of §4 contain the same family completely, and an invariant that takes distinct values within the family matches value by value; (4) agreement of invariants only.

All 27 lattices lie in S_{gen} of Theorem B: 25 of them have $|\mathrm{Aut}\ L|$ divisible by an odd prime, and the remaining two have 2-groups of orders 2^{18} and 2^{14}, larger than the orders allowed in $S_{C_2^2}$ and S_{CQ}. Their 27 values of $|\mathrm{Aut}\ L|$ are pairwise distinct [C32, C33]. The lattice of S with $|\mathrm{Aut}\ L| = 2^{17}3^45$, the tenth of the ten with $|\mathrm{Aut}\ L| > 10^7$ (§6.3), is not among them. Of the twelve rows from [KN93], the lattices S_1 and S_2 have neighbour defect 0 and are two of the five code lattices of [KV89, Satz 10]; the other ten, S_3, $C_1, ..., C_5$ and $G_1, ..., G_4$, are exactly the extremal lattices of neighbour defect 8 [KN93, Thm 1].

Table A1. The 27 previously known lattices of S, by decreasing $|\mathrm{Aut}\ L|$, with the grade of each identification.

lattice	$\lvert\mathrm{Aut}\ L\rvert$	identification (grade)
BW_{32}	$2^{31}3^55^27\cdot 17\cdot 31$	[NS] BW32, dim32f1 (1); the code lattice of RM$(2,5)$, one of the five codes of [KV89, Satz 10], by definition (2); [Neb98q, §21] (via dim32f1); [Hen13, Table 4, no. 70] (3); [Jür15, Satz 3.6.9, L_4] (3); [BN97, Thm 5.1, Λ_{32}''] (4); [BCDYY25, §3.3] (4)
code lattice of one of the five doubly even self-dual codes of length 32 and minimum weight 8 [KV89, Satz 10]	$2^{31}3^55^27$	[NS] KV32F (1); [BCDYY25, §3.3] (4)
[KN93] S_1	$2^{28}3^27$	[NS] KV32U (1); [KN93, Table 1] (2); [BCDYY25, §3.3] (4)
[KN93] S_2	$2^{25}3^25$	[NS] KV32G (1); [KN93, Table 1] (2); [BCDYY25, §3.3] (4)
Λ_{32}'	$2^{13}3^55^4$	[NS] BN32q, dim32f3 (1); [Neb98q, §21] (via dim32f3); [Hen13, no. 71] (3); [BN97, Thm 5.1] (3); [BCDYY25, §3.3] (4)
code lattice of the extended quadratic residue code of length 32	$2^{21}3\cdot 5\cdot 31$	[NS] KV32QR (1); (2) by definition; [BCDYY25, §3.3] (4)
Λ_{32}	$2^{17}3^47$	[NS] BN32p (1); [Hen13, no. 72] (3); [BN97, Thm 5.1] (3)
[KN93] C_1	$2^{21}3\cdot 7$	[NS] KV32C1 (1); [KN93, Table 1] (2)†
[KN93] S_3	$2^{16}3^45$	[NS] KV32S3 (1); [KN93, Table 1] (2); [Hen13, no. 73] (3)
[Hen13] no. 74	$2^{13}3^37$	(3)
[KN93] G_4	$2^{16}3^2$	[NS] KV32G4 (1); (2)
[KN93] C_2	$2^{17}3$	[NS] KV32C2 (1); (2)
[Neb98q] $[4.L_3(4).2^2]_{32,2}$	$2^{10}3^25\cdot 7$	[NS] dim32f5 (1); [Neb98q, §21, Thm 21.1] (via dim32f5)
[KN93] G_3	$2^{15}3^2$	[NS] KV32G3 (1); (2)
[KN93] C_3	2^{18}	[NS] KV32C3 (1); (2)
[KN93] G_1	$2^{14}3\cdot 5$	[NS] KV32G1 (1); (2)

lattice	$\lvert\mathrm{Aut}\ L\rvert$	identification (grade)
[KN93] C_4	$2^{16}3$	[NS] KV32C4 (1); (2)
[Jür15] L_3 = [Hen13] no. 75	$2^7 3^3 17$	[NS] L_32.2, dim32f6 (1); [BB92, §2, $m = 51$] (1); [Neb98q, §21] (via dim32f6); (3) for both
[Hen13] no. 76	$2^8 3^4$	(3)
[Jür15] L_2	$2^7 3^2 17$	(3)
[KN93] C_5	2^{14}	[NS] KV32C5 (1); (2)†
[Hen13] no. 77	$2^8 3^2 7$	(3)
[KN93] G_2	$2^{12}3$	[NS] KV32G2 (1); (2); [BCDYY25, §3.3] (4)
[Hen13] no. 78	$2^8 3^2 5$	(3)
[Hen13] no. 79	$2^{10}3^2$	(3)
[Hen13] no. 80	$2^6 3^4$	(3)
[Jür15] L_1	$2^5 3^2 17$	(3)

Notes. [Hen13, Thm 1, Table 4, Rem. 3(a)] shows that there are exactly eleven extremal lattices with an automorphism of order 3 without fixed points, and gives the orders $|\mathrm{U}|$ of their unitary groups; our list of type $3-(16,0)-0$ has eleven lattices, its completeness is proved by the rootless Hermitian mass (§4.2), and $|C(h)| = |\mathrm{U}|$ holds for all eleven with eleven distinct values. Our representatives of this type were constructed without the Gram matrices of [Hen13], by neighbour methods and from codes; [Hen13, Table 4] was used only to compare the orders $|\mathrm{U}|$. [Hen13, Table 4] also names rows 70, 71, 72 as $\mathrm{BW}_{32} = \Lambda_{32}''$, Λ_{32}', Λ_{32}, which agrees with the identification through [BN97].

[Jür15, Satz 3.6.9] classifies a genus of Hermitian $\mathbb{Z}[\zeta_{17}]$-lattices of rank 2, shows that six Hermitian classes have extremal trace lattices, representing four integer-lattice classes, and gives the orders $|O(L_i)|$, which are distinct. Our four lattices of type $17-(2,0)-0$ were obtained by Kneser neighbours inside that genus, starting from its lattice with minimum 2 and preserving the $\mathbb{Z}[\zeta_{17}]$-structure at each step [C32]; they are therefore lattices of the genus, our list is complete because the six Hermitian classes they carry have $\sum 1/\ |\mathrm{U}(Z)| = 1/12$, the rootless mass of the genus (§4.2), and they are identified with $L_1, ..., L_4$ by $|\mathrm{Aut}\ L| = |O(L_i)|$.

For [KN93], the twelve codes were reconstructed from Table 1 there and matched by canonical forms with the codes from which our lattices are built; the lattice is determined by the code [KN93, Thm 2(b)]. †For C_1 and C_5 the printed word c does not give a code with the stated weight enumerator; after changing two coordinates it does, and the matching uses the corrected word. The column $|\mathrm{Aut}\ C|$ of [KN93, Table 1] differs from our recomputation in six rows.

[BN97, Thm 5.1] classifies the eleven classes of rank-8 lattices over the Hurwitz order in the relevant genus, complete by the mass formula [BN97, Rem. 5.5]; our representatives of Λ_{32} and Λ_{32}' were constructed by neighbours in that genus, so they carry a structure over the Hurwitz order by construction, and the orders of their unitary groups are those in [BN97, Table 4]. This gives grade (3). For BW_{32} we use only that it is one of the three extremal lattices with such a structure [BN97, Rem. 5.2].

[BCDYY25, §3.3] gives seven orders $|\mathrm{Aut}\ L|$ of the 19 extremal lattices of rank 32 in the Magma database: those of the five code lattices, without saying which order belongs to which code; $2^{13}3^5 5^4$, the only other order above $2^{21}3 \cdot 5 \cdot 31$; and the least order, $2^{12} \cdot 3$. In Table A1 they are the orders of the five code lattices, of Λ_{32}' and of G_2 of [KN93]; this is why the comparison with [BCDYY25] is only at grade (4).

An entry "[NS] *name* (1)" means that the Gram matrix on the page *name* of the catalogue [NS] was mapped onto our lattice by an isometry found with bliss and checked in integers [C33]; the 25 extremal pages of rank 32 of the catalogue give 19 distinct lattices, all in Table A1, which names 22 of the pages; the other three are further pages of BW_{32}, and eight lattices of the table have no page there. The catalogue names the pages dim32f1, dim32f3, dim32f5, dim32f6 by the groups $\left[2_+^{1+10}.O_{10}^+(2)\right]_{32}$, $\left[((\mathrm{SL}_2(5) \circ \mathrm{SL}_2(5)) \boxtimes (\mathrm{SL}_2(5) \circ \mathrm{SL}_2(5))) : S_4\right]_{32,2}$, $\left[4.L_3(4).2^2\right]_{32,2}$,

$[\mathrm{SL}_2(17)\square S_3]_{32,1}$ of [Neb98q, §21]; there these are listed among the rational irreducible maximal finite subgroups of $\mathrm{GL}_{32}(\mathbb{Q})$ with an invariant lattice L of determinant 1 and minimum 4, and $G = \mathrm{Aut}\ L$ [Neb98q, Thm 21.1 and its proof]. Of these, $[4.L_3(4).2^2]_{32,2}$ appears in no other source we know.

[BB92, §2] gives, for $m = 51$, the Gram matrix of a unimodular trace form lattice over $\mathbb{Z}[\zeta_{51}]$ by its first row; it is isometric to our lattice L_3 of [Jür15] [C33].

Appendix B. Component lists

What this appendix does. §B.1 lists the quadruples excluded in Proposition 3.3. §B.2 proves that the components of the involution types of ranks 8 to 20 are reached by chains of index-2 overlattices, which makes their enumeration complete. §B.3 reduces the type $(8, 24)$ to three Niemeier lattices and proves the list of 14,482 lattices complete. §B.4 treats the components of the types of odd prime order.

B.1 Excluded types of odd prime order

Table B1. The 74 quadruples $p - (z, f) - s$ excluded in the proof of Proposition 3.3, with the reason. The density exclusions are grouped by p [C2].

type	reason
$3-(2,28)-0$	no genus
$3-(6,20)-0$	no genus
$3-(7,18)-5$	no rootless U
$3-(7,18)-7$	no rootless U
$3-(8,16)-6$	linear programming
$3-(9,14)-5$	linear programming
$3-(10,12)-0$	no genus
$3-(11,10)-7$	linear programming
$3-(11,10)-9$	linear programming
$3-(12,8)-6$	linear programming
$3-(14,4)-0$	no genus
$3-(14,4)-4$	no genus
$7-(2,20)-0$	no genus
$7-(3,14)-3$	linear programming
$11-(2,12)-0$	no genus
$p=3$: $(1,30)-1$, $(2,28)-2$, $(3,26)-1$, $(3,26)-3$, $(4,24)-0$, $(4,24)-2$, $(4,24)-4$, $(5,22)-1$, $(5,22)-3$, $(5,22)-5$, $(6,20)-2$, $(6,20)-4$, $(7,18)-1$, $(7,18)-3$, $(8,16)-0$, $(8,16)-2$, $(8,16)-4$, $(9,14)-1$, $(9,14)-3$, $(10,12)-2$, $(10,12)-4$, $(11,10)-1$, $(11,10)-3$, $(11,10)-5$, $(12,8)-0$, $(12,8)-2$, $(12,8)-4$, $(13,6)-1$, $(13,6)-3$, $(14,4)-2$, $(15,2)-1$	density
$p=5$: $(1,28)-1$, $(2,24)-0$, $(2,24)-2$, $(3,20)-1$, $(3,20)-3$, $(4,16)-0$, $(4,16)-2$, $(5,12)-1$, $(5,12)-3$, $(6,8)-0$, $(6,8)-2$, $(7,4)-1$	density
$p=7$: $(1,26)-1$, $(2,20)-2$, $(3,14)-1$, $(4,8)-0$, $(4,8)-2$, $(5,2)-1$	density
$p=11$: $(1,22)-1$, $(2,12)-2$, $(3,2)-1$	density
$p=13$: $(1,20)-1$, $(2,8)-0$, $(2,8)-2$	density
$p=17$: $(1,16)-1$	density
$p=19$: $(1,14)-1$	density

type	reason
$p = 23$: $(1, 10) - 1$	density
$p = 29$: $(1, 4) - 1$	density

B.1.1 Exact exclusion of the six remaining types

The six types excluded by linear programming in Table B1 can be checked by finite rational certificates. Here we specify the lattice-to-linear-system step and the bases used in that calculation [C3].

One coset. Let X have minimum at least 4. Distinct vectors $v_1, ..., v_r$ of norm α, $0 < \alpha < 2$, in the same coset of $X^\#/X$ satisfy $(v_i, v_j) \le \alpha - 2$. Hence

$$0 \le \left\| \sum_i v_i \right\|^2 \le r\alpha + r(r-1)(\alpha - 2) = r(2 - r(2 - \alpha)).$$

There are therefore at most $m(\alpha) = \lfloor 2/(2-\alpha) \rfloor = 1 + \lfloor \alpha/(2-\alpha) \rfloor$ such vectors. For $\alpha < 1$ the bound is 1. Vectors are counted with both signs.

Gluing inequality. Write $N(j)$ for the number of discriminant classes of F of value $2j/p$ modulo $2\mathbb{Z}$, and $c_F(j), c_Z(j)$ for the coefficients of q^j in the theta series of $\sqrt{p}F^\#$ and $\sqrt{p}Z^\#$. For $1 \le j < p$,

$$c_F(j)/m(2j/p) + c_Z(p-j)/m(2 - 2j/p) \le N(j). \tag{B.LP1}$$

Indeed the classes containing the first kind of vector and the inverse images of the classes containing the second kind are disjoint: otherwise their gluing would give a root. The coset bound applied to each set proves the inequality.

The spaces and Fricke action. For an even p-elementary lattice X of rank $n = 2k$ and determinant p^s, put $\theta_X(\tau) = \sum_x e^{\pi i \tau (x,x)}$. The lattice $\sqrt{p}X^\#$ is even: $p(x,x)$ is integral and $p^2(x,x)$ is even. The theta series lies in $M_k(\Gamma_0(p), \chi)$ with $\chi(a) = \left((-1)^k p^s / a\right)$ [BV01, Thm 2.1]. Our Fricke convention is

$$\left(f|W_p\right)(\tau) = p^{k/2}(p\tau)^{-k} f(-1/(p\tau)).$$

Poisson summation gives $\theta_X|W_p = i^{-k} p^{(k-s)/2} \theta_{\sqrt{p}X^\#}$. The matrix in [BV01, Prop. 3.1] is the negative of this convention, accounting for its phase i^k. Thus a basis element coming from determinant p^{s_i} has normalized dual component $p^{(s-s_i)/2}\theta_{\sqrt{p}X_i^\#}$. The exponent is integral since $s_i \equiv s$ modulo 2.

The following bases have the stated dimension. Their 64-coefficient matrices have full rank over $\mathbb{Q}$. Since every listed theta belongs to the specified space and the rank equals its dimension, they span that space; no converse inference from a truncated identity is used. Dimensions are computed with SageMath; the theta coefficients use PARI enumeration and the specified code. These are explicit computation assumptions.

p	rank	s	dimension	basis of component theta series; determinant exponent
3	8	6	2	$A_2 \perp \sqrt{3}E_6^\#$ $(s_i = 6)$; $A_2 \perp E_6$ $(s_i = 2)$
3	10	7	2	$(A_2)^2 \perp \sqrt{3}E_6^\#$ $(s_i = 7)$; $(A_2)^2 \perp E_6$ $(s_i = 3)$
3	10	9	2	$(A_2)^2 \perp \sqrt{3}E_6^\#$ $(s_i = 7)$; $(A_2)^2 \perp E_6$ $(s_i = 3)$
3	14	5	3	$A_2 \perp \left(\sqrt{3}E_6^\#\right)^2$ $(s_i = 11)$; $A_2 \perp E_6 \perp \sqrt{3}E_6^\#$ $(s_i = 7)$; $A_2 \perp (E_6)^2$ $(s_i = 3)$
3	16	6	3	$(A_2)^2 \perp \left(\sqrt{3}E_6^\#\right)^2$ $(s_i = 12)$; $(A_2)^2 \perp E_6 \perp \sqrt{3}E_6^\#$ $(s_i = 8)$; $(A_2)^2 \perp (E_6)^2$ $(s_i = 4)$
3	18	5	4	$\left(\sqrt{3}E_6^\#\right)^3$ $(s_i = 15)$; $E_6 \perp \left(\sqrt{3}E_6^\#\right)^2$ $(s_i = 11)$; $(E_6)^2 \perp \sqrt{3}E_6^\#$ $(s_i = 7)$; $(E_6)^3$ $(s_i = 3)$

p	rank	s	dimension	basis of component theta series; determinant exponent
3	22	7	4	$(A_2)^2 \perp \left(\sqrt{3}E_6^{\#}\right)^3$ $(s_i = 17)$; $(A_2)^2 \perp E_6 \perp \left(\sqrt{3}E_6^{\#}\right)^2$ $(s_i = 13)$; $(A_2)^2 \perp (E_6)^2 \perp \sqrt{3}E_6^{\#}$ $(s_i = 9)$; $(A_2)^2 \perp (E_6)^3$ $(s_i = 5)$
3	22	9	4	$(A_2)^2 \perp \left(\sqrt{3}E_6^{\#}\right)^3$ $(s_i = 17)$; $(A_2)^2 \perp E_6 \perp \left(\sqrt{3}E_6^{\#}\right)^2$ $(s_i = 13)$; $(A_2)^2 \perp (E_6)^2 \perp \sqrt{3}E_6^{\#}$ $(s_i = 9)$; $(A_2)^2 \perp (E_6)^3$ $(s_i = 5)$
3	24	6	5	$\left(\sqrt{3}E_6^{\#}\right)^4$ $(s_i = 20)$; $E_6 \perp \left(\sqrt{3}E_6^{\#}\right)^3$ $(s_i = 16)$; $(E_6)^2 \perp \left(\sqrt{3}E_6^{\#}\right)^2$ $(s_i = 12)$; $(E_6)^3 \perp \sqrt{3}E_6^{\#}$ $(s_i = 8)$; $(E_6)^4$ $(s_i = 4)$
7	14	3	5	$(Q_7)^7$ $(s_i = 7)$; $\sqrt{7}A_6^{\#} \perp (Q_7)^4$ $(s_i = 9)$; $\left(\sqrt{7}A_6^{\#}\right)^2 \perp Q_7$ $(s_i = 11)$; $A_6 \perp (Q_7)^4$ $(s_i = 5)$; $A_6 \perp \sqrt{7}A_6^{\#} \perp Q_7$ $(s_i = 7)$
7	18	3	7	$(Q_7)^9$ $(s_i = 9)$; $\sqrt{7}A_6^{\#} \perp (Q_7)^6$ $(s_i = 11)$; $\left(\sqrt{7}A_6^{\#}\right)^2 \perp (Q_7)^3$ $(s_i = 13)$; $\left(\sqrt{7}A_6^{\#}\right)^3$ $(s_i = 15)$; $A_6 \perp (Q_7)^6$ $(s_i = 7)$; $A_6 \perp \sqrt{7}A_6^{\#} \perp (Q_7)^3$ $(s_i = 9)$; $A_6 \perp \left(\sqrt{7}A_6^{\#}\right)^2$ $(s_i = 11)$

Here $\sqrt{3}E_6^{\#}$ has Gram $3G(E_6)^{-1}$, $\sqrt{7}A_6^{\#}$ has Gram $7G(A_6)^{-1}$, and Q_7 has Gram $\left(\begin{smallmatrix} 2 & 1 \\ 1 & 4 \end{smallmatrix}\right)$. Powers denote orthogonal sums of identical factors. Positive definiteness, evenness, determinants, evenness of the p-scaled duals, and equality of the characters are checked in integers and rationals. The machine names of the two scaled duals and of Q_7 are `rt3E6d`, `rt7A6d`, and `Q7`, respectively.

The discriminant sign. At the odd prime p, write $F_p = U_0 \perp pU_1$, rank $U_1 = s$, and $\varepsilon = (\det U_1/p)$. The other odd places have excess zero and the even unimodular 2-adic part has oddity zero. The oddity formula [SPLAG, Ch. 15, §7.7, (30), (35)] gives

$$d + s(p-1) + 4[\varepsilon = -1] \equiv 0 \pmod 8, \quad \varepsilon = (-1)^{(d+s(p-1))/4}.$$

On $F^{\#}/F$ define $Q(a) = j$ when $q(a) = 2j/p$ modulo $2\mathbb{Z}$. Its matrix is $2^{-1}U_1^{-1}$ over $\mathbb{F}_p$, so its determinant character is $(2/p)^s\varepsilon$. For $p = 3, 7$ this equals $(-1)^{(d-2s)/4}$. Counting values in a diagonal form with this determinant gives $N(j)$; only the zero vector is removed from $N(0)$.

Finite system and certificates. The variables are the rational coefficients in the displayed bases for F and Z. Their theta and normalized dual series have constant term 1; the original series have zero q^1 coefficient and non-negative coefficients. The dual series are non-negative, and Lemma 4.2 makes the dual Z coefficients at $q^1, ..., q^{p-2}$ zero. Add (B.LP1). The implementation initially imposes these conditions through degree 39. Each certificate below uses only the indicated equations and inequalities, of degree at most 6.

Each equation is represented by its two inequalities; "reverse" denotes the negative of that equation. `F:0:j` and `F:1:j` mean the equations for the coefficients of q^j of the original and of the normalized dual series of F, respectively, and `F:dual:j>=0` the non-negativity of the coefficient of q^j of the normalized dual series; similarly for Z. `gluing:j` means (B.LP1). Multiplying the indicated rows by the non-negative rational multipliers gives zero for every variable and right side -1.

type $p-z-f-s$	multiplier	row
3-8-16-6	1/2	F:0:0=1
3-8-16-6	1/12	F:0:1=0
3-8-16-6	3/2	F:1:0=1 (reverse)
3-8-16-6	1/4	F:dual:1>=0
3-9-14-5	1/24	F:0:1=0
3-9-14-5	14/9	F:1:0=1 (reverse)

type $p-z-f-s$	multiplier	row
3-9-14-5	1/8	F:dual:1>=0
3-9-14-5	1/162	Z:dual:1>=0
3-9-14-5	1/162	gluing:2
3-11-10-7	1/4	F:0:0=1 (reverse)
3-11-10-7	1/12	F:0:1=0
3-11-10-7	3/4	F:1:0=1 (reverse)
3-11-10-9	1/2	F:0:0=1 (reverse)
3-11-10-9	1/6	F:0:1=0
3-11-10-9	1/2	F:1:0=1 (reverse)
3-12-8-6	1/6	F:0:1=0
3-12-8-6	40/27	F:1:0=1 (reverse)
3-12-8-6	1/486	Z:dual:1>=0
3-12-8-6	1/486	gluing:2
7-3-14-3	541/464	F:0:0=1 (reverse)
7-3-14-3	1/58	F:0:1=0
7-3-14-3	315/464	F:1:0=1 (reverse)
7-3-14-3	13/58	F:dual:1>=0
7-3-14-3	2/29	F:dual:2>=0
7-3-14-3	7/464	Z:dual:1>=0
7-3-14-3	7/464	gluing:6

Thus a necessary system for each of the six types implies $0 \le -1$, excluding that type. The rational verification needs no linear-programming solver. The full matrices, coefficients, basis Grams and code hashes are fixed with the certificates [C3]. This finite contradiction does not independently revalidate the software computations of the theta coefficients or space dimensions.

B.2 Components of involutions

Let $\mathcal{M}(d)$ be the set of isometry classes of even 2-elementary lattices M of rank d with $\min M \ge 4$.

Lemma B.1. Let $M \in \mathcal{M}(d)$ with $M^\#/M \cong \mathbb{F}_2^s$. There are a unimodular lattice U of rank d without vectors of norm 1 and a chain $\sqrt{2}U = Y_0 \subset Y_1 \subset \ldots \subset Y_m = M$, $m = (d-s)/2$, with $\left[Y_{j+1} : Y_j\right] = 2$, such that every Y_j lies in $\mathcal{M}(d)$ and $Y_{j+1} = Y_j + \mathbb{Z}v_j$ for some $v_j \in Y_j^\#$ with $\left(v_j, v_j\right) \in 2\mathbb{Z}$.

Proof. The proof of Proposition 3.2 gives $T = \sqrt{2}U \subset M$ with U unimodular and $\min U \ge 2$, and M/T is elementary abelian of order 2^m, since $2M \subset 2M^\# \subset T$. Take any chain of subgroups of M/T with successive quotients of order 2, and let Y_j be their preimages. Each Y_j is a sublattice of M, so it is even with minimum at least 4. Since $Y_j \supset T$, $Y_j^\# \subset T^\# = U/\sqrt{2}$ and $2Y_j^\# \subset \sqrt{2}U = T \subset Y_j$, so Y_j is 2-elementary. Finally $Y_{j+1} \subset M \subset M^\# \subset Y_j^\#$, so $Y_{j+1} = Y_j + \mathbb{Z}v_j$ with $v_j \in Y_j^\#$, and $\left(v_j, v_j\right) \in 2\mathbb{Z}$ because M is even. □

So $\mathcal{M}(d)$, or its part with $s \ge s_0$, is obtained as follows: start from the lattices $\sqrt{2}U$, where U runs over the unimodular lattices of rank d without vectors of norm 1 [SPLAG, Table 16.7], and at each step replace every Y by the lattices $Y + \mathbb{Z}v$, where $v + Y$ runs over representatives of the $\mathrm{Aut}\, Y$-orbits of elements of order 2 of $Y^\#/Y$ with $(v, v) \in 2\mathbb{Z}$, keeping those with minimum at least 4 and merging isometric ones. Since s drops by 2 at each step, the part with $s \ge s_0$ is reached by chains staying in

$s \geq s_0$. The generators of $\mathrm{Aut}\, Y$ used for the orbits are checked to be automorphisms in integers, and merging uses explicit isometries checked in integers; two lattices are kept apart only if `qfisom` finds no isometry between them; in rank 20 this answer was also confirmed by different isometry invariants (§7.2). Hence no class is lost, and the list has no repetition [C5, C7].

Table B2. The components of involution types. "Classes" is the number of classes in $\mathcal{M}(d)$ with the given (s, δ) [C5, C6, C7].

rank d	(s, δ)	classes	$\lvert\mathrm{Aut}\, M\rvert$	used in type
8	$(8, 0)$	1	696,729,600 ($\sqrt{2}E_8$)	$(8, 24)$
12	$(12, 1)$	1	980,995,276,800 ($\sqrt{2}D_{12}^+$)	$(12, 20)$
14	$(14, 1)$	1	16,855,282,483,200 ($\sqrt{2}(E_7^2)^+$)	$(14, 18)$
15	$(15, 1)$	1	41,845,579,776,000 ($\sqrt{2}A_{15}^+$)	$(15, 17)$
16	all	10	Table 6	$(16, 16)$
17	$(15, 1)$	1	8,957,952,000	$(15, 17)$
18	$(14, 1)$	5	7,077,888, 11,197,440, 477,757,440, 566,231,040, 2,717,908,992	$(14, 18)$
20	$(12, 1)$	66	from 64 to 79,272,345,600	$(12, 20)$
24	$(8, 0)$, $\min M^\# \geq 2$	14,482	from 2 to 253,671,505,920	$(8, 24)$

For ranks 8, 12, 14, 15 and 16 the enumeration was run for all s; it shows that $\sqrt{2}E_8$, $\sqrt{2}D_{12}^+$, $\sqrt{2}(E_7^2)^+$ and $\sqrt{2}A_{15}^+$ are the only lattices in $\mathcal{M}(d)$ for $d = 8, 12, 14, 15$, and gives the ten classes of rank 16. Since F and K have the same (s, δ), this determines which (s, δ) are needed in ranks 17, 18, 20 and 24; in ranks 17, 18 and 20 the enumeration was stopped at $s_0 = 15, 14, 12$ respectively. In rank 20 it gives 297 classes with $s \geq 12$, of which 66 have $s = 12$, $\delta = 1$. Rank 24 is treated in §B.3.

Two checks, not used in the proofs, compare these lists with computations that do not use chains. In rank 16, a classification of all seventeen genera of even 2-elementary lattices by Kneser 3-neighbours, completed when $\sum 1/\lvert\mathrm{Aut}\rvert$ reached the mass of the genus, contains exactly the ten classes of minimum at least 4 [C7]. For the 57 genera of rank at most 18 and for $(20, 12, 1)$, $\sum 1/\lvert\mathrm{Aut}\, M\rvert$ over the list equals, as a fraction, the rootless mass of the genus computed by King's inclusion–exclusion over root systems [Kin03] with the local densities at 2 for 2-elementary lattices [C5]. For $(20, 12, 1)$ this second computation found the same 66 classes, matched one-to-one with the chain enumeration by explicit isometries.

B.3 The type $(8, 24)$

Let σ be an involution of type $(8, 24)$ of an even unimodular lattice L of rank 32, with $F = L^\sigma \cong \sqrt{2}E_8$ (§B.2) and $K = L^{-\sigma}$. Then K is even, of rank 24, 2-elementary with $s = 8$ and $\delta = 0$, so $(y, y) \in \mathbb{Z}$ for $y \in K^\#$.

Proposition B.2. *L is extremal if and only if $\min K \geq 4$ and $\min K^\# \geq 2$.*

Proof. If L is extremal, $\min K \geq 4$. By the criterion of §4.1.3, L has a root if and only if some class $a \in F^\#/F$ and its image $\varphi(a)$ both contain a vector of norm 1. Since $F^\# = E_8/\sqrt{2}$, a class of $F^\#/F = E_8/2E_8$ contains a vector of norm 1 if and only if it is the class of a root of E_8, that is, if and only if $q_F(a) \equiv 1$. A vector $y \in K^\#$ of norm 1 lies in a class b with $q_K(b) \equiv 1$, so $\varphi^{-1}(b)$ has $q_F \equiv -1 \equiv 1$ and contains a vector of norm 1. Hence L has a root if and only if $\min K < 4$ or $K^\#$ has a vector of norm 1; and $K^\#$ has no non-zero vector of norm less than 1 (§4.1.3). □

Let $\mathcal{K}$ be the set of isometry classes of such K. Let $T' \subset K^\#/K \cong \mathbb{F}_2^8$ be a maximal totally singular subspace for q_K; it has dimension 4, and $P = K + T'$ is an even unimodular lattice of rank 24, a Niemeier lattice. There are 270 such T', and each of the 135 non-zero singular elements lies in $270 \cdot 15/135 = 30$ of them, since $O(q_K)$ is transitive on these elements.

Theorem B.3. For every $K \in \mathcal{K}$, $K^\#$ has exactly 720 vectors of norm 2, the 270 Niemeier lattices $K + T'$ have on average exactly 80 roots, and hence at least one of them is the Leech lattice, A_1^{24} or A_2^{12}.

Strategy of the proof. The lattices K of rank 24 cannot be listed by a direct search, so we look at the 270 Niemeier lattices $K + T'$ that contain them. The vectors of norm 4 of L form a spherical 7-design, and this fixes the number of vectors of norm 2 of $K^\#$. That number fixes the average number of roots of the Niemeier lattices $K + T'$. A Niemeier lattice with at most this many roots is the Leech lattice, A_1^{24} or A_2^{12}, so every K lies in one of these three lattices, where the search is finite.

Proof. By Proposition B.2, L is extremal. Let X be the set of its vectors of norm 4; there are 146,880 of them, the coefficient of q^2 in the theta series of an extremal lattice of rank 32, which is the unique modular form of weight 16 for $\mathrm{SL}_2(\mathbb{Z})$ of the form $1 + O(q^2)$. By [BV01, Cor. 4.1], X is a spherical 7-design. For $x \in X$ write $x = x_+ + x_-$ with $x_+ \in F^\#$, $x_- \in K^\#$. For a 7-design on the sphere of radius 2 in $\mathbb{R}^{32}$ and the 8-dimensional space $F \otimes \mathbb{R}$, the averages of the polynomials $|x_+|^{2j}$, $j \le 3$, are those of the uniform measure:

$$\sum_{x \in X} (x_+, x_+)^j = |X| \cdot 4^j \prod_{i=0}^{j-1} \frac{8 + 2i}{32 + 2i}.$$

Since $(x_+, x_+) \in \{0, 1, 2, 3, 4\}$ and the 240 vectors with $(x_+, x_+) = 4$ are those of F, the four equations $j = 0, ..., 3$ determine the numbers N_a of x with $(x_+, x_+) = a$: $N_0 = 12{,}240$, $N_1 = 122{,}880$, $N_2 = 11{,}520$, $N_3 = 0$. A vector $y \in K^\#$ of norm 2 is not in K and lies in a class b with $q_K(b) = 0$, so $\varphi^{-1}(b)$ is a class of $E_8/2E_8$ of vectors of norm 4, which contains exactly 16 vectors of norm 2 of $F^\#$ (in E_8, two vectors of norm 4 that are congruent modulo $2E_8$ are equal up to sign or orthogonal, since their difference lies in $2E_8$ and so has norm 0 or at least 8; hence a class contains at most 16 of them. The same comparison of norms shows that two congruent roots are equal up to sign and that no vector of norm 4 is congruent to 0 or to a root. The zero class and the 120 classes $\pm r$ of roots leave 135 of the 256 classes, and $2{,}160 = 135 \cdot 16$ forces exactly 16 vectors of norm 4 in each); each gives a vector $x_+ + y \in X$. So $N_2 = 16n$, where n is the number of vectors of norm 2 in $K^\#$, and $n = 720$.

The roots of $P = K + T'$ are the vectors of norm 2 of $K^\#$ lying in the classes of T', since K has none. Each of the 720 vectors lies in a non-zero singular class and is counted in 30 of the T', so the average number of roots is $720 \cdot 30/270 = 80$. The number of roots of a Niemeier lattice is $24h$, where h is the Coxeter number of its root system, and the only values at most 80 are 0, 48 and 72, attained only by the Leech lattice, A_1^{24} and A_2^{12} [SPLAG, Table 16.1]. □

Conversely, for a Niemeier lattice P and a 4-dimensional subspace $W \subset P/2P$, totally singular for $x \mapsto (x, x)/2 \bmod 2$, with preimage $\widetilde{W} \subset P$, put $K_W = \left\{x \in P : \left(x, \widetilde{W}\right) \subset 2\mathbb{Z}\right\}$. Then $K_W^\# = P + \widetilde{W}/2$, so K_W is even, 2-elementary with $s = 8$ and $\delta = 0$, and every $K \in \mathcal{K}$ is K_W for $P = K + T'$ and $W = 2K^\#/2P$. Moreover $\min K_W \ge 4$ if and only if no root of P is orthogonal modulo 2 to $\widetilde{W}$, and $\min K_W^\# \ge 2$ if and only if every non-zero class of W has minimum at least 8 in P. We call such W *good*.

Enumeration. For P the Leech lattice and A_1^{24}, all Aut P-orbits of good W were enumerated, subspace by subspace of dimension $1, \ldots, 4$, with an orbit-canonical choice at each step; there are 1,244 and 376,306 orbits, and they give 1,216 and 13,231 classes of K with a T' of 0 or 48 roots [C6]. By Theorem B.3 every other $K \in \mathcal{K}$ has a T' with $K + T' \cong A_2^{12}$. For $P = A_2^{12}$ we use the identity

$$\sum_{[W]} \frac{1}{|\mathrm{Stab}_{\mathrm{Aut}\ P}(W)|} = \sum_{K \in \mathcal{K}} \frac{a(K,P)}{|\mathrm{Aut}\ K|}, \tag{B.1}$$

where the left side runs over the Aut P-orbits of good W and $a(K,P)$ is the number of T' with $K + T' \cong P$. It holds because $(P,W) \mapsto (K_W, P/K_W)$ is a bijection between the isometry classes of the two kinds of pairs which preserves automorphism groups, and both sides are the orbit sums of this set of pairs. Call a 3-dimensional subspace $W_3 \subset P/2P$ good if it is totally singular and every non-zero class of W_3 has minimum at least 8 in P; the condition on roots is not imposed, since it does not pass to subspaces. The left side was computed from the 106,624 orbits of good W_3, enumerated as for the Leech lattice: for each, $g(W_3)$ is the number of non-zero singular $x \in W_3^{\perp}/W_3$ such that every class of $x + W_3$ has minimum at least 8 and every root of P orthogonal modulo 2 to $\widetilde{W}_3$ has odd inner product with x, that is, the number of good $W \supset W_3$. Since every hyperplane of a good W is a good W_3, counting the pairs $W_3 \subset W$ gives $\sum_{[W]} 1/\,|\mathrm{Stab}(W)| = \frac{1}{15} \sum_{[W_3]} g(W_3)/\,|\mathrm{Stab}(W_3)| = 57{,}341{,}069{,}029\ /\ 69{,}120$. The right side over the 14,447 lattices already found, with $a(K, A_2^{12})$ counted over all 270 subspaces T', falls short by 429,583 /768. Every lattice not yet found has $a(K, A_2^{12}) \geq 1$, so it contributes a positive term to the difference. A search adapted to A_2^{12} (a good W maps onto each of the twelve summands $A_2/2A_2$ of $A_2^{12}/2A_2^{12}$, so it is described by twelve lines in W) found 35 further lattices, and their terms fill the difference exactly. Hence no lattice is missing, and $\mathcal{K}$ has 14,482 elements. How the 35 lattices were found plays no role in this argument. Identity (B.1) holds exactly also for the Leech lattice and A_1^{24}, where both sides were computed independently [C6].

Table B3. The 14,482 lattices K of type $(8, 24)$, by the smallest number of roots of a Niemeier lattice $K + T'$ [C6].

P	good W (orbits)	classes K	completeness
Leech	1,244	1,216	all orbits
A_1^{24}	376,306	13,231	all orbits
A_2^{12}	—	35	identity (B.1)
total		14,482	

The classes of K are separated by the canonical form (§7.2) of the graph on the 720 vectors of norm 2 of $K^{\#}$, with inner products as colours. For 14,462 of the lattices these vectors generate $K^{\#}$, so Aut K is the automorphism group of the graph and the canonical form is a complete invariant. For the remaining 20, let $J' \subset J = \sqrt{2}K^{\#}$ be the sublattice spanned by the 720 vectors (of norm 4 in J); the automorphism group A of the graph acts on $J'^{\#}/J'$, Aut K is the stabilizer in A of the subgroup J/J', so $|\mathrm{Aut}\ K| = |A|\ /\ |A \cdot (J/J')|$, and the orbit, written in the canonical coordinates, completes the invariant (16 lattices); for 4 lattices the orbit was too large and Aut K was computed with `qfauto`; each of these four has an order $|\mathrm{Aut}\ K|$ that no other of the 14,482 lattices has, which separates them from each other and from the rest [C6].

B.4 Components of elements of odd prime order

Let h have type $p-(z,f)-s$ with p odd. We need all F of minimum at least 4 in the genus of even p-elementary lattices of rank f and determinant p^s, and all Z of minimum at least 4 in the genus of Hermitian $\mathcal{O}$-lattices of rank z with $Z^\#/Z \cong \mathbb{F}_p^s$ (Proposition D.1). Table B4 gives the result.

The lattices F. Except in two genera, the whole genus was classified by Kneser q-neighbours for a prime $q \neq p$, which stay in the genus; classes were identified by explicit isometries checked in integers and separated by invariants, and the classification is complete when $\sum 1/\,|\mathrm{Aut}\ F|$ over the classes found equals the mass of the genus given by the Smith–Minkowski–Siegel mass formula [SPLAG, Ch. 16] [C2]. The lattices of minimum at least 4 were then selected. For the genera $(20,6)$ ($p=3$) and $(16,4)$ ($p=5$) only rootless lattices were searched for, and the list is complete when its total weight equals the rootless mass of the genus (Appendix D).

The lattices Z. If $s=z$, Z is unimodular for $\tilde{H}$ (Lemma 4.4), and the list of rootless Z is complete when its total weight equals the rootless Hermitian mass of rank z, obtained from the mass of the genus and (4.2) (Appendix D). If $s<z$, let ℓ be a line of the symplectic space Z/Z^* of Lemma 4.4 and $Z_+ \subset Z$ the preimage of its orthogonal $\ell^\perp$. Then $[Z:Z_+]=p$, $Z_+/Z_+^* \cong \ell^\perp/\ell$, so Z_+ lies in the genus $(z,s+2)$, and $Z = Z_+ + \mathcal{O}y$ for an isotropic element $y+Z_+$ of the discriminant group of Z_+; if Z has minimum at least 4, so has Z_+. Each Z contains exactly $n=(p^{z-s}-1)/(p-1)$ such Z_+. Let $g(Z_+)$ be the number of isotropic lines of the discriminant group of Z_+ that give a lattice of minimum at least 4. Counting pairs $Z_+ \subset Z$ in two ways gives

$$\sum_{Z} \frac{n}{|\mathrm{U}(Z)|} = \sum_{Z_+} \frac{g(Z_+)}{|\mathrm{U}(Z_+)|}, \tag{B.2}$$

the sums over the lattices of minimum at least 4 in the two genera. Given the complete list for $(z,s+2)$, the right side is computed from it, and the list for (z,s) is complete when its left side reaches this value. For $3-(13,6)-5$ and for $p=5,7$ the lists were completed instead against the rootless mass of the genus (z,s) (Appendix D). For $p=5$, (B.2) was also checked at every step of the chains that lead to the genera of Table B4, from $(6,6)$ to $(6,4)$ and from $(7,7)$ through $(7,5)$ to $(7,3)$ (for $p=7$ the genus $(4,4)$ is itself the start of a chain, and the list for $5-(8,0)-0$ was found directly, §4.2). For $z=13$ ($p=3$) it was checked only from $s=13$ to $s=11$ [C9, C2]. As a check not used in the proofs, for $p=3$ each list of a genus (z,s) with $s<z$ obtained by (B.2) also has $\sum 1/\,|\mathrm{U}(Z)|$ equal to the rootless Hermitian mass of the genus computed by the method of Appendix D.4 [C16].

Table B4. Components of the 19 types of odd prime order: the genus (f,s) of F and the number of its classes of minimum at least 4 (in brackets, the number of classes of the whole genus when it was classified); the genus (z,s) of Z and the number of its classes of minimum at least 4. "Mass" means the mass of the genus, "rootless" the rootless mass of Appendix D, "(B.2)" the identity above, starting from the rootless unimodular lattices [C8, C9, C2].

type	F	Z	F complete by	Z complete by
$3-(10,12)-6$	$(12,6)$: 1 [10], K_{12}	$(10,6)$: 1	mass	(B.2)
$3-(9,14)-7$	$(14,7)$: 1 [29]	$(9,7)$: 1	mass	(B.2)
$3-(10,12)-8$	$(12,8)$: 1 [6]	$(10,8)$: 1	mass	(B.2)
$3-(8,16)-8$	$(16,8)$: 6 [163]	$(8,8)$: 1	mass	rootless
$3-(12,8)-8$	$(8,8)$: 1, $\sqrt{3}E_8$	$(12,8)$: 51	$s=f$	(B.2)
$3-(9,14)-9$	$(14,9)$: 2 [21]	$(9,9)$: 1	mass	rootless

type	F	Z	F complete by	Z complete by
$3-(10,12)-10$	$(12,10)$: 1 [3]	$(10,10)$: 2	mass	rootless
$3-(13,6)-5$	$(6,5)$: 1 [1], $\sqrt{3}E_6^{\#}$	$(13,5)$: 105	mass	rootless
$3-(6,20)-6$	$(20,6)$: 24	$(6,6)$: 1, K_{12}	rootless	rootless
$3-(16,0)-0$	—	$(16,0)$: 11	—	[Hen13]; rootless
$5-(4,16)-4$	$(16,4)$: 1	$(4,4)$: 1	rootless	rootless
$5-(5,12)-5$	$(12,5)$: 1 [41]	$(5,5)$: 1	mass	rootless
$5-(6,8)-4$	$(8,4)$: 1 [5]	$(6,4)$: 5	mass	rootless; (B.2)
$5-(6,8)-6$	$(8,6)$: 1 [2]	$(6,6)$: 4	mass	rootless
$5-(7,4)-3$	$(4,3)$: 1 [1], $\sqrt{5}A_4^{\#}$	$(7,3)$: 23	mass	rootless; (B.2)
$5-(8,0)-0$	—	$(8,0)$: 9	—	rootless
$7-(4,8)-4$	$(8,4)$: 1 [8]	$(4,4)$: 3	mass	rootless
$17-(2,0)-0$	—	$(2,0)$: 6 Hermitian classes; 4 trace lattices	—	[Jür15]; rootless
$31-(1,2)-1$	$(2,1)$: 1	$(1,1)$: 8	direct	class number

For $s=f$ the lattice F is $\sqrt{p}M$ with M even unimodular, so $F=\sqrt{3}E_8$ for $3-(12,8)-8$.

Appendix C. The rootless mass over $\mathbb{Z}[i]$

What this appendix does. It proves Theorem 4.5. §C.1 sets up the problem and states the strategy (King's inclusion–exclusion over root systems). §C.2 classifies the root lattices over $\mathbb{Z}[i]$ and §C.3 their root sublattices, which gives the coefficients of the inclusion–exclusion. §C.4 proves a Siegel-type formula for the average number of embeddings of a lattice T, and §C.5 computes its local factors, the essential new point being the ramified prime 2. §C.6 explains how the 2-adic sums were computed and §C.7 lists the checks. The appendix can be read without the rest of the paper except §2.4.

C.1 Setting and strategy

Let $\mathcal{O}=\mathbb{Z}[i]$, $\varpi=1+i$, and $\chi=\chi_{-4}$ the character of $\mathbb{Q}(i)$. A *theta lattice* of rank r is a free $\mathcal{O}$-module L of rank r with a positive definite Hermitian form h (linear in the first variable) such that $h(L,L)\subset\mathcal{O}$, $h(x,x)\in 2\mathbb{Z}$ for all x, and $L=\{x\in L\otimes\mathbb{Q} : h(x,L)\subset\mathcal{O}\}$ [HKN12]. With $(x,y)=\operatorname{Re}h(x,y)$, theta lattices of rank r are exactly the even unimodular lattices of rank $2r$ with an automorphism g ($x\mapsto ix$) satisfying $g^2=-1$; this is the normalization $H'=dH$, $d=2$, of the proof of Lemma 2.2. A *root* is a vector with $h(x,x)=2$. The mass $\mu_r=\sum_L 1/\,|\mathrm{U}(L)|$ over isometry classes is known [HKN12, Thm 2]; in the notation of (C.6) below, $\mu_r=S'_r$ for $r\equiv 0\pmod 4$, the case of [HKN12, Thm 2] and the only one we use. We want the rootless mass $\mu_r^{\emptyset}$; then $M_2=\mu_{16}^{\emptyset}$.

In the equivalent Möbius form of [Kin03, Prop. 1], for a lattice T generated by roots let $r(T,L)$ be the number of embeddings $T\to L$ (isometric $\mathcal{O}$-linear maps) and

$$A_r(T)=\sum_L \frac{r(T,L)}{|\mathrm{U}(L)|}. \tag{C.1}$$

§C.4–§C.5 compute $A_r(T)$ from local data without knowing the classes L, and §C.3 expresses $\mu_r^{\emptyset}$ as a finite linear combination of the $A_r(T)$.

C.2 Root lattices over $\mathbb{Z}[i]$

The roots of a theta lattice form a root system in the trace form, preserved by g. Since $g^2 = -1$ preserves every irreducible component, g acts on the set of components as an involution.

Proposition C.1. The irreducible root lattices over $\mathcal{O}$ are the following, where $C[i] = C \otimes_{\mathbb{Z}} \mathcal{O}$ and $G_m = \{z \in \mathcal{O}^m : \sum z_k \in \varpi\mathcal{O}\}$ with the standard form.

name	root system over $\mathbb{Z}$	rank over $\mathcal{O}$	$\lvert\mathrm{U}\rvert$
$A_n[i]$, $n \geq 1$	A_n^2	n	4 for $n = 1$, else $4(n+1)!$
$D_n[i]$, $n \geq 4$	D_n^2	n	2,304 for $n = 4$, else $2^{n+1}n!$
$E_n[i]$, $n = 6, 7, 8$	E_n^2	n	$2 \lvert\mathrm{Aut}\ E_n\rvert$
G_m, $m \geq 2$	D_{2m}	m	96 for $m = 2$, else $4^m m!$
E_8^g	E_8	4	46,080

Proof. If g interchanges two components C and Cg, identifying them by g gives $C[i]$. If g preserves a component C, then $g \in \mathrm{Aut}\ C$ with $g^2 = -1$. For A_n ($n \geq 2$) and E_6, $\mathrm{Aut}\ C = W \times \{\pm 1\}$ with $-1 \notin W$, so $g^2 = w^2 = -1$ is impossible; A_1, E_7 and D_n with n odd have odd rank. For D_{2m}, $m \geq 3$, $\mathrm{Aut}\ C$ is the group of signed permutations, and $g^2 = -1$ forces a fixed-point-free involutive permutation with each transposition carrying sign product -1; these form one conjugacy class, giving G_m. For D_4 the 12 elements with $g^2 = -1$ in $W(F_4)$ form one class ($|W(F_4)|\ /96 = 12$), giving G_2. For E_8, the elements of $W(E_8)$ with characteristic polynomial Φ_4^4 form one conjugacy class, of size 15,120 and with centralizer of order 46,080 [C11, C20]; this gives E_8^g. The orders $|\mathrm{U}|$ in the table are centralizers of g. If g interchanges C and Cg, the centralizer of g in $\mathrm{Aut}(C \perp Cg) = \mathrm{Aut}\ C \wr C_2$ consists of the elements acting by the same $a \in \mathrm{Aut}\ C$ on both components, together with their products with g; hence $|\mathrm{U}(C[i])| = 2\ |\mathrm{Aut}\ C|$, which gives the first three rows ($|\mathrm{Aut}\ A_1| = 2$, $|\mathrm{Aut}\ A_n| = 2(n+1)!$ for $n \geq 2$, $|\mathrm{Aut}\ D_4| = 1{,}152$, $|\mathrm{Aut}\ D_n| = 2^n n!$ for $n \geq 5$). For G_m, $m \geq 3$, the centralizer of g in the group of signed permutations is $(\mathbb{Z}/4)^m \rtimes S_m$, of order $4^m m!$; the values for G_2 and E_8^g are the centralizer orders above. □

Different ways of letting g act on the same root system over $\mathbb{Z}$ are different types over $\mathcal{O}$; for instance E_8^3 gives the types $\left(E_8^g\right)^3$ and $E_8[i] \perp E_8^g$. The numbers of multisets of irreducible root lattices over $\mathcal{O}$ of total rank exactly $0, 4, 8, 12, 16$ are $1, 12, 113, 802, 4{,}663$; over all total ranks at most 16 there are 12,934.

C.3 Inclusion–exclusion

Proposition C.2. For an irreducible root lattice Y define $c(Y) = -\sum_{Q \subsetneq Y} c(Q)$, the sum over the sublattices Q of Y generated by roots (as $\mathcal{O}$-modules), with $c(0) = 1$, and extend c multiplicatively to orthogonal sums. Then

$$\mu_r^{\emptyset} = \sum_T c(T) \frac{A_r(T)}{|\mathrm{U}(T)|}, \tag{C.2}$$

the sum over isomorphism classes of root lattices T over $\mathcal{O}$ of rank at most r.

Proof. Let R be the root lattice of L. Every sublattice $Q \subset L$ generated by roots lies in R, and is the orthogonal sum of its intersections with the irreducible components Y_k of R, since every root lies in one component: each nonzero component in an orthogonal sum of even positive definite lattices has norm at least 2. Hence $\sum_{Q \subset R} c(Q) = \prod_k \sum_{Q_k \subset Y_k} c(Q_k)$, which is 1 if $R = 0$ and 0 otherwise

by the definition of c. The number of sublattices of L isomorphic to T and generated by roots is $r(T,L)/\,|\mathrm{U}(T)|$. Weight by $1/\,|\mathrm{U}(L)|$ and sum over L. □

To compute c we need all root sublattices of each irreducible Y. For $0 \le \kappa \le 2$ let $\mathcal{O}_\kappa = \mathbb{Z}[\zeta_{2^\kappa}]$ ($\mathcal{O}_0 = \mathcal{O}_1 = \mathbb{Z}$), μ_{2^i} the group of 2^i-th roots of unity, and $\varpi_i = 1 - \zeta_{2^i}$ for $i \ge 2$, $\varpi_1 = 2$, $\varpi_0 = 0$. For $\kappa \ge 1$ put $G_k^{(\kappa)} = \{z \in \mathcal{O}_\kappa^k : \sum z_p \in \varpi_\kappa \mathcal{O}_\kappa\}$; with the standard form, $G_k^{(\kappa)}$ is $D_{2^{\kappa-1}k}$, $G_k^{(1)} = D_k$ and $G_k^{(2)} = G_k$; put $G_k^{(0)} = \{z \in \mathbb{Z}^k : \sum z_p = 0\} = A_{k-1}$. Up to roots of unity, the roots of $G_k^{(\kappa)}$ are the *edges* $e_p - \varepsilon e_q$ ($p \ne q$, $\varepsilon \in \mu_{2^\kappa}$) and the *half-edges* $(1-\eta)e_p$ ($\eta \in \mu_{2^\kappa}$, $\eta \ne \pm 1$); for $\kappa = 0$ only the edges with $\varepsilon = 1$.

Proposition C.3. Let $0 \le \kappa \le 2$ and let Q be an $\mathcal{O}_\kappa$-submodule of $G_k^{(\kappa)}$ generated by roots. There are a partition of the k coordinates into blocks V and, for each block, roots of unity u_p ($p \in V$) and $0 \le i \le \kappa$, such that Q is the orthogonal sum of

$$Q_V = \left\{ \sum_{p \in V} z_p u_p e_p : z_p \in \mathcal{O}_\kappa, \sum_p z_p \in \varpi_i \mathcal{O}_\kappa \right\},$$

where for $|V| = 1$ either $Q_V = 0$ or $i \ge 2$. The pair (V, i) is determined by Q, and for fixed (V, i) there are $\left(2^{\kappa-i}\right)^{|V|-1}$ such Q_V. The type of Q_V is the base change to $\mathcal{O}_\kappa$ of $A_{|V|-1}$ for $i = 0$ and of $D_{|V|}$ for $i = 1$ (for $\kappa = 2$ these are $A_{|V|-1}[i]$ and $D_{|V|}[i]$), and $G_{|V|}^{(i)} \otimes \mathcal{O}_\kappa$ for $i \ge 2$.

Proof. Let V be a connected component of the graph on the coordinates whose edges are the edges of Q. Choose a spanning tree and rescale the basis vectors by roots of unity ($e_q \mapsto u_q e_q$, an isometry of $G_k^{(\kappa)}$) so that the tree edges become $e_p - e_q$; then $e_p - e_q \in Q$ for $p, q \in V$. Let $\Gamma = \{\eta \in \mu_{2^\kappa} : (1-\eta)e_p \in Q\}$. Since $(1-\eta)e_q = (1-\eta)e_p - (1-\eta)\big(e_p - e_q\big)$, Γ does not depend on $p \in V$, and $1 - \eta\eta' = (1-\eta) + \eta(1-\eta')$ shows that it is a subgroup, $\Gamma = \mu_{2^i}$. An edge $e_p - \varepsilon e_q = \big(e_p - e_q\big) + (1-\varepsilon)e_q$ lies in Q only if $\varepsilon \in \Gamma$. So Q_V is generated by the $e_p - e_q$ and by $(1-\eta_i)e_p$ for a generator η_i of Γ; since $(1-\eta_i)\mathcal{O}_\kappa = \varpi_i \mathcal{O}_\kappa$, Q_V has the stated form (for $i = 1$, $2e_p$ is not a root but lies in the span of $e_p \pm e_q$). Conversely, for η of order 2^j the valuation of $1 - \eta$ at the prime of $\mathcal{O}_\kappa$ above 2 is $2^{\kappa-j}$, so $1 - \eta \in \varpi_i \mathcal{O}_\kappa$ if and only if $\eta \in \mu_{2^i}$; hence the roots of Q_V are those with $\eta, \varepsilon \in \mu_{2^i}$, i is determined by Q_V, and no new roots arise. The count is the number of classes of the ratios u_q/u_p along the tree modulo μ_{2^i}. □

For $1 \le \kappa \le 2$, if C is a lattice over $\mathcal{O}_b$, $0 \le b \le \kappa$, the roots of $C \otimes_{\mathcal{O}_b} \mathcal{O}_\kappa$ split into $2^{\kappa - \max(b,1)}$ orthogonal copies of the roots of C, permuted cyclically by g; so the root sublattices of $C \otimes \mathcal{O}_\kappa$ correspond bijectively to those of C. In particular those of $A_k[i]$ and $D_k[i]$ correspond to the root sublattices of A_k and D_k over $\mathbb{Z}$, which are given by Proposition C.3 with $\kappa = 0, 1$. For $\kappa = 2$: a root sublattice of G_m is given by a partition of the coordinates and, on each block V, one of $A_{|V|-1}[i]$ ($4^{|V|-1}$ ways), $D_{|V|}[i]$ ($2^{|V|-1}$ ways) or $G_{|V|}$ (one way), singletons being unused or $G_1 = A_1[i]$. If $w(n)$ denotes the total coefficient of a block of n coordinates (for instance $w(n) = 2^{n-1}c(A_{n-1}) + c(D_n)$ for $\kappa = 1$), Proposition C.3 says that the sum of $\prod_{V \in \mathcal{P}} w(|V|)$ over the set partitions $\mathcal{P}$ of k coordinates is the sum of $c(Q)$ over all root sublattices Q of $G_k^{(\kappa)}$, which is 0 for $k \ge 2$ (and for every k in the G family) and 1 for $k = 1$ otherwise; so $\exp \sum_n w(n)x^n/n! = 1 + x$ for the A, D families and $= 1$ for the G family, that is $w(n) = (-1)^{n-1}(n-1)!$ or 0. Solving for the last term of $w(n)$ gives

$$c(A_n) = (-1)^n n!, \quad c(D_n) = (-1)^n (n-1)!(2^{n-1} - 1), \quad c(G_m) = (-1)^m 2^{m-1}(m-1)!.$$

The D formula includes $D_2 = A_1^2$, $D_3 = A_3$; the G formula includes $G_1 = A_1[i]$. Base change gives $c(C[i]) = c_{\mathbb{Z}}(C)$. For $E_n[i]$ and E_8^g the root sublattices were counted from explicit roots ($c(E_8[i]) =$

$-28{,}183{,}680$, $c(E_8^g) = -5{,}184$). For ADE systems a symmetric root subset closed under root sums is exactly the set of roots in its integer span, so it represents a root-generated submodule after imposing g-stability. The closed formulas agree with explicit enumeration of these subsets for small (κ, k) [C11].

The exceptional coefficients. For $E_6[i], E_7[i], E_8[i]$, enumerate all ADE root types of rank at most that of the real root system. Count the ordered root tuples with each type's positive definite simple-root Gram and divide by the full automorphism-group order of the source root lattice. This counts each root sublattice once. At each recursive step the program uses reflections in roots orthogonal to the already selected tuple, splits the candidates into their orbits, and multiplies one continuation count by the orbit size. These reflections fix the selected tuple pointwise, so they give bijections between all continuations in an orbit; the subgroup need not be the full stabilizer.

For E_8^g, take the 240 explicit real roots and the rotation g of successive coordinate pairs. Test both real and imaginary parts of the Hermitian Gram. The Hermitian reflection in a root x is the product of the real reflections in the orthogonal roots x, gx; it preserves the root set, commutes with g, and fixes the preceding Hermitian-orthogonal tuple. Enumerating the possible Gaussian root types of rank at most 4 gives the same orbit-counting argument. Omitting the ambient exceptional lattice itself and taking the negative coefficient sum gives $c(E_6[i]) = 7{,}680$, $c(E_7[i]) = -233{,}280$, $c(E_8[i]) = -28{,}183{,}680$, and $c(E_8^g) = -5{,}184$ [C11, C19].

C.4 A Siegel-type formula

Let Λ_n be the lattice of even Hermitian $n \times n$ matrices over $\mathcal{O}$ (diagonal in $2\mathbb{Z}$). For a theta lattice with Gram matrix S over $\mathcal{O}_p = \mathcal{O} \otimes \mathbb{Z}_p$ and $T \in \Lambda_n$ positive definite, the local density is

$$\alpha_p(T) = \lim_{a \to \infty} p^{-a(2rn-n^2)} \#\{X \in M_{r\times n}(\mathcal{O}/p^a) : X^* S X \equiv T \pmod{p^a \Lambda_n}\}.$$

All theta lattices of rank r lie in one genus (by Lemma C.9(iii) at $p = 2$ and the classical theory at odd p), so $\alpha_p(T)$ does not depend on L.

Theorem C.4. For $n < r$,

$$A_r(T) = \mu_r \cdot 2^n \frac{\pi^{n(2r-n+1)/2}}{\prod_{j=0}^{n-1} \Gamma(r-j)} \det(T)^{r-n} \prod_p \alpha_p(T), \tag{C.3}$$

and the product converges absolutely.

Strategy of the proof. This is Siegel's formula in the form "an average number of representations is a product of local densities", proved by unfolding an adelic integral. By Witt's theorem the unitary group acts transitively on the solutions of $X^* S X = T$, so the sum over embeddings becomes an integral over the solutions, which factors into local volumes. The Tamagawa numbers of the unitary group and of the stabilizer are both 2 and cancel.

Proof. Let $V = L \otimes \mathbb{Q}$, $G = \mathrm{U}(V)$ and $\Omega = \{X \in V^n : X^* S X = T\}$. $\Omega(\mathbb{Q})$ is non-empty: with $\delta = \det S \det T > 0$, the form $T \perp \langle \delta, 1, \ldots, 1 \rangle$ has the same dimension as V, is positive definite, and has determinant $\det S (\det T)^2$, which lies in the class of $\det S$ modulo norms since $\det T \in \mathbb{Q}^\times$; so it is isometric to V, because a Hermitian space over a number field is determined by its dimension, its indices at the real places and its determinant modulo norms [Shi08, Thm 2.2(i)] (Landherr). By Witt's theorem G acts transitively on Ω, over $\mathbb{Q}$ and over every completion, with stabilizer $H = \mathrm{U}(W)$, $W = T^\perp$, of rank $r - n \geq 1$. Let $K_f \subset G(\mathbb{A}_f)$ be the stabilizer of $L \otimes \hat{\mathbb{Z}}$ and φ the characteristic function of $\left(L \otimes \hat{\mathbb{Z}}\right)^n$. Unfolding,

$$\int_{G(\mathbb{Q})\backslash G(\mathbb{A})} \sum_{x\in\Omega(\mathbb{Q})} \varphi(g^{-1}x)\,\mathrm{d}g = \operatorname{vol}(H(\mathbb{Q})\setminus H(\mathbb{A}))\int_{\Omega(\mathbb{A})}\varphi\,\mathrm{d}\mu,$$

which converges since $G(\mathbb{R})$ is compact. The left side is $\operatorname{vol}\big(K_f G(\mathbb{R})\big)A_r(T)$, and for $T=\emptyset$ it gives $\operatorname{vol}\big(K_f G(\mathbb{R})\big)\mu_r = \operatorname{vol}(G(\mathbb{Q})\setminus G(\mathbb{A}))$. With Tamagawa measures $\omega_G=\omega_\Omega\wedge\omega_H$, the convergence factors $L_p(1,\chi)$ of G and H cancel, and $\tau(H)=\tau(G)$ [Wei65, n°53–54] ($\tau(\mathrm{U}(m))=2$ for all $m\geq 1$, including the norm-one torus $m=1$). The local volumes of $\omega_\Omega=\mathrm{d}X/\,\mathrm{d}T$, normalized so that $\mathcal{O}_p^{rn}$ and $\Lambda_n\otimes\mathbb{Z}_p$ have volume 1, are the $\alpha_p(T)$, and at infinity $\alpha_\infty(T)=2^n c\det(T)^{r-n}$ with c from Lemma C.5: the covolume of $\mathcal{O}^{rn}$ in $\mathbb{C}^{rn}$ is 1 since $\det S=1$, and that of Λ_n is 2^n. □

Lemma C.5. Let $\mathrm{d}X$ be Lebesgue measure on $M_{r\times n}(\mathbb{C})=\mathbb{R}^{2rn}$ and $\mathrm{d}T$ Lebesgue measure on Hermitian matrices in the coordinates t_{kk}, $\operatorname{Re} t_{kl}$, $\operatorname{Im} t_{kl}$ ($k<l$). Under $X\mapsto T=X^*X$ ($n\leq r$), $\mathrm{d}X$ pushes forward to $c\det(T)^{r-n}\,\mathrm{d}T$ on positive definite T, with $c=\pi^{rn-n(n-1)/2}/\prod_{j=0}^{n-1}\Gamma(r-j)$.

Proof. For $g\in\mathrm{GL}_n(\mathbb{C})$, $X\mapsto Xg$ multiplies $\mathrm{d}X$ by $|\det g|^{2r}$ and $T\mapsto g^*Tg$ multiplies $\mathrm{d}T$ by $|\det g|^{2n}$; since $\mathrm{GL}_n(\mathbb{C})$ acts transitively on positive definite matrices, the density has the form $c\det(T)^{r-n}$. To find c, compare $\pi^{rn}=\int e^{-\operatorname{tr} X^*X}\,\mathrm{d}X = c\int_{T>0} e^{-\operatorname{tr} T}\det(T)^{r-n}\,\mathrm{d}T$. Write $T=B^*B$ with B upper triangular with positive diagonal; then $\mathrm{d}T = 2^n\prod_k b_{kk}^{2(n-k)+1}\,\mathrm{d}B$ and $\det T=\prod_k b_{kk}^2$, and the integral is $\pi^{n(n-1)/2}\prod_{k=1}^n\Gamma(r-k+1)$. □

For $n=r$ we do not use (C.3). In that case $r(T,L)/\,|\mathrm{U}(L)|$ summed over L counts the even unimodular overlattices $T'\supset T$ in $T\otimes\mathbb{Q}$, because an embedding of a lattice of full rank is determined by its image and the only automorphism of T' fixing T pointwise is the identity. Overlattices correspond to submodules $C\subset T^*/T$ with $C^\perp=C$ that are isotropic for $q(x)=h(x,x)\bmod 2$, and T^*/T is the direct sum of its p-parts; hence

$$A_r(T)=\prod_p \#\big\{C\subset D_p(T) : C \text{ isotropic}, C^\perp=C\big\}\quad (n=r), \tag{C.4}$$

with $D_p(T)=T^*/T\otimes\mathbb{Z}_p$.

C.5 Local densities

An embedding X is *primitive* at p if X has rank n modulo every prime of $\mathcal{O}$ above p.

Lemma C.6. If X is primitive, the differential of $X\mapsto X^*SX$ maps onto $\Lambda_n\otimes\mathbb{Z}_p$. Consequently the primitive part $\alpha_{p,\,\mathrm{prim}}(T)$ of $\alpha_p(T)$ is determined by solutions modulo p for odd p and modulo 2 for $p=2$.

Proof. The differential is $A\mapsto X^*SA+(X^*SA)^*$. X^*S maps $\mathcal{O}_p^r$ onto $\mathcal{O}_p^n$, and $Y\mapsto Y+Y^*$ maps $M_n(\mathcal{O}_p)$ onto $\Lambda_n\otimes\mathbb{Z}_p$, because $\operatorname{Tr}_{\mathcal{O}/\mathbb{Z}}(\mathcal{O})=2\mathbb{Z}$ covers the even diagonal. Replacing X by $X+2A$ changes X^*SX by $2\,\mathrm{d}F_X(A)+4A^*SA$, and $4A^*SA\in 4\Lambda_n$ since L is even; so each solution modulo $2\Lambda_n$ lifts to exactly 2^{2rn-n^2} solutions at every higher level. The odd case is the usual Hensel argument. □

On $\overline{V}=L/\varpi L\cong\mathbb{F}_2^r$ the form $\omega(x,y)=h(x,y)\bmod\varpi$ is non-degenerate and alternating (proof of Lemma 2.2). For T let $2a$ be the rank of ω on $T/\varpi T$ and $d=n-2a$ the dimension of its radical; d is the number of generators of the $\mathcal{O}$-module $D_2(T)$.

Proposition C.7. $\alpha_{2,\,\mathrm{prim}}(T)=\prod_{k=0}^{(n+d)/2-1}\big(1-4^k2^{-r}\big)$. For odd p, with d the number of generators of $D_p(T)$, $\alpha_{p,\,\mathrm{prim}}(T)=\prod_{j=r-n-d+1}^{r}\big(1-\chi(p)^jp^{-j}\big)$.

Proof. Write the columns of X modulo 2 as $x_j = \hat{w}_j + \varpi u_j$ with $w_j \in \overline{V}$ independent (primitivity) and $u_j \in \overline{V}$. Since $\overline{\varpi} \equiv \varpi$ modulo 2,

$$h(x_j, x_k) \equiv h(\hat{w}_j, \hat{w}_k) + \varpi\big(\omega(w_j, u_k) + \omega(u_j, w_k)\big), \quad h(x_j, x_j)/2 \equiv h(\hat{w}_j, \hat{w}_j)/2 + \omega(w_j, u_j) \pmod 2.$$

For fixed w these are linear conditions on u: the map $u \mapsto B = \big(\omega(w_k, u_j)\big)_{k,j} \in M_n(\mathbb{F}_2)$ is onto, and the conditions prescribe B_{jj} and $B_{jk} + B_{kj}$, which are $n(n+1)/2$ independent linear forms. So there are $2^{rn-n(n+1)/2}$ choices of u for each admissible w. The admissible w are the isometric injections of $(T/\varpi T, \omega)$ into $\big(\overline{V}, \omega\big)$, whose number is $\prod_{k=0}^{a-1}(2^{r-2k}-1)2^{r-2k-1} \cdot \prod_{j=0}^{d-1}(2^{r-2a-j} - 2^j)$. Multiplying and dividing by 2^{2rn-n^2} gives the formula. For odd p the same count over $\mathcal{O}/p$ (a Hermitian space over $\mathbb{F}_{p^2}$ if $p \equiv 3 \pmod 4$, a pair of dual spaces over $\mathbb{F}_p$ if $p \equiv 1$) gives the second formula. □

Theorem C.8. Let $t = r - n$. For each isotropic $\mathcal{O}$-submodule $C \subset D_p(T)$ let $T_C \supset T$ be the corresponding even overlattice and $d(C)$ the number of generators of $C^\perp/C$. Then

$$\alpha_p(T) = \sum_C |C|^{-t}\ \alpha_{p,\,\mathrm{prim}}(T_C), \tag{C.5}$$

and, with $S_t = 2^{1-t}\prod_{j=1}^{t}|L(1-j,\chi^j)|$ ($\chi^j = \chi$ for odd j, the trivial character for even j, so that $L(1-j,\chi^j) = \zeta(1-j)$), $S'_t = 2^{-t[t\text{ odd}]}S_t$ and $S'_0 = 1$,

$$A_r(T) = S'_t \det(T)^t \prod_p f_p(T,t), \quad f_p(T,t) = \sum_C |C|^{-t} \prod_{j \in J_p(t,d(C))} (1 - \chi(p)^j p^{-j}), \tag{C.6}$$

where $J_p(t,d) = \{t-d+1, ..., t\}$ for odd p, $J_2(t,d) = \{j \text{ even} : t-d+2 \le j \le t\}$, and $\chi(2)^j$ is read as 1.

Strategy of the proof. An embedding of T need not be primitive: the closure of its image is an overlattice $T_C \supset T$, given by an isotropic submodule C. (C.5) splits the local density according to this overlattice, and Proposition C.7 gives each primitive density in closed form. For (C.6), the factors that do not depend on T are divided out and, with the functional equations of the L-functions, combined into the constant S'_t.

Proof. (C.5): the closure of the image of X is some T_C; the embeddings with closure T_C are the primitive embeddings of T_C composed with the inclusion, and the change of basis multiplies the volume on the X side by $|C|^{-r}$ and on the Λ_n side by $|C|^{-n}$.

(C.6) for $n < r$: if $p \nmid \det T$ then $D_p(T) = 0$ and $\alpha_p(T) = \prod_{j=t+1}^{r}(1 - \chi(p)^j p^{-j})$ by Proposition C.7 (for $p = 2$, the factors with $j = r - 2k$ even). Dividing $\alpha_p(T)$ by this product gives $f_p(T,t)$ for every p, again by Proposition C.7. The product of the divisors over all p is $\prod_{j=t+1}^{r} L(j,\chi^j)^{-1}$. By the functional equations, $|L(1-j,\chi^j)| = 2(j-1)! \cdot 2^{(2j-1)[j\text{ odd}]}(2\pi)^{-j}L(j,\chi^j)$. Inserting this and $\mu_r = S'_r$ into (C.3), the factors π and $(j-1)!$ cancel against those of (C.3), and the remaining power of 2 is 2^E with $E = \sum_{j=t+1}^{r}(1 - j + (2j-1)[j\text{ odd}]) - r[r\text{ odd}] + t[t\text{ odd}]$; the summand is j for odd j and $1-j$ for even j, so consecutive pairs (odd j, $j+1$) cancel, and the parity terms cancel the unpaired end terms. Hence $E = 0$ and (C.6) holds.

(C.6) for $n = r$: $t = 0$, $S'_0 = 1$, and in $f_p(T,0)$ the product is 0 unless $d(C) = 0$ (it contains the factor $j = 0$, and $d(C)$ is even at $p = 2$), so $f_p(T,0)$ counts the C with $C^\perp = C$, which is (C.4). □

The formula expresses the averaged embedding number through the complement rank $t = r - n$ and local gluing data. It need not be the ordinary mass of a single complement genus.

C.6 Computing the 2-adic sums

For odd p, $D_p(T)$ is small and the isotropic submodules are counted in closed form. At $p = 2$, $D_2(T)$ can be large ($D_2(A_1[i]^{12})$ has about 2.2×10^{14} isotropic submodules), but only the distribution of $(|C|, d(C))$ is needed. We compute it by adding the irreducible components of T one at a time. The state is the isometry type of the discriminant module of an overlattice of the part already added, recorded by its Jordan decomposition:

Lemma C.9. Let L be an even lattice over $\mathcal{O}_2 = \mathbb{Z}_2[i]$.

1. L is an orthogonal sum of Jordan components of rank 1 and 2.
2. A component $\langle 2^e u\rangle$ of rank 1 is determined by $u \bmod 4$.
3. A plane of scale ϖ^{2e} is $2^e H$, H the hyperbolic plane; in particular an even unimodular lattice is $H^{r/2}$.
4. A plane of scale ϖ^{2e+1} is one of the two planes $P(e,t)$, $t = 0, 1$, with Gram matrix $\left(\begin{smallmatrix} 2^{e+1}t & \varpi^{2e+1} \\ \overline{\varpi}^{2e+1} & 2^{e+1}t \end{smallmatrix}\right)$. $P(e,0)$ has a primitive isotropic vector and $P(e,1)$ has none, so they are not isometric.
5. $\langle 2^e\rangle^2 \cong \langle 3 \cdot 2^e\rangle^2$ and $P(e,1)^2 \cong P(e,0)^2$.

Proof. We use that the norms of units are $N(\mathcal{O}_2^\times) = 1 + 4\mathbb{Z}_2$: the inclusion holds modulo 4 since $u^2 + v^2 \equiv 1$ for $u + vi$ a unit, and $5 = N(1 + 2i)$ generates $1 + 4\mathbb{Z}_2$ topologically while the norm group is closed. Let s be the ϖ-adic scale of L.

(i) If a diagonal entry attains the scale, it splits off a component of rank 1; otherwise only off-diagonal entries attain it, and a plane spanned by two basis vectors splits off.

(ii) $\langle 2^e u\rangle \cong \langle 2^e u'\rangle$ if and only if u'/u is a norm of a unit, that is $u' \equiv u$ modulo 4.

(iii) Let the plane be $2^e\left(\begin{smallmatrix} 2a & \varepsilon \\ \overline{\varepsilon} & 2b \end{smallmatrix}\right)$ with ε a unit; rescaling the basis we may take $\varepsilon = 1$. For $x = e_1 + \lambda e_2$, $\lambda = u + vi$, half the norm divided by 2^e is $a + b(u^2 + v^2) + u$, whose derivative in u is odd, so Hensel's lemma applies; a root modulo 2 always exists ($(0,0)$ if a is even, $(1,1)$ if a and b are odd, $(1,0)$ if b is even). So there is a primitive isotropic x with $h(x, e_2)$ a unit, and e_2 can be moved by a multiple of x to an isotropic vector, since $\mathrm{Tr}(\mathcal{O}) = 2\mathbb{Z}$. Hence the plane is $2^e H$.

(iv) Let the plane be $\left(\begin{smallmatrix} 2^{e+1}a & \varpi^s\varepsilon \\ \overline{\varpi^s\varepsilon} & 2^{e+1}b \end{smallmatrix}\right)$ with $s = 2e + 1$. For $x = e_1 + \lambda e_2$ the norm is $2^{e+1}(a + b(u^2 + v^2) + cu + dv)$ with c, d odd. If a is even, Hensel's lemma at $(0,0)$ gives an isotropic x and, as in (iii), the plane is $P(e,0)$ (the plane with an isotropic primitive vector); if b is even, exchange e_1 and e_2. If a and b are odd, Hensel's lemma from the root $(0,0)$ of $a - 1$ modulo 2 gives x of norm 2^{e+1}; rescale e_2 so that $h(x, e_2) = \varpi^s$. With $\rho = 2^{e+1}\varpi^{-s}$ and $y = \eta e_2 + \mu x$, $\eta = 1 - \mu\overline{\rho}$, we keep $h(x,y) = \varpi^s$, and the norm of y divided by 2^{e+1} is a polynomial in μ that is odd at $\mu = 0$ with a derivative having odd coefficients; Hensel's lemma gives y of norm 2^{e+1}. This plane is $P(e,1)$.

(v) In $\langle 2^e\rangle^2$ the vector $(1, \varpi)$ has norm $3 \cdot 2^e$, equal to the scale, so it splits off and its complement is determined by the determinant. In $P(e,1)^2$ with basis e_1, f_1, e_2, f_2, the vectors $e_1 + e_2$ and f_1 span a plane with an isotropic primitive vector, hence $P(e,0)$; with ρ as above, $z = -\rho e_1 + f_1 + \rho e_2 - f_2$ is a primitive vector of its orthogonal complement of norm $2 \cdot 2^{e+1}$, so the complement is also $P(e,0)$. For $P(e,1)$, $h(x,x)/2^{e+1} \equiv 1 \pmod 2$ for every primitive x, by the computation above with a, b odd. □

Closed sums at odd primes. Put $R_p = \mathbb{Z}_{p[i]}$ and $t = r - n$. At odd primes the root inputs of Gaussian rank at most 16 have

$$D_p(T) \cong \left(R_p/p\right)^a \perp \left(R_p/p^2\right)^b,$$

where $b = 0, 1$, or $(p, a, b) = (3, 0, 2)$. Only $A_8[i]$ contributes a p^2 factor. For a residual module requiring d generators, abbreviate

$$\ell_p(d) = \prod_{j=t-d+1}^{t} (1 - \chi(p)^j p^{-j}).$$

Every submodule contributes $|C|^{-t}\ell_p(d(C))$ to (C.6); $d > t$ gives the zero factor at $j = 0$.

If $p \equiv 1 \pmod 4$, then $b = 0$ and the discriminant splits into two mutually dual a-dimensional spaces over $\mathbb{F}_p$. An isotropic submodule is a pair (C_+, C_-) with $C_- \subset C_+^\perp$. For dimensions k_1, k_2 its multiplicity is

$$\binom{a}{k_1}_p \binom{a-k_1}{k_2}_p,$$

where $\binom{n}{k}_p$ counts k-subspaces over $\mathbb{F}_p$. It has $|C| = p^{k_1+k_2}$ and $d(C) = a - k_1 - k_2$.

If $p \equiv 3 \pmod 4$, the residue field is $\mathbb{F}_{p^2}$. A non-degenerate Hermitian a-space has

$$N(a,k) = \left(\prod_{j=0}^{k-1} (p^{a-2j} - (-1)^{a-2j})(p^{a-2j-1} - (-1)^{a-2j-1})\right) / \left(\prod_{j=1}^{k} (p^{2j} - 1)\right)$$

totally isotropic k-subspaces. This is zero when $2k > a$ and one when $k = 0$. At step j, each chosen nonzero isotropic vector of the orthogonal quotient has p^{2j} lifts. These factors cancel the same powers of p in the number of ordered bases of a k-space over $\mathbb{F}_{p^2}$, leaving the displayed denominator. For $b = 0$ these subspaces have $|C| = p^{2k}$ and $d(C) = a - 2k$.

For $b = 1$, apply the graph transition to the elementary summand and the cyclic summand $B = R_p/p^2$. After choosing C_0 of dimension k in the elementary summand, its quotient A_0 has dimension $a' = a - 2k$. The projection Z to B is 0, pB or B. The last case is impossible: the image of a generator in A_0 has norm denominator at most p, which cannot cancel the unit denominator p^2 on B. The remaining choices are

projection and map	multiplicity	$\log_p\lvert C\rvert$	$d(C)$
$Z = 0$	1	$2k$	$a' + 1$
$Z = pB$, zero map	1	$2k + 2$	a'
$Z = pB$, nonzero isotropic image	$N(a', 1)(p^2 - 1)$	$2k + 2$	$a' - 1$

The final quotient in the last row replaces one hyperbolic plane of A_0 and B by a cyclic p^2 factor; the other $a' - 2$ elementary factors remain. Multiplying each row by $N(a, k)$ and its local weight gives the three terms used in the code.

For $(p, a, b) = (3, 0, 2)$, $D_3(A_8[i]^2) = (R_3/9)^2$ with the form $\mathrm{diag}(8/9, 8/9)$; since 8 is a unit at 3, the implementation uses $\mathrm{diag}(1/9, 1/9)$, which has the same isotropic submodules and the same orthogonals $C^\perp$, hence the same $|C|$ and $d(C)$, which are all that (C.6) uses. It works with the actual finite set $(R_3/9)^2$. Starting with zero, it adds x, ix for isotropic $x \in C^\perp$ and takes their additive span, discarding duplicate subsets. Every isotropic submodule is reached by adjoining its generators in this way. The subgroup of $C^\perp/C$ killed by 3 has size $9^{d(C)}$, which determines the required generator number. This finite enumeration, the closed odd-prime sums and the 2-adic transition below are the local operations of [C11].

Table C1. The 2-adic Jordan blocks of the finite transition. $u \in \{1, 3\}$ modulo 4; the discriminant module is that of the block, with its generator(s).

block	Gram matrix	discriminant module	occurs in the root lattices
$\langle 2^e u\rangle$, $e \geq 1$	$(2^e u)$	$\mathcal{O}/2^e$, generator $e_1/2^e$, $q = u/2^e$	$A_n[i]$ $(n+1 = 2^e m,\ m$ odd, $u \equiv mn)$, $D_n[i]$ $(n$ odd: $e = 2,\ u \equiv n$; $n \equiv 2 \bmod 4$: two blocks with $e = 1,\ u \equiv n/2)$, $E_7[i]$ $(e = 1,\ u = 3)$, G_m (m odd, $e = 1$, $u \equiv m$)
$2^e H$, $e \geq 1$	$2^e \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$	$(\mathcal{O}/2^e)^2$	$D_n[i]$, $n \equiv 0 \bmod 4$ $(e = 1)$
$P(e,t)$, $e \geq 0$, $t = 0, 1$	$\left(\begin{smallmatrix} 2^{e+1}t & \varpi^{2e+1} \\ \varpi^{2e+1} & 2^{e+1}t \end{smallmatrix}\right)$	$(\mathcal{O}/\varpi^{2e+1})^2$	G_m, m even ($e = 0$; $t = 1$ iff $m \equiv 2 \bmod 4$)

The state of the programme is a pair (ℓ, E): $\ell = \log_2|C|$, and E is the multiset of non-unimodular Jordan blocks of the current overlattice, normalized by Lemma C.9(v) ($P(e,1)^2 \to P(e,0)^2$, $\langle 3 \cdot 2^e\rangle^2 \to \langle 2^e\rangle^2$); unimodular blocks are dropped. The blocks $P(0,t)$ are processed first, in closed form (below), then the planes $2^e H$, then the blocks of rank 1. A transition adds $c = \log_2|Z|$ to ℓ and replaces E by the normalized Jordan decomposition of $\Gamma_\varphi^\perp/\Gamma_\varphi$; at the end the weights are aggregated by $(\ell, d(E))$, $d(E)$ the sum of the ranks of the blocks of E. $E_6[i]$, $E_8[i]$ and E_8^g contribute no block at 2. For G_m the discriminant group of D_{2m} is $\{0, v, s, c\}$ with $q(v) = 1$, $q(s) = q(c) = m/2$; g has m sign changes, so for even m it lies in $W(D_{2m})$, fixes s and c, and the discriminant module is $(\mathcal{O}/\varpi)^2$ with $t = 0$ or 1 according as $m \equiv 0$ or 2 modulo 4, while for odd m it exchanges s and c and the module is $\mathcal{O}/2$.

The finite transition. Write $\mathcal{O}_2 = \mathbb{Z}_2[i]$ and $\varpi = 1 + i$. A state after adding blocks $B_1, ..., B_a$ is (ℓ, E), with integer weight $W_a(\ell, E)$ equal to the number of isotropic submodules C of their direct sum for which $|C| = 2^\ell$ and $C^\perp/C \cong E$. The initial state is $W_0(0,0) = 1$.

Lemma C.10 (graph transition). Let A, B be finite discriminant modules. Isotropic submodules $C \subset A \perp B$ are in bijection with triples consisting of an isotropic $C_0 \subset A$, a submodule $Z \subset B$, and an $\mathcal{O}_2$-linear map $\varphi : Z \to E = C_0^\perp/C_0$ whose graph $\Gamma_\varphi \subset E \perp B$ is isotropic. The map need not be injective. Then $|C| = |C_0| \cdot |Z|$ and $C^\perp/C \cong \Gamma_\varphi^\perp/\Gamma_\varphi$.

Proof. Take $C_0 = C \cap A$ and project C onto B. Orthogonality to C_0 puts C inside $C_0^\perp \perp B$. Modulo C_0 the projection onto its image Z is injective, so it is the graph of a unique linear map. Conversely, the inverse image of any such isotropic graph is the unique corresponding C. The order and residual-form identities follow from the same quotient. □

For each actual submodule Z of the new block, fix one generating set and count each allowed map once. There is no division by $|\mathrm{Aut}\ Z|$. Put $c = \log_2|Z|$ and $E' = \Gamma_\varphi^\perp/\Gamma_\varphi$; this pair adds $W_a(\ell, E)$ to $W_{a+1}(\ell + c, E')$. Isometric representatives of E have bijective sets of these transitions. It is enough that each normalization preserves isometry; a complete canonical classification of state keys is not required.

Conditions on maps. A rank-1 block has $B \cong \mathcal{O}_2/\varpi^s$, and all its submodules are $\varpi^j B$, $0 \leq j \leq s$. With fixed generator $z = \varpi^{j-s}$, its image y must satisfy $\varpi^{s-j} y = 0$ and $q_E(y) + q_B(z) = 0$. A rank-2 block has $B \cong (\mathcal{O}_2/\varpi^s)^2$. Its submodules are at most two-generated; enumerate cyclic submodules, then their pairwise sums, retaining each resulting subset once. For fixed generators z_1, z_2 the candidate images y_1, y_2 must satisfy every relation

$$a_1 z_1 + a_2 z_2 = 0 \quad \Rightarrow \quad a_1 y_1 + a_2 y_2 = 0,$$

as well as $\mathrm{ann}(z_i) y_i = 0$, $q_E(y_i) + q_B(z_i) = 0$ in $\mathbb{Q}_2/2\mathbb{Z}_2$, and $h_E(y_1, y_2) + h_B(z_1, z_2) = 0$ in $\mathbb{Q}_2(i)/\mathcal{O}_2$. These conditions are necessary and sufficient for an isotropic graph. Relations with coefficients in $\mathcal{O}_2/\varpi^s$

then suffice, since $\varpi^s y_i = 0$. Individual orders or norms alone do not replace the relations or the mixed Hermitian condition.

Orbit compression. In each Jordan block use a subgroup of its actual isometry group. For a single image use its orbits on elements; for two images use its diagonal orbits on ordered pairs, preserving their relative position. If a block occurs m times, an orbit pattern with k_j copies of an orbit of size v_j represents

$$m!/\left(\prod_j k_{j!}\right)\prod_j v_j^{k_j}, \quad \sum_j k_j = m$$

candidate images. Permuting equal blocks and applying their isometries takes any two candidates of this pattern to each other, preserving relations, isotropy and the residual form. Thus one representative has this multiplicity. A subgroup in place of the whole block isometry group refines the orbits and increases work; it does not omit candidates. The implementation retains joint ordered-pair orbits, not just the two marginal element orbits.

New states and endpoints. Add the graph generators to the local lattice and compute its Jordan decomposition. In a one-generator step, replace the basis vector in the coordinate of least ϖ-adic valuation by the new vector. Every old basis vector is still integral over this new local basis; its index is the annihilator exponent of the added class. Repeating this gives the index 2^c and the residual form. Every transition has $|E'| = |E| \cdot |B| \ /2^{2c}$. The $P(0,t)$ blocks are processed first. Their discriminant forms are the binary planes $q(x,y) = xy$ for $t = 0$ and $q(x,y) = x + y + xy$ for $t = 1$. For m such planes with sign $\varepsilon = (-1)^{\sum t}$, the number of totally singular k-subspaces is

$$\left(\prod_{j=0}^{k-1}(2^{m-j} - \varepsilon)(2^{m-j-1} + \varepsilon)\right)/\left(\prod_{j=1}^{k}(2^j - 1)\right).$$

It is zero for $k > m$ and for $\varepsilon = -1$, $k = m$, and one for $k = 0$. The residual space has dimension $2(m-k)$ and sign ε. Count ordered bases: given $V_j = \langle v_1, ..., v_j\rangle$ totally singular, v_{j+1} is a lift of a non-zero singular vector of $V_j^\perp/V_j$, a non-degenerate space of dimension $2(m-j)$ and sign ε with $(2^{m-j} - \varepsilon)(2^{m-j-1} + \varepsilon)$ such vectors, each with 2^j lifts, all singular; dividing by $|\operatorname{GL}_k(\mathbb{F}_2)| = 2^{k(k-1)/2}\prod_{j=1}^k(2^j - 1)$ cancels the factors 2^j. These are the closed initial weights before the H and rank-1 blocks are added. Finally aggregate by $(\ell, d(E))$, where $d(E)$ is the sum of the ranks of the remaining non-unimodular blocks, and substitute the counts into (C.6). In the present even ambient rank, t and $d(E)$ have the same parity. If $d(E) > t$, the product contains its zero factor at $j = 0$; for $t = 0$ only $d(E) = 0$ survives. This includes the full-rank, Lagrangian endpoint. The finite computation and its local basis operations are part of [C11]; the small checks support their implementation and do not replace this transition argument.

The full root-type index and all terms of (C.2) are also recorded in [C19]. This reconstruction regenerates the exceptional root coefficients, checks the index against coefficient extraction from $\prod_Y \left(1 - x^{\operatorname{rank} Y}\right)^{-1}$, and sums in exact arithmetic. It reuses fixed local DP statistics for 1,498 block multisets; those transition counts are not independently recomputed by this aggregation.

C.7 Checks and the value

For ranks $r = 4, 8, 12$ the rootless masses are known independently, and (C.2) reproduces them exactly (Table C2). At rank 12 the only rootless theta lattice is the Leech lattice with its unique class of automorphisms g with $g^2 = -1$ in Co_0, whose centralizer has order 2,012,774,400 (computed from the

character table of $2.\mathrm{Co}_1$ in GAP); this value is obtained independently of (C.2). At rank 8 three root systems have non-zero mass, and $1/m(R)$ equals the order of the unitary group of the corresponding lattice for all three; at rank 12 the 25 root systems with non-zero mass map to exactly the 19 Niemeier root systems that admit a structure over $\mathbb{Z}[i]$ [HKN12, Remark 4(d)].

Table C2. Rootless masses of theta lattices computed by (C.2) [C11], with the independent values.

r	root types	$\mu_r^{\emptyset}$ from (C.2)	independent value
4	22	0	0 (the only theta lattice is E_8 [HKN12, Cor. 1])
8	254	0	0 (no rootless even unimodular lattice of rank 16)
12	2,035	1/ 2,012,774,400	1/ $\lvert C_{\mathrm{Co}_0}(g)\rvert$ (Leech lattice)
16	12,934	4,128,748,922,231,417 / 120,324,096,000	Proposition 4.6

At rank 16 the value is $\mu_{16}^{\emptyset} \approx 34{,}313.567$, with denominator $2^{21} \cdot 3^3 \cdot 5^3 \cdot 17$. Of the 12,934 root types R, the mass $m(R)$ of the theta lattices whose root system is exactly R, obtained from the values $A_{16}(T)$ by solving $A_{16}(T) = \sum_R m(R) r(T,R)$, is non-zero for 1,154 types, exactly 0 for the other 11,780, and never negative; an error in a single value $A_r(T)$ would make some mass that is truly 0 non-zero, except for 240 of the 12,933 non-empty types T. These are consistency checks; the value rests on the proofs above and is confirmed by Proposition 4.6.

Appendix D. Rootless masses for elements of odd prime order

What this appendix does. It proves the values of the rootless masses used in §4.2 and Appendix B.4. §D.1 shows that the Hermitian lattices Z of a given type form one genus and recalls its mass. §D.2 derives the rootless masses of unimodular Hermitian lattices from (4.2). §D.3 computes the rootless masses of the genera (z,s) over $\mathbb{Z}[\zeta_p]$, $p = 5, 7, 17$, by inclusion–exclusion over the root lattices A_{p-1} and E_8, with the method of Appendix C. §D.4 treats the $\mathbb{Z}$-lattices of the genera $(20,6)$ for $p = 3$ and $(16,4)$ for $p = 5$, and the Eisenstein lattices of the genus $(13,5)$.

Throughout, p is an odd prime, $\mathbb{K} = \mathbb{Q}(\zeta_p)$, $\mathbb{K}^+$ its real subfield, $\mathcal{O} = \mathbb{Z}[\zeta_p]$, $\mathcal{O}^+ = \mathcal{O} \cap \mathbb{K}^+$, $\varpi = 1 - \zeta_p$ and $\nu = \varpi\overline{\varpi}$. An *$\mathcal{O}$-lattice of type* (z,s) is a torsion-free $\mathcal{O}$-module Z of rank z with a positive definite Hermitian form whose trace form $(x,y) = \mathrm{Tr}_{\mathbb{K}/\mathbb{Q}} H(x,y)$ is integral, with $Z^\#/Z \cong \mathbb{F}_p^s$ and ζ_p acting trivially on $Z^\#/Z$; we use the rescaled form $\tilde{H} = \nu^{(p-3)/2} H$ of Lemma 4.4, written h in this appendix, so that $Z^* = \varpi Z^\#$ and $Z^* \subset Z \subset \varpi^{-1} Z^*$.

D.1 The genus and its mass

Proposition D.1. The $\mathcal{O}$-lattices of type (z,s) form at most one genus, and their trace forms are even. Here $0 \le s \le z$, since $Z^* = \varpi Z^\# \subset Z$ gives $[Z^\# : Z] \le [Z^\# : \varpi Z^\#] = p^z$. The genus exists if and only if $z \equiv s \pmod 2$, and $s > 0$ or $(-1/p)^{z/2} = 1$. For $s = z$ the quadratic space $U/\varpi U$ with the form $h \bmod \varpi$ has square discriminant, so for even z it is hyperbolic if and only if $(-1/p)^{z/2} = 1$.

Proof. Evenness. $\mathbb{K}/\mathbb{K}^+$ has different $\varpi\mathcal{O}$, so $\mathbb{K}^+/\mathbb{Q}$ has different $\varpi^{p-3}\mathcal{O} \cap \mathcal{O}^+$. For $x \in Z$, $h(x,x) \in \mathbb{K}^+ \cap \varpi^{-1}\mathcal{O} = \mathcal{O}^+$ (elements of $\mathbb{K}^+$ have even ϖ-adic valuation), so $H(x,x) = \nu^{-(p-3)/2} h(x,x)$ lies in the inverse different of $\mathbb{K}^+/\mathbb{Q}$, and $(x,x) = 2\mathrm{Tr}_{\mathbb{K}^+/\mathbb{Q}} H(x,x) \in 2\mathbb{Z}$.

Places other than ϖ. There $Z_v = Z_v^*$. If v splits in $\mathbb{K}/\mathbb{K}^+$, a unimodular Hermitian lattice has a basis in which h is the identity. If v is inert with residue extension $\mathbb{F}_{q^2}/\mathbb{F}_q$, $h(x,x)$ is not identically 0 modulo v (otherwise $h(x,y) + \overline{h(x,y)} \equiv 0$ for all x, y, and replacing y by cy gives $h \equiv 0$), so some $h(x,x)$ is a unit, $\langle h(x,x)\rangle$ splits off, and since the extension is unramified every unit of $\mathcal{O}_v^+$ is a norm. Hence $Z_v \cong \langle 1, ..., 1\rangle$.

The place ϖ. Let $\Lambda = Z_\varpi$. The form $a(x,y) = \varpi h(x,y) \bmod \varpi$ on Λ/Λ^* is non-degenerate and alternating (Lemma 4.4), so $z - s = 2m$. Take x, y with $h(x,y) = \varpi^{-1}$; then $\mathcal{O}x + \mathcal{O}y$ has Gram matrix ϖ^{-1} times a unimodular matrix and is an orthogonal summand. Replacing x by $x + r\varpi y$ with $r \in \mathcal{O}_\varpi^+$ chosen by Hensel's lemma (the derivative of the norm in r is -2 modulo ϖ), we get $h(x,x) = 0$, and similarly $h(y,y) = 0$. So $\Lambda = R \perp P^m$ with P the plane with Gram matrix $\left(\begin{smallmatrix} 0 & \varpi^{-1} \\ \varpi^{-1} & 0 \end{smallmatrix}\right)$ and R unimodular of rank s. The form $h \bmod \varpi$ on $R/\varpi R$ is a non-degenerate symmetric form over $\mathbb{F}_p$; it can be diagonalized, $\langle u\rangle$ depends only on the square class of $u \bmod \varpi$ ($\text{units} \equiv 1 \bmod \varpi$ are norms), and $\langle u, v\rangle \cong \langle 1, uv\rangle$. So $R \cong \langle 1, ..., 1, d\rangle$, and Λ is determined by z, s and $\chi_\varpi(d)$, where χ_ϖ is the norm residue symbol of $\mathbb{K}_\varpi/\mathbb{K}_\varpi^+$ on units, $\chi_\varpi(u) = (u \bmod \varpi/p)$.

Product formula. Let $\delta = \det h \in \mathbb{K}^+$. At the real places $\delta > 0$ is a norm, and at the finite places $v \neq \varpi$, $\chi_v(\delta) = 1$ by the above. By the product formula $\chi_\varpi(\delta) = 1$. Since $\det P = -\nu^{-1}$ and ν is a norm, $\chi_\varpi(\det P) = \chi_\varpi(-1) = (-1/p)$, hence $\chi_\varpi(d) = (-1/p)^m$. If $s > 0$ this determines Λ; if $s = 0$ it requires $(-1/p)^{z/2} = 1$. So all completions are determined by (z, s), and two such lattices lie in one genus. (Their spaces are isometric: they are locally isometric everywhere, so their determinants agree modulo local norms at every place, hence modulo global norms by the Hasse norm theorem for the cyclic extension $\mathbb{K}/\mathbb{K}^+$, and [Shi08, Thm 2.2(i)] applies.) Conversely, if the conditions hold, the local norm classes of the determinants satisfy the product formula, so by global class field theory (the norm residue symbols of an element of $\mathbb{K}^+$ can be prescribed at all places, equal to 1 at almost all, subject only to the product formula) there is $\delta \in \mathbb{K}^+$, totally positive, with these local classes; [Shi08, Thm 2.2(ii)] gives a positive definite Hermitian space with determinant class δ, and a lattice with the prescribed completions exists in it. For $s = z$, $m = 0$ and $\chi_\varpi(d) = 1$, so the discriminant of $U/\varpi U$ is a square. □

The standard mass of [Jür15, §3.3.2] is

$$\mathrm{std}_z = 2^{1-zr} \prod_{j=1}^{z} |L(1-j, \chi^j)| \cdot \begin{cases} 1 & (z \text{ even}) \\ 2^{-1} & (z \text{ odd}) \end{cases}, \quad r = \frac{p-1}{2}, \tag{D.1}$$

where χ is the quadratic character of $\mathbb{K}/\mathbb{K}^+$, χ^j is the trivial character for even j, and $L(1-j, \chi^j)$ is the L-function of $\mathbb{K}^+$. By [Jür15, Satz 3.3.15] it is the mass of the genus $G_z^{\mathbb{K}}$ of [Jür15, §3.3.1], whose local shape at the ramified place is hyperbolic for even z and hyperbolic plus one rank-1 summand for odd z; in our notation this is the genus of type $(z, z \bmod 2)$, whenever that genus exists.

For other s we count pairs. Let U be of type (z, z) and Z of type (z, s), $z - s = 2m$. Then $U \subset Z$ with $Z^* \subset U$ if and only if $W = \varpi Z/\varpi U$ is an m-dimensional totally isotropic subspace of the quadratic space $U/\varpi U$: indeed $h(\varpi^{-1}u, \varpi^{-1}u') = h(u,u')/\nu$ lies in $\varpi^{-1}\mathcal{O}$ if and only if $h(u,u') \equiv 0 \bmod \varpi$, and then $Z^* = \{x \in U : h(x,u) \in \varpi\mathcal{O} \text{ for all } u \in W\}$ has index p^m in U, so $[Z : Z^*] = p^{2m}$ and Z has type (z, s). Conversely every Z contains $\prod_{i=1}^m (p^i + 1)$ such U, one for each Lagrangian subspace of Z/Z^* (Lemma 4.4). By Proposition D.1 all these Z lie in one genus, so with $I(z, m)$ the number of m-dimensional totally isotropic subspaces of $U/\varpi U$, which depends only on z, m and the type of $U/\varpi U$ given by Proposition D.1,

$$\operatorname{mass}(z,s)\cdot\prod_{i=1}^{m}(p^i+1)=\operatorname{mass}(z,z)\cdot I(z,m). \tag{D.2}$$

For a non-degenerate quadratic space of dimension n over $\mathbb{F}_p$, $I(n,m)=\prod_{i=0}^{m-1}\big(p^{2(k-i)}-1\big)/\big(p^{i+1}-1\big)$ if $n=2k+1$, and $I(n,m)=\prod_{i=0}^{m-1}\big(p^{k-i}-\varepsilon\big)\big(p^{k-i-1}+\varepsilon\big)/\big(p^{i+1}-1\big)$ if $n=2k$ and the space has type $\varepsilon=\pm1$. Taking $s=z \bmod 2$ in (D.2) gives $\operatorname{mass}(z,z)$ from (D.1) whenever the genus $(z,z \bmod 2)$ exists, that is, unless $z\equiv 2 \pmod 4$ and $p\equiv 3 \pmod 4$; and then (D.2) gives every $\operatorname{mass}(z,s)$. For $p=5,17,31$ this covers all genera we need. For $p=7$ the only genus we need is $(4,4)$ (Table D1), and (D.1)–(D.2) give $\operatorname{mass}(z,s)$ for $z=1,3,4$. But the triangular system (D.3) for $z=3,4$ needs $\operatorname{mass}^{\emptyset}(2)$, which (D.3) for $z=2$ takes from $\operatorname{mass}(2,2)$, and (D.5) for $(4,4)$ uses $\operatorname{mass}(2,2)$ for the complement of A_6^2; the genus $(2,0)$ is empty. We determine the genus $(2,2)$ directly. Its trace lattices are even, of rank 12 and determinant 7^2. A rootless one would have minimum at least 4 and so centre density at least $7^{-1}\approx 0.143$, more than the bound 0.06279 of [CE03, Table 3] in dimension 12. So every U of type $(2,2)$ has a root, and by (4.2) $U=A_6^k\perp U'$ with $k\geq 1$ and U' rootless of type $(2-k,2-k)$. If $k=1$, the trace lattice of U' is even, of rank 6 and determinant 7, with minimum at least 4, so its centre density is at least $7^{-1/2}\approx 0.378$, more than the bound 0.08084 of [CE03, Table 3] in dimension 6. Hence $k=2$, the genus $(2,2)$ consists of A_6^2 alone, and by (4.2) $\operatorname{mass}(2,2)=1/((2\cdot 7)^2\cdot 2!)=1/392$. The computation [C2, C9] takes $\operatorname{mass}(z,z)$ for even z from the closed form $2^{-zr}\prod_{j=1}^{z}|L(1-j,\chi^j)|\cdot\Big(p^{z/2}+\big(\frac{-1}{p}\big)^{z/2}\Big)$; for $p\geq 5$ it agrees with (D.1)–(D.2) in every case we use in which the genus $(z,0)$ exists, and for $p=7$, $z=2$ it gives $1/392$, the value just proved. For $p=3$ and $z\equiv 2 \pmod 4$ we use instead Theorem D.5 of §D.4, which computes the masses of all genera over the Eisenstein integers from the masses of maximal lattices [GHY01]; for $z=6$ and $z=10$ it gives $\operatorname{mass}(z,z)=13/\ 235{,}146{,}240$ and $8{,}899\ /\ 31{,}345{,}665{,}638{,}400$. As an independent check, these values agree with the mass formula for unimodular Hermitian lattices over $\mathbb{Z}[\zeta_3]$ of [HK89, Thm 5.6] (also quoted in [Jür15, §3.7.1]), and [Jür15, Table 4.8], reproducing Feit's classification, has exactly one rootless class for $z=6$, the Coxeter–Todd lattice, and exactly two for $z=10$, as (D.3) predicts (the misprints in the mass column of that table, at $N=2$ and $N=11$, do not concern $z=6,10$).

D.2 Rootless unimodular lattices

By (4.2) every U of type (z,z) is uniquely $A_{p-1}^k\perp U'$ with U' rootless of type $(z-k,z-k)$, and conversely every such sum is of type (z,z) and so lies in the genus. Since $|\mathrm{U}(U)|=(2p)^k k!\ |\mathrm{U}(U')|$, summing over the genus gives

$$\operatorname{mass}(z,z)=\sum_{k=0}^{z}\frac{\operatorname{mass}^{\emptyset}(z-k)}{(2p)^k k!}, \tag{D.3}$$

where $\operatorname{mass}^{\emptyset}(n)$ is the sum of $1/\ |\mathrm{U}(U')|$ over the rootless U' of type (n,n) ($\operatorname{mass}^{\emptyset}(0)=1$). This triangular system gives $\operatorname{mass}^{\emptyset}(z)$ from the masses of §D.1 [C2]. In particular $\operatorname{mass}^{\emptyset}(z)=0$ for the ranks used in reason 3 of Proposition 3.3, and for $p=31$, $z=1$, $\operatorname{mass}(1,1)=9/62$ and $\operatorname{mass}^{\emptyset}(1)=9/62-1/62=4/31$.

D.3 Rootless masses for $p=5,7,17$

Let Z be an $\mathcal{O}$-lattice of type (z,s) with $z(p-1)\leq 32$, and $p\geq 5$. The roots of Z (vectors of trace norm 2) span a root lattice on which ζ_p acts without fixed vectors.

> **Lemma D.2.** The irreducible components of this root lattice are A_{p-1} (of rank 1 over $\mathcal{O}$) and, for $p=5$ only, E_8 (of rank 2).

Proof. If ζ_p permuted p components cyclically, the sum of the orbit of a root would be a non-zero fixed vector. So each component C is preserved, and $\zeta_p\,|_C$ is an automorphism of order p without fixed vector; its characteristic polynomial is a power of Φ_p, so $p-1$ divides the rank n of C. For $C = A_n$, $\mathrm{Aut}\ C = S_{n+1} \times \{\pm 1\}$, an element of order p is a permutation, and its fixed space in A_n has dimension one less than its number of cycles, so it is a single p-cycle and $n = p-1$. For $C = D_n$ an element of order p is a signed permutation ($|\mathrm{Aut}\ D_4| = 2^7 3^2$ has no element of order $p \geq 5$) which, after conjugation, has all signs $+$ on its p-cycles, and it fixes the sum of the coordinates of a cycle. For E_6, $p-1 \mid 6$ gives $p = 7$, but 7 does not divide $|\mathrm{Aut}\ E_6|$; $p-1 \mid 7$ is impossible for E_7; for E_8, $p-1 \mid 8$ gives $p = 5$, and such elements exist (characteristic polynomial Φ_5^2). □

On $A = A_{p-1}$ every root generates A as an $\mathcal{O}$-module, so its root sublattices are 0 and A, and $c(A) = -1$ in the notation of Proposition C.2; $|\mathrm{U}(A)| = 2p$. In E_8 with an element of order 5 of characteristic polynomial Φ_5^2, the $\mathcal{O}$-submodules generated by roots are 0, twelve copies of A_4, six copies of A_4^2 (the orthogonal complement of A_4 in E_8 is again A_4), and E_8, so $c(E_8) = -(1 - 12 + 6) = 5$, and $|\mathrm{U}(E_8)| = 600$. The proof of Proposition C.2 applies verbatim over $\mathcal{O}$ and gives

$$\mathrm{mass}^{\emptyset}(z,s) = \sum_{a,b} (-1)^a 5^b \frac{A\big(A^a E_8^b\big)}{(2p)^a a! \cdot 600^b b!}, \tag{D.4}$$

where $A(T) = \sum_Z r(T,Z)/\,|\mathrm{U}(Z)|$ over the genus (z,s), $r(T,Z)$ is the number of $\mathcal{O}$-linear isometric embeddings $T \to Z$, and $b = 0$ unless $p = 5$.

Computing $A(T)$. An embedding $\iota : T \to Z$ extends to $T \otimes \mathbb{Q}$, and $\iota^{-1}(Z)$ is an overlattice T_C of T given by an isotropic subspace C of $D(T) = \mathbb{F}_p^a$ [Nik79, Prop. 1.4.1] (all subspaces are $\mathcal{O}$-stable, since $\mathcal{O}$ acts on $D(T)$ through $\mathcal{O}/\varpi = \mathbb{F}_p$). So $A(T) = \sum_C A_{\mathrm{prim}}(T_C)$ with A_{prim} counting primitive embeddings. For a primitive $T' \subset Z$ of rank n and $D(T') = \mathbb{F}_p^{s'}$, the complement $Y = T'^{\perp}$ is an $\mathcal{O}$-lattice of type $(z-n, s-s'+2k)$ (by the argument given in §D.4.4 for $p = 3$, which applies to every odd p) and Z is the gluing of $T' \perp Y$ along an anti-isometry γ from a k-dimensional subspace $\mathcal{H} \subset D(T')$ onto a subspace of $D(Y)$ [Nik79, §1.5, Prop. 1.5.1]; conversely every such gluing is of type (z,s), hence in the genus (Proposition D.1). Counting the data $(\mathcal{H}, \gamma, Y)$ with weights $1/\,|\mathrm{U}(Y)|$ as in Proposition 4.1,

$$A_{\mathrm{prim}}(T') = \sum_{\mathcal{H}} \frac{\mathrm{emb}(\mathcal{H}, D(T')) \cdot \mathrm{emb}(\mathcal{H}^-, D(Y_{\mathcal{H}}))}{|O(\mathcal{H})|} \cdot \mathrm{mass}(z-n, s-s'+2\dim\mathcal{H}), \tag{D.5}$$

where $\mathcal{H}$ runs over the isometry types of (possibly degenerate) quadratic spaces over $\mathbb{F}_p$, $\mathrm{emb}(\mathcal{H}, D)$ is the number of isometric embeddings of $\mathcal{H}$ into D, computed by Witt's theorem, $\mathcal{H}^-$ is $\mathcal{H}$ with the form negated, and $D(Y_{\mathcal{H}})$ is the discriminant form of the genus of Y, determined by Proposition D.1. With the masses of §D.1 this gives every $\mathrm{mass}^{\emptyset}(z,s)$ in exact arithmetic [C9].

Table D1. Rootless masses of the genera (z,s) over $\mathbb{Z}[\zeta_p]$ used in §4.2 and Appendix B.4 [C9, C2].

p, (z,s)	$\mathrm{mass}^{\emptyset}(z,s)$	approx.	used for
5, $(4,4)$	1/ 72,000	1.39×10^{-5}	$5-(4,16)-4$
5, $(5,5)$	1/ 150,000	6.67×10^{-6}	$5-(5,12)-5$
5, $(6,6)$	191/ 7,200,000	2.65×10^{-5}	$5-(6,8)-6$
5, $(6,4)$	4,567 / 1,440,000	3.17×10^{-3}	$5-(6,8)-4$
5, $(7,7)$	10,358,729 / 11,340,000,000	9.13×10^{-4}	$5-(7,4)-3$ (first step)
5, $(7,5)$	16,927 / 28,800	0.588	$5-(7,4)-3$ (second step)
5, $(7,3)$	799,867 / 1,440,000	0.5555	$5-(7,4)-3$

p, (z,s)	$\mathrm{mass}^{\emptyset}(z,s)$	approx.	used for
5, $(8,0)$	168,947 / 138,240,000	1.22×10^{-3}	$5-(8,0)-0$
7, $(4,4)$	439/ 395,136	1.11×10^{-3}	$7-(4,8)-4$
17, $(2,0)$	1/12	0.0833	$17-(2,0)-0$
31, $(1,1)$	4/31	0.129	$31-(1,2)-1$

Checks. For $p=5$, $(2,0)$ and $(4,0)$, (D.4) gives that E_8 and E_8^2 are the only lattices, with masses $1/600$ and $1/\ 720{,}000 = 1/(600^2\cdot 2)$. For $s=z$, (D.4) agrees with (D.3). Three values are compared with the Leech lattice Λ: an element $g\in \mathrm{Co}_0$ with $g^p=1$ and no fixed vector makes Λ a rootless lattice of type $(24/(p-1),0)$, and the genus contains no other rootless lattice since Λ is the only rootless even unimodular lattice of rank 24; so the rootless mass is $\#\{\text{such } g \text{ up to conjugacy, weighted}\} = \sum 1/\ |C_{\mathrm{Co}_0}(g)|$, read from the character table of $2.\mathrm{Co}_1$ in GAP. The values $1/\ 6{,}048{,}000$ ($p=5$, $(6,0)$), $1/\ 35{,}280$ ($p=7$, $(4,0)$) and $1/312$ ($p=13$, $(2,0)$) agree with (D.4) [C9]. Finally, for $17-(2,0)-0$ the value $1/12$ is the sum of $1/\ |\mathrm{U}(Z)|$ over the six Hermitian classes whose trace lattices are the four lattices of [Jür15, Satz 3.6.9] (§4.2).

D.4 The case $p=3$ and the genera of F

What this section does. For $p=3$ the root lattices over $\mathcal{O}=\mathbb{Z}[\omega]$ include D_4, whose discriminant group has a 2-part, so the orthogonal complements of root sublattices of Z leave the genera of §D.1 and acquire a 2-adic level. §D.4.1 describes these genera (N,s,b) and §D.4.2 proves a closed formula for their masses (Theorem D.5), anchored at the maximal lattices of [GHY01]; in particular it gives the masses of the genera (z,z) for $z=6,10$, where the genus $(z,0)$ is empty. §D.4.3 lists the root lattices over $\mathbb{Z}[\omega]$ and §D.4.4 computes the rootless masses as in §D.3; §D.4.5 gives the value for $(13,5)$ and the checks. §D.4.6 treats the $\mathbb{Z}$-lattices F of the genera $(20,6)$ for $p=3$ and $(16,4)$ for $p=5$ by the same inclusion–exclusion over root lattices, with the masses of the complements given by Siegel's mass formula; the new point there is a local formula (Theorem D.10) that expresses the local factor of a complement at a prime where F is unimodular by counting on F/pF.

Throughout §D.4.1–§D.4.5, $p=3$, $\mathcal{O}=\mathbb{Z}[\omega]$ with $\omega=\zeta_3$, $\varpi=1-\omega$ and $\chi=\chi_{-3}$; for $p=3$ the form $\tilde H$ of Lemma 4.4 is H, written h below, and $Z^{\#}=\{x : h(x,Z)\subset \varpi^{-1}\mathcal{O}\}$. The prime 2 is inert in $\mathbb{Q}(\omega)$ and $\mathcal{O}/2=\mathbb{F}_4$. A *root* is a vector of trace norm 2, that is, with $h(x,x)=1$.

D.4.1 Genera with a 2-adic level

An *$\mathcal{O}$-lattice of type* (N,s,b) is a free $\mathcal{O}$-module Λ of rank N with a positive definite Hermitian form h such that $h(\Lambda,\Lambda)\subset\varpi^{-1}\mathcal{O}$, and, with $\Lambda^{\#}=\{y : h(y,\Lambda)\subset\varpi^{-1}\mathcal{O}\}$:

- at 3: $\varpi\Lambda_3^{\#}\subset\Lambda_3$ and $\Lambda_3^{\#}/\Lambda_3\cong\mathbb{F}_3^{s}$;
- at 2: $2\Lambda_2^{\#}\subset\Lambda_2$ and $\Lambda_2^{\#}/\Lambda_2\cong\mathbb{F}_4^{b}$;
- $\Lambda_v^{\#}=\Lambda_v$ at all other finite places.

Type $(N,s,0)$ is type (N,s) of §D.1. As there, the trace form is even. On $D_2(\Lambda)=\Lambda_2^{\#}/\Lambda_2$ the form $\beta_2(x,y)=2h(x,y) \bmod 2\mathcal{O}_2$ is a non-degenerate Hermitian form over $\mathbb{F}_4$ ($\beta_2\left(x,\Lambda_2^{\#}\right)=0$ means $h\left(x,\Lambda_2^{\#}\right)\subset\mathcal{O}_2$, that is $x\in\Lambda_2$); on $D_3(\Lambda)=\Lambda_3^{\#}/\Lambda_3$ the form $\beta_3(x,y)=3h(x,y) \bmod \varpi$ is a non-degenerate symmetric form over $\mathbb{F}_3$. Since the trace form is $(x,y)=2\ \mathrm{Re}\ h(x,y)$, the discriminant form of the trace lattice is $q(x)=2h(x,x) \bmod 2\mathbb{Z}$, and $\beta_3(x,x)=(3/2)q(x) \bmod 3$ is the value $Q(x)$ of the convention of §B.1.1 (Table 3).

Lemma D.3. Let Λ be a lattice over $\mathcal{O}_2 = \mathbb{Z}_2[\omega]$ with a Hermitian form, $2\Lambda^\# \subset \Lambda \subset \Lambda^\#$ and $\Lambda^\#/\Lambda \cong \mathbb{F}_4^b$. Then $\Lambda \cong \langle 1\rangle^{N-b} \perp \langle 2\rangle^b$.

Proof. On $W_2 = \Lambda/2\Lambda^\#$ the form $h \bmod 2$ is a non-degenerate Hermitian form over $\mathbb{F}_4$. If $W_2 \neq 0$, some $h(x,x)$ is a unit: otherwise $h(x,y) + h(y,x) \equiv 0$ for all x, y, and replacing y by ωy gives $h \equiv 0$. Since $\mathbb{Q}_2(\omega)/\mathbb{Q}_2$ is unramified, every unit of $\mathbb{Z}_2$ is a norm of a unit, so $\langle 1\rangle$ splits off. When $W_2 = 0$, $\Lambda = 2\Lambda^\#$ and $h/2$ is unimodular on Λ; the same argument gives $\langle 2\rangle^b$. □

Proposition D.4. The $\mathcal{O}$-lattices of type (N, s, b) form at most one genus. The genus exists if and only if $0 \le s \le N$, $N \equiv s \pmod 2$, $0 \le b \le N$, and $s > 0$ or $(-1)^{N/2+b} = 1$. Write $m = (N-s)/2$. The form β_3 on $D_3(\Lambda)$ has determinant $(-1)^{m+b}$ modulo squares; for even s its type (hyperbolic $+1$, anisotropic part of dimension 2 -1) is $\tau = (-1)^{N/2+b}$.

Proof. At 3, the proof of Proposition D.1 gives $\Lambda_3 \cong R \perp P^m$ with $R \cong \langle 1, \dots, 1, d\rangle$ unimodular of rank s, determined by N, s and $\chi_\varpi(d)$; the form $h \bmod \varpi$ on $R/\varpi R$ is isometric to β_3 (send $x \in R$ to $x/\varpi \in R^\#$; then $3h(x/\varpi, y/\varpi) = h(x,y)$). At 2, Lemma D.3 determines Λ_2 by b. At the other places Λ_v is unimodular, hence unique (proof of Proposition D.1). For the product formula, let $\delta = \det h$. At 2, $\delta \in 2^b\mathbb{Z}_2^\times$, and since $\mathbb{Q}_2(\omega)$ is unramified the norms are the elements of even valuation, so $\chi_2(\delta) = (-1)^b$. The other places $v \neq 3$ contribute 1, and at 3, $\chi_\varpi(\delta) = \chi_\varpi(d) \cdot (-1/3)^m$ as in the proof of Proposition D.1. Hence $\chi_\varpi(d) = (-1)^{m+b}$, which is the determinant of β_3 modulo squares; for $s = 2l$, $\tau = (-1)^l \det \beta_3 = (-1)^{l+m+b} = (-1)^{N/2+b}$. For $s = 0$ this is the condition $(-1)^{N/2+b} = 1$. Uniqueness of the genus and existence follow as in Proposition D.1, with [Shi08, Thm 2.2]. □

D.4.2 The masses of the genera (N, s, b)

For $m \ge 0$ let $u_m = \prod_{i=1}^m (2^i - (-1)^i)$ and $g_m = \prod_{i=1}^m (9^i - 1)$. Put

$$\lambda_2(N,b) = \frac{u_N}{u_{N-b}u_b}, \quad \lambda_3(N,s) = \frac{2g_{\lfloor N/2\rfloor}}{g_m o_s}, \quad o_0 = 1, \quad o_{2l+1} = 2g_l, \quad o_{2l} = 2\big(3^l - \tau\big)g_{l-1} \quad (l \ge 1),$$

with $m = (N-s)/2$ and τ as in Proposition D.4. Here $|\mathrm{U}_b(\mathbb{F}_4)| = 2^{b(b-1)/2}u_b$, and o_s is the order of the orthogonal group of (D_3, β_3) divided by its power of 3.

Theorem D.5. Let $N \ge 1$. If the genus of type (N, s, b) exists, its mass is

$$\mathrm{mass}(N,s,b) = 2^{-N} \prod_{j=1}^{N} |L(1-j, \chi^j)| \cdot \lambda_3(N,s) \cdot \lambda_2(N,b). \tag{D.6}$$

At rank zero we instead set $\mathrm{mass}(0,0,0) = 1$, the mass of the zero lattice, and give non-existing types mass zero. The empty-product expression (D.6) is not applied at rank zero. These conventions also apply to the zero-rank complements in (D.9).

For $b = 0$ and $s = N \bmod 2$ this is (D.1): $\lambda_3 = 2$ for $s = 0$ and $\lambda_3 = 1$ for $s = 1$. Theorem D.5 covers every existing type (N, s, b) of positive rank, including the genera (z, z) with $z \equiv 2 \pmod 4$, where it is anchored at the maximal lattices with $s = 2$, $\tau = -1$; for instance $\mathrm{mass}(6,6,0) = 13/\ 235{,}146{,}240$ and $\mathrm{mass}(10,10,0) = 8{,}899\ /\ 31{,}345{,}665{,}638{,}400$. Being the true masses, the values (D.6) with $b = 0$ satisfy (D.2).

The proof has two steps. The genera with $b \le 1$ and $s \le 2$ that contain maximal lattices are given by [GHY01]; the others are reached from them by counting pairs of lattices of index 3 (Lemma D.6) and of index 4 (Lemma D.7).

The maximal lattices. [GHY01] computes the mass of a genus of lattices that are maximal among lattices with $h(x,x) \in \mathbb{Z}$ (for $k = \mathbb{Q}$, $d = 1$, absolute rank N):

$$\mathrm{Mass}(\Lambda) = 2^{-N}\ |L(M)| \cdot \tau(G) \cdot \prod_{v \in S} \lambda_v, \quad L(M) = \prod_{j=1}^{N} L(1-j, \chi^j), \quad \tau(G) = 2, \tag{D.7}$$

[GHY01, (2.12), Prop. 2.13, (4.2), (4.3), Prop. 4.4, 4.5], with $\lambda_v = q^{-N(\overline{H}^0)}\ |\overline{H}^0(\mathbb{F}_q)|\ / q^{-N(\overline{G})}\ |\overline{G}(\mathbb{F}_q)|$ and S the set of places where the reduction $\overline{G}$ of the stabilizer differs from that of the quasi-split model. The mass is positive while $L(M) < 0$ for $N \equiv 2 \pmod 4$, so the product of L-values is read in absolute value. The local factors needed are [GHY01, Tables 1, 2]:

Table D2. The maximal lattices and their local factors in [GHY01]. The last column is the factor $\lambda_3(N,s)$, $\lambda_2(N,b)$ of (D.6).

local type	$\overline{G}$	λ_v of [GHY01]	factor in (D.6)
$p=2$, $b=0$	U_N	1 ($2 \notin S$)	$\lambda_2(N,0) = 1$
$p=2$, $b=1$, N odd	$\mathrm{U}_{N-1} \times \mathrm{U}_1$	$(2^N+1)/3$	$\lambda_2(N,1)$
$p=2$, $b=1$, N even	$\mathrm{U}_{N-1} \times \mathrm{U}_1$	$(2^N-1)/3$	$\lambda_2(N,1)$
$p=3$, N odd, $s=1$	$\mathbb{Z}/2 \times \mathrm{Sp}_{N-1}$	$1/2$	$\lambda_3 = 2 \cdot 1/2 = 1$
$p=3$, N even, $s=0$, $\tau=+1$	Sp_N	1 ($3 \notin S$)	$\lambda_3 = 2$
$p=3$, N even, $s=2$, $\tau=-1$	$\mathrm{Sp}_{N-2} \times O_2^-$	$(3^N-1)/8$	$\lambda_3 = (3^N-1)/4$

The factor 2 in λ_3 is $\tau(G)$. To identify these lattices with our types it suffices, by Lemma D.3 and Proposition D.4, to compare the numerical invariants. The lattice $\Delta = A_E e \oplus \mathfrak{D}^{-1} f$ of [GHY01, (3.2)] is the plane P at 3 (after replacing f by $-f$) and is unimodular at 2. The lattice A_D at 3 with $\alpha = -1$ is $\mathcal{O} \oplus \mathcal{O}z$ with the form $\langle 1, 1\rangle$: the unit parts of $N(x_1)$ and $N(x_2)$ are $1 \bmod 3$, so their sum does not cancel the denominator; $R/\varpi R$ is anisotropic, so $s = 2$, $\tau = -1$. At 2 with $\alpha = 2$ it is $\mathcal{O}_2 \oplus \mathcal{O}_2 z$ with the form $\langle 1, -2\rangle \cong \langle 1, 2\rangle$, so $b = 1$. Conversely the lattices with $b \ge 2$ ($\langle 2, 2\rangle$ lies in a unimodular lattice), $s \ge 3$, or $s = 2$, $\tau = +1$ (an isotropic vector of D_3 gives an overlattice) are not maximal; [GHY01] treats only these extremal vertices (p. 102). In each row of Table D2 the last column equals the corresponding factor of (D.6), so (D.6) holds for the genera (N, s_0, b_0) with $b_0 \in \{0, 1\}$ and $s_0 = 1$ (N odd), $s_0 = 0$ (N even, $\tau = +1$), $s_0 = 2$ (N even, $\tau = -1$).

For a non-degenerate quadratic space over $\mathbb{F}_3$ of dimension n (and type τ if $n = 2l$) let $I_O(n)$ be its number of isotropic lines, $(3^{n-1}-1)/2$ for odd n and $\big(3^l - \tau\big)\big(3^{l-1} + \tau\big)/2$ for $n = 2l$; for a non-degenerate Hermitian space over $\mathbb{F}_4$ of dimension n let $I_{\mathrm{U}}(n) = (2^n - (-1)^n)(2^{n-1} - (-1)^{n-1})/3$ be its number of isotropic lines.

Lemma D.6. If $N - s \ge 2$, then $\mathrm{mass}(N, s+2, b) \cdot I_O(s+2) = \mathrm{mass}(N, s, b) \cdot (3^{N-s} - 1)/2$, where $I_O(s+2)$ is taken for the form of type $(N, s+2, b)$.

Proof. Count the pairs $\Lambda' \subset \Lambda$ of $\mathcal{O}$-lattices of index 3 with Λ of type (N, s, b) and Λ' of type $(N, s+2, b)$, weighted by $1/\ |\mathrm{U}(\Lambda)|$ and $1/\ |\mathrm{U}(\Lambda')|$; by the orbit–stabilizer relation both weighted counts are equal. Only the completions at 3 change.

Sublattices. A sublattice of index 3 is the kernel of a non-zero linear form f on $\Lambda/\varpi\Lambda$. If Λ' satisfies $\varpi\Lambda'^{\#} \subset \Lambda'$, then $\varpi\Lambda^{\#} \subset \varpi\Lambda'^{\#} \subset \Lambda'$, so f vanishes on $\varpi\Lambda^{\#}/\varpi\Lambda$ and is a form on $W = \Lambda/\varpi\Lambda^{\#}$. Conversely, for such f, non-degeneracy of a on W (Lemma 4.4) gives $f = a(\cdot, v)$ with $v \in \Lambda \setminus \varpi\Lambda^{\#}$, and $\Lambda' = \{x \in \Lambda : h(x,v) \in \mathcal{O}\}$. Then $v/\varpi \in \Lambda'^{\#} \setminus \Lambda^{\#}$, comparing indices $\Lambda'^{\#} = \Lambda^{\#} + \mathbb{Z}_3 v/\varpi$, and $\varpi\Lambda'^{\#} = \varpi\Lambda^{\#} + \mathbb{Z}_3 v \subset \Lambda'$ because $h(v,v) \in \mathbb{Z}_3$; so $D_3(\Lambda') \cong \mathbb{F}_3^{s+2}$. The forms f correspond to the hyperplanes of W, of which there are $(3^{N-s}-1)/2$.

Overlattices. If $\Lambda' \subset \Lambda$, then $\Lambda \subset \Lambda^{\#} \subset \Lambda'^{\#}$ and $I = \Lambda/\Lambda'$ is a line in $D' = D_3(\Lambda')$. For $x \in \Lambda'^{\#}$, $\Lambda' + \mathbb{Z}_3 x$ is an $\mathcal{O}$-module since $(\omega - 1)x \in \varpi\Lambda'^{\#} \subset \Lambda'$, and it is integral if and only if $h(x,x) \in \mathbb{Z}_3$, that is $\beta_3(x,x) = 0$. Then $\varpi\Lambda^{\#} \subset \Lambda$ holds automatically, and $D_3(\Lambda) = I^{\perp}/I$ has dimension s. So the overlattices correspond to the isotropic lines of D'.

Types. $D' \cong D_3(\Lambda) \perp$ (hyperbolic plane), so for even s the type τ is unchanged; b is unchanged. By Proposition D.4 the lattices obtained lie in the genera (N, s, b) and $(N, s+2, b)$. □

Lemma D.7. If $N - b \geq 2$, then $\mathrm{mass}(N, s, b+2) \cdot I_{\mathrm{U}}(b+2) = \mathrm{mass}(N, s, b) \cdot I_{\mathrm{U}}(N-b)$.

Proof. Count pairs of index 4 at 2 as in Lemma D.6. A sublattice Λ' of index 4 is the kernel of a form $f = h(\cdot, v) \bmod 2$ on $W_2 = \Lambda/2\Lambda^{\#}$ ($v \in \Lambda \setminus 2\Lambda^{\#}$); the condition $2\Lambda'^{\#} = 2\Lambda^{\#} + \mathcal{O}_2 v \subset \Lambda'$ holds if and only if $h(v,v) \in 2\mathbb{Z}_2$, that is the line of v is isotropic in W_2, which has dimension $N - b$: there are $I_{\mathrm{U}}(N-b)$ of them. The overlattices of Λ' of index 4 with $2\Lambda^{\#} \subset \Lambda$ correspond to the isotropic lines of $D_2(\Lambda')$, of dimension $b+2$. By Lemma D.3 and Proposition D.4 the lattices lie in the stated genera; s and τ are unchanged since the Hermitian space is the same. □

Proof. (of Theorem D.5) The right side of (D.6) satisfies the relations of Lemmas D.6 and D.7: $\lambda_3(N, s+2)/\lambda_3(N,s) = (g_m o_s)/(g_{m-1} o_{s+2})$, where $g_m/g_{m-1} = 9^m - 1 = 3^{N-s} - 1$ and o_{s+2}/o_s equals $9^{l+1} - 1$ for $s = 2l+1$, $\big(3^{l+1} - \tau\big)\big(3^l + \tau\big)$ for $s = 2l \geq 2$ and 4 for $s = 0$ ($\tau = +1$), in each case $2I_O(s+2)$; and $\lambda_2(N, b+2)/\lambda_2(N,b) = u_{N-b}u_b/(u_{N-b-2}u_{b+2}) = I_{\mathrm{U}}(N-b)/I_{\mathrm{U}}(b+2)$. It agrees with the maximal lattices (Table D2). Every existing genus is reached from one of these: from (N, s_0, b_0) with $b_0 = b \bmod 2$, Lemma D.6 reaches every $s \leq N$ with $N - s$ even (each step needs $N - s \geq 2$, and the case $s = 2$, $\tau = +1$ is one step from $s = 0$ with $I_O(2) = 2$; the case $s = 0$, $\tau = -1$ is empty, consistent with $I_O(2) = 0$ for $\tau = -1$), and Lemma D.7 then reaches every $b \leq N$ with $b \equiv b_0$ (each step needs $N - b \geq 2$). In every step the genus on the far side is non-empty, since $I_O(n) > 0$ for $n \geq 3$ and $I_{\mathrm{U}}(n) > 0$ for $n \geq 2$. The genus (N, s_0, b_0) itself is non-empty: take maximal lattices at every place in $V = \langle 1 \rangle^{N-1} \perp \langle c \rangle$ with $c = 1$ for $b_0 = 0$ and $c = 2$ for $b_0 = 1$. □

D.4.3 Root lattices over $\mathbb{Z}[\omega]$

Lemma D.8. The irreducible components of the root lattice of an $\mathcal{O}$-lattice are A_2, D_4, E_6, E_8, each with a unique $\mathcal{O}$-structure, with the data of Table D3.

Proof. For a root r, $(r, \omega r) = \mathrm{Tr}\,\overline{\omega} = -1$, so r and ωr lie in the same component and ω preserves each component Y, acting without fixed vectors ($1 + \omega + \omega^2 = 0$). For A_n, $n \geq 2$, $\mathrm{Aut}\, Y = W \times \{\pm 1\}$ and an element of order 3 is a permutation in S_{n+1} whose fixed space has dimension one less than the number of cycles; so it is one 3-cycle and $n = 2$. A_1, E_7 and D_n with n odd have odd rank. For D_n, $n \geq 5$, $\mathrm{Aut}\, Y = W(B_n)$, and an element of order 3 is a signed permutation whose cycles have sign product $+1$, so it fixes a vector on each cycle. The four remaining lattices carry $\mathcal{O}$-structures (explicitly inside $E_8 = \{x \in \mathcal{O}^4 : x \bmod \varpi \in \text{tetracode}\}$ with $h(x,y) = \frac{1}{3}\sum x_i \overline{y_i}$). Each such structure

is an $\mathcal{O}$-lattice of the type in Table D3, and by Theorem D.5 the masses of these genera are $1/6$, $1/72$, $1/\ 1{,}296$, $1/\ 155{,}520$, equal to $1/\ |\mathrm{U}(Y)|$; so each genus consists of one class and the structure is unique. □

Table D3. Irreducible root lattices over $\mathbb{Z}[\omega]$ and their root sublattices (as $\mathcal{O}$-modules), counted from explicit roots [C16].

Y	rank	$\lvert\mathrm{U}(Y)\rvert$	type (N,s,b)	$c(Y)$	root sublattices
A_2	1	6	$(1,1,0)$	-1	$\emptyset$: 1, A_2: 1
D_4	2	72	$(2,0,1)$	3	$\emptyset$: 1, A_2: 4, D_4: 1
E_6	3	1,296	$(3,1,0)$	-24	$\emptyset$: 1, A_2: 12, A_2^2: 12, D_4: 9, A_2^3: 4, E_6: 1
E_8	4	155,520	$(4,0,0)$	324	$\emptyset$: 1, A_2: 40, A_2^2: 240, D_4: 90, A_2^3: 160, D_4A_2: 360, E_6: 40, A_2^4: 40, E_6A_2: 40, D_4^2: 45, E_8: 1

The discriminant modules are $D_3(A_2)\cong\mathbb{F}_3$ with $\beta_3=\langle 1\rangle$, $D_3(E_6)\cong\mathbb{F}_3$ with $\beta_3=\langle -1\rangle$, and $D_2(D_4)\cong\mathbb{F}_4$; so for $T=A_2^aD_4^bE_6^cE_8^d$, $D_3(T)$ has dimension $a+c$ and determinant $(-1)^c$, and $D_2(T)\cong\mathbb{F}_4^b$. The proof of Proposition C.2 applies verbatim over $\mathcal{O}$ and gives

$$\mathrm{mass}^{\emptyset}(z,s)=\sum_{a,b,c,d}\frac{(-1)^a3^b(-24)^c324^d}{6^a a!\cdot 72^b b!\cdot 1{,}296^c c!\cdot 155{,}520^d d!}A\big(A_2^aD_4^bE_6^cE_8^d\big),\tag{D.8}$$

the sum over $a+2b+3c+4d\le z$ (194 terms for $z=13$), with $A(T)=\sum_Z r(T,Z)/\ |\mathrm{U}(Z)|$ over the genus (z,s).

D.4.4 The terms $A(T)$

As in §D.3, $A(T)=\sum_C A_{\mathrm{prim}}(T_C)$, now with $C=C_2\oplus C_3$, C_3 a totally singular subspace of $(D_3(T),\beta_3)$ and C_2 a totally isotropic subspace of $(D_2(T),\beta_2)$; both are automatically $\mathcal{O}$-stable, and every such C gives an integral overlattice. For a primitive $T'\subset Z$ of rank n with $D_3(T')\cong\mathbb{F}_3^{s'}$ and $D_2(T')\cong\mathbb{F}_4^{b'}$ let $Y=T'^{\perp}$. Then:

- $D_3(Y)$ is killed by ϖ: for $y\in Y^{\#}$ choose $w=y+v\in Z^{\#}$ with $v\in T'^{\#}$; then $\varpi w\in Z$ and $\varpi v\in T'\subset Z$, so $\varpi y\in Z\cap\mathbb{Q}Y=Y$;
- at 2, Z_2 is unimodular, so Z is glued along an anti-isometry of all of $D_2(T')$ onto all of $D_2(Y)$, and Y has type $(z-n,\,s-s'+2\dim\mathcal{H},\,b')$ for the subspace $\mathcal{H}\subset D_3(T')$ glued at 3;
- conversely every such gluing is an $\mathcal{O}$-lattice of type $(z,s,0)$ (subspaces of $\mathbb{F}_3$- and $\mathbb{F}_4$-spaces are $\mathcal{O}$-stable since $\omega\equiv 1 \bmod\varpi$ and $\mathcal{O}/2=\mathbb{F}_4$), hence lies in the genus.

The anti-isometries $D_2(T')\to D_2(Y)$ form a torsor under $\mathrm{U}(D_2(T'))\cong\mathrm{U}_{b'}(\mathbb{F}_4)$. Counting as in (D.5),

$$A_{\mathrm{prim}}(T')=|\mathrm{U}_{b'}(\mathbb{F}_4)|\sum_{\mathcal{H}}\frac{\mathrm{emb}(\mathcal{H},D_3(T'))\cdot\mathrm{emb}(\mathcal{H}^-,D_3(Y_{\mathcal{H}}))}{|O(\mathcal{H})|}\cdot\mathrm{mass}(z-n,s-s'+2\dim\mathcal{H},b'),\tag{D.9}$$

with the masses of Theorem D.5 and $D_3(Y_{\mathcal{H}})$ the form of that genus (Proposition D.4). The numbers of totally singular subspaces of $D_3(T)$ and of totally isotropic subspaces of $D_2(T)$ and the embedding numbers in (D.9) are given by Witt's theorem and the orders of the finite orthogonal and unitary groups; all terms are exact rationals [C16].

D.4.5 The genus $(13,5)$ and checks

Theorem D.9. $\mathrm{mass}^{\emptyset}(13,5)=27{,}547{,}211\ /\ 6{,}635{,}520\approx 4.1515$.

The mass of the genus $(13,5)$ is $\mathrm{mass}(13,5,0)\approx 6.9221$, so the rootless part is 0.59974 of it. Of the 194 root types, 51 have non-zero mass (the masses of the classes with a given root system are obtained from

the $A(T)$ by the triangular inversion of (D.8) over all root systems), and none is negative [C16]. The 105 classes of type $(13,5)$ listed in Appendix B.4 are pairwise non-isometric and the sum of $1/\,|\mathrm{U}(Z)|$ over them equals Theorem D.9 [C16]; as for Proposition 4.6, this shows that the list is complete. For type $3-(13,6)-5$, this component mass also determines the gluing-pair mass: $F=\sqrt{3}E_6^{\#}$ and $\mathrm{Aut}\ F$ maps onto $O(q_F)$.

Table D4. Checks of (D.6)–(D.9), in exact arithmetic [C16]. They are consistency checks; the values rest on the proofs above.

check	result
(D.6) with $b=0$ against (D.1)–(D.2), $N\le 20$	equal
$\mathrm{mass}^{\emptyset}(z,z)$ by (D.8) against (D.3), $z\le 16$	equal ($1/$ 39,191,040 for $z=6$, $109/$ 451,480,780,800 for $z=10$)
$A(T)$ against direct counts inside A_2, A_2^2, A_2^3, E_6, E_6A_2, E_8	44 values equal
$(9,5)$, $(9,7)$, $(10,6)$, $(10,8)$, $(11,9)$, $(12,8)$, $(12,10)$, $(13,13)$: sum over the complete lists of Appendix B.4	equal (e.g. 65,919,403,483 / 2,612,138,803,200 for $(12,8)$, 51 classes)
$(12,0)$: the 5 root systems with non-zero mass	$\emptyset$: $1/\,\lvert 6.\mathrm{Suz}\rvert$; E_8^3, E_6^4, D_4^6, A_2^{12}: $1/\,\lvert\mathrm{U}\rvert$ of the lattices E_8^3, E_6^4, D_4^6, A_2^{12}
$(16,0)$: rootless mass 373,348,469 / 360,972,288,000	equal to the sum over the eleven lattices of [Hen13, Table 4]
masses of the classes with a given root system, all (z,s) with $z\le 16$	none negative

For $(12,0)$ the only rootless lattice is the Leech lattice with an automorphism g of order 3 without fixed vectors; in $2.\mathrm{Co}_1$ these form the single class $3a$ (character value -12 on the 24-dimensional representation), with centralizer of order $2{,}690{,}072{,}985{,}600 = 6\,|\mathrm{Suz}|$ (character table library of GAP), and (D.8) gives exactly $1/\,2{,}690{,}072{,}985{,}600$. A random walk on 2-neighbours of the genus $(13,5)$, which does not use (D.8), gives the rootless fraction 0.5975 ± 0.0047 against 0.59974; this is a statistical check only.

D.4.6 The p-elementary $\mathbb{Z}$-lattices F

Let $p_0\in\{3,5\}$, and let F be an even $\mathbb{Z}$-lattice of rank d with $F^{\#}/F\cong\mathbb{F}_{p_0}^{s}$; these lattices form one genus [Nik79, Cor. 1.9.4]. We need the rootless mass $\mu_F^{\emptyset}(d,s)=\sum_{\min F\ge 4} 1/\,|\mathrm{Aut}\ F|$ for $(p_0;d,s)=(3;20,6)$ and $(5;16,4)$.

Inclusion–exclusion. With the coefficients c of King's inclusion–exclusion over the root lattices of type ADE [Kin03, Prop. 1] (Proposition C.2 over $\mathbb{Z}$),

$$\mu_F^{\emptyset}(d,s)=\sum_T c(T)\frac{A(T)}{|\mathrm{Aut}\ T|},\quad A(T)=\sum_F \frac{r(T,F)}{|\mathrm{Aut}\ F|}, \tag{D.10}$$

the sum over the 10,632 root lattices T of rank at most 20 (for $d=20$). As before $A(T)=\sum_C A_{\mathrm{prim}}(T_C)$ over the isotropic subgroups C of the discriminant form of T, and for a primitive $T'\subset F$ with complement Y of rank $N=d-\mathrm{rank}\ T'$, F is glued from $T'\perp Y$ [Nik79, §1.5, Prop. 1.5.1]. At a prime $p\ne p_0$, F_p is unimodular, so the gluing at p is an anti-isometry of all of $D_p(T')$ onto all of $D_p(Y)$, and Y_p is determined by T'_p [Nik79, Cor. 1.9.3]; at p_0 the gluing is along a subgroup $\mathcal{H}\subset D_{p_0}(T')$ as in (D.5). By Siegel's mass formula in the form [CS88, (2), (3)], the mass of the genus of Y is $c_\infty(N)\prod_p 2m_p(Y)$ with $c_\infty(N)=2\pi^{-N(N+1)/4}\prod_{j=1}^{N}\Gamma(j/2)$ and m_p depending only on Y_p. Since both the gluing data and the local factors are local, the sum over C and over the genera of Y factors over the primes; each choice

of local data comes from one global genus, since the rational complement exists globally by the Hasse–Minkowski theorem and Witt cancellation:

$$A(T) = c_\infty(N) \prod_p \Psi_p(T), \quad \Psi_p(T) = \sum_{C_p} \sum_{(\mathcal{H},\gamma) \text{ at } p} 2m_p\big(Y_{C_p,\mathcal{H}}\big), \tag{D.11}$$

where for $p \notin S = \{2, p_0\} \cup \{p \mid \det T\}$ the only term is $C_p = 0$ with Y_p unimodular, and the product of these factors with $c_\infty(N)$ is the standard mass of [CS88, (6)–(8)] (written there with ζ_D; we use the primitive $L\big(s, \chi_{D_0}\big)$ and remove the factors at S, which gives the same value). For $N = 1$ the leading 2 of [CS88, (2)] is replaced by 1 [CS88, §6], and $N = 0$ is counted directly by overlattices.

At $p = p_0$, Ψ_p is computed from [CS88, (3), (5), Table 2], since Y_p is a Jordan sum of components of scales 1, p and (when T has a component A_8 or A_{17}) p^2. At $p \neq p_0$ we avoid the Jordan decomposition of Y_p, which at $p = 2$ involves the type I and bound components of [CS88, Table 1], by the following local formula.

Theorem D.10. Let L_p be a unimodular $\mathbb{Z}_p$-lattice of rank d (even if $p = 2$), $T' \subset L_p$ a primitive sublattice of rank n with orthogonal complement Y of rank $N \geq 1$, and $v = v_p(\det T')$. With m_p as in [CS88, §12],

$$2m_p(Y) \cdot \big|O\big(D_p(T')\big)\big| = p^{v(N-1)/2} \cdot 2m_p\big(L_p\big) \cdot \alpha_{\mathrm{prim}}\big(T', L_p\big), \tag{D.12}$$

where, with $Q(x) = (x,x)/2$,

$$\alpha_{\mathrm{prim}}\big(T', L_p\big) = \varepsilon_p \cdot p^{-(dn - n(n+1)/2)} \cdot \#\big\{\text{isometric injections } (T'/pT', Q) \to \big(L_p/pL_p, Q\big)\big\},$$

with $\varepsilon_p = 1$ for odd p and $\varepsilon_p = 2^n$ for $p = 2$.

Strategy of the proof. Both sides of (D.12) are local volumes. The primitive embeddings of T' into L_p form finitely many orbits of $O\big(L_p\big)$, one for each anti-isometry $D_p(T') \to D_p(Y)$ modulo the image of $O(Y)$. Computing the volume of the set of primitive embeddings in two ways, by these orbits and by counting solutions modulo p and lifting them, gives (D.12).

Proof. Measures. For a symmetric matrix S of size k let $\omega_S = \mathrm{d}X / \mathrm{d}S$ be the measure on $O(S)$ given by the fibres of $X \mapsto X^t S X$, with $\mathrm{d}S$ normalized by $\mathrm{vol}\big(\mathrm{Sym}_k\big(\mathbb{Z}_p\big)\big) = 1$. By the definition of the density, $\mathrm{vol}_{\omega_S} O(\Lambda) = \alpha_p(\Lambda)$, and [CS88, §12] defines $m_p(\Lambda) = p^{v_p(\det \Lambda)(k+1)/2} / \alpha_p(\Lambda)$. If $S' = C^t S C$ then $\mathrm{d}S' = |\det C|^{k+1}\, \mathrm{d}S$, so $\omega^0 = |\det S|^{(k+1)/2}\, \omega_S$ is independent of the basis and $m_p(\Lambda) = 1/\mathrm{vol}_{\omega^0} O(\Lambda)$.

Decomposition. Let $X = \{A : A^t S A = S_{T'}\}$ with $\omega_X = \mathrm{d}A / \mathrm{d}S_{T'}$ and $G_Y = O\big(Y \otimes \mathbb{Q}_p\big)$. In a basis adapted to $S_0 = S_{T'} \perp S_Y$ write $g = (A, B)$; near $A = (I, 0)^t$, with $B = (B_1, B_2)^t$, $A^t S_0 B = S_{T'} B_1$, so $\mathrm{d}(A^t S_0 B) = |\det S_{T'}|^N\, \mathrm{d}B_1$ and $\omega_{S_0} = \omega_{X,S_0} \wedge |\det S_{T'}|^{-N}\, \omega_{S_Y}$. Returning to a basis of L_p, $S_0 = C^t S C$ with $|\det C| = |\det S_{T'}| = |\det S_Y|$ (as L_p is unimodular and T' primitive), $\omega_S = |\det C|^{d+1}\, \omega_{S_0}$ and $\omega_{X,S} = |\det C|^n\, \omega_{X,S_0}$. Collecting the factors, $\omega^0_{O(L)} = |\det S_{T'}|^{(1-N)/2}\, \omega_X \wedge \omega^0_{G_Y}$.

Orbits. Let $X_{\mathrm{prim}} \subset X$ be the primitive embeddings. The complement of each is isometric to Y, since it has the same rank, determinant class and discriminant form, which determine a p-adic (even, if $p = 2$) lattice [Nik79, Cor. 1.9.3]. The $O\big(L_p\big)$-orbits on X_{prim} correspond to the classes of anti-isometries $D_p(T') \to D_p(Y)$ modulo the image of $O(Y)$, and the stabilizer of an embedding is the kernel of $O(Y) \to O\big(D_p(Y)\big)$. Hence

$$\mathrm{vol}_{\omega_X}\big(X_{\mathrm{prim}}\big) = |\det S_{T'}|^{(N-1)/2}\, \big|O\big(D_p(T')\big)\big|\, \frac{\mathrm{vol}^0 O\big(L_p\big)}{\mathrm{vol}^0 O(Y)} = p^{-v(N-1)/2}\, \big|O\big(D_p(T')\big)\big|\, \frac{m_p(Y)}{m_p\big(L_p\big)}.$$

The left side is the primitive density.

Lifting. For odd p, $B \mapsto A^tSB + B^tSA$ maps onto all symmetric matrices when A is primitive and S unimodular, and Hensel's lemma gives the count modulo p. For $p = 2$ the image is the set Sym^e of even symmetric matrices, whose volume for $\mathrm{d}S$ is 2^{-n}; measuring with $\mathrm{vol}(\mathrm{Sym}^e) = 1$ instead multiplies by 2^n. In that normalization the solutions are counted modulo 2^r (diagonal modulo 2^{r+1}); for $r = 1$ they are exactly the injections preserving Q, and since $B \mapsto A^tSB + B^tSA$ maps onto Sym^e, each lifts by the same factor $2^{dn-n(n+1)/2}$ at every level. □

For a unimodular $\mathbb{Z}_p$-lattice L_p of rank d (even if $p = 2$) put $\varepsilon = \big((-1)^{d/2} \det L_p/p\big)$ (Kronecker symbol) when d is even; for $p = 2$, $\varepsilon = 1$ exactly when L_2 is an orthogonal sum of hyperbolic planes. Put

$$\hat{s}(p,d,\varepsilon) = \prod_{i=1}^{\lceil d/2\rceil - 1} \big(1 - p^{-2i}\big)^{-1} \cdot \begin{cases} \big(1 - \varepsilon p^{-d/2}\big)^{-1} & (d \text{ even}) \\ 1 & (d \text{ odd}) \end{cases},$$

so that $\hat{s}(p,d,\varepsilon) = 2p^{\dim \mathrm{GO}_d} / \,|\mathrm{GO}_d^{\varepsilon}(\mathbb{F}_p)|$ for odd p, with $\dim \mathrm{GO}_d = d(d-1)/2$; this is twice the standard p-mass of [CS88, (6)–(8)]. Formally, for $n = d$ (so $T' = L_p$, outside the hypothesis $N \geq 1$) the formula gives $2m_p\big(L_p\big) = \hat{s}(p,d,\varepsilon)$ for odd p and $\hat{s}(2,d,\varepsilon)/2^d$ for $p = 2$; and for $N = 1$, $2m_p(\langle a\rangle) = 1$ (p odd) or $1/2$ ($p = 2$), in agreement with [CS88, §6]. The numbers of isometric injections are given by Witt's theorem in terms of the radical of T'/pT' and the type of its non-degenerate part; for $p = 2$ the invariants are the dimension of the radical, whether Q vanishes on it, and if so the Arf invariant of the non-degenerate part.

The sums over C at $p = 2$. As in §C.6, the distribution of the isotropic subgroups $C_2 \subset D_2(T)$ by the invariants entering (D.12) is computed by adding the irreducible components of T one at a time. Write $D_2(T) = E \perp C_{\mathrm{cyc}}$, where E is 2-elementary and C_{cyc} is a sum of cyclic blocks $\mathbb{Z}/2^e$, $e = 2,3,4$, each with $q = u/2^e$ (u odd, modulo 2^{e+1}). The 2-elementary part is recorded by (ρ,δ,σ): $\rho = \dim_{\mathbb{F}_2} E$, δ as in §2.3, and $\sigma \in \mathbb{Z}/8\mathbb{Z}$ its Brown invariant, additive under orthogonal sums, with $\sigma = 1,7,0,4$ for $\langle 1/2\rangle$, $\langle 3/2\rangle$, the hyperbolic plane $q(x,y) = xy$ and the anisotropic plane $q(x,y) = x^2 + xy + y^2$. For $\delta = 0$, $\sigma \in \{0,4\}$ is the type ($+1$ or -1) of the quadratic space; for $\delta = 1$, $E \cong \langle 1/2\rangle^a \perp \langle 3/2\rangle^b$ with $a + b = \rho$ and $a - b \equiv \sigma \pmod 8$. The isotropic subgroups of E are counted in closed form ($\delta = 0$: totally singular subspaces of a quadratic space; $\delta = 1$: split by the subspace D_0 where q is integral and the characteristic vector c); the cyclic blocks are then added one at a time by a dynamic programme on the graph of a map from a subgroup of the new block, counting images by orbits of the block automorphisms. The state of the programme is the pair $\big((\rho,\delta,\sigma), C_{\mathrm{cyc}}\big)$ of the residual form $C^\perp/C$, with integer weights indexed by $\log_2|C|$. At the end only three invariants of $T' = T_C$ enter (D.12) at $p = 2$: the number ρ' of generators of $D_2(T')$ (the dimension of the radical of $(T'/2T', Q)$), $\delta' = \delta$ (whether Q vanishes on that radical), and, if $\delta' = 0$, the type $\eta = ((w/2))s_E \prod \varepsilon(b)$ of the non-degenerate part, where w is the odd part of $\det T$, $s_E = \pm 1$ the type of E, $\varepsilon(b) = ((u/2))$ for a cyclic block and ± 1 for a hyperbolic or anisotropic plane, and $((\cdot\,/2))$ is the Kronecker symbol. The programme agrees with brute-force enumeration for 42 small discriminant forms and for the 515 forms with $|D_2| \leq 2^9$ that occur [C17].

The 2-adic finite transition. The sum over primitive closures uses the same graph decomposition as in §C.6, now for finite quadratic modules over $\mathbb{Z}_2$. For an added cyclic factor, enumerate its actual subgroups and every possible image of a fixed generator in the preceding quotient. Require the annihilator relation and cancellation of the quadratic values. Reducing the isotropic graph gives the next quotient; replacing a coordinate of least valuation gives an overlattice basis. Orbits of actual isometries

may be used with their sizes because they preserve all these conditions and the residual form. Equal repeated factors have the corresponding multinomial multiplicity.

The elementary part requires a separate orbit justification. For $\delta = 0$, use Witt's theorem on the non-degenerate quadratic space over $\mathbb{F}_2$. For $\delta = 1$, diagonalize as

$$E = \langle 1/2 \rangle^a \perp \langle 3/2 \rangle^b.$$

To absorb a binary integer-valued plane with basis u, v, $b(u, v) = 1/2$, take an orthogonal vector c of norm $1/2$. The three vectors $c + u, c + v, c + u + v$ are an orthogonal basis. Their norms are $1/2, 1/2, 3/2$ for the hyperbolic plane, and $3/2, 3/2, 3/2$ for the other plane. The case $q(c) = 3/2$ reverses the signs. On four coordinates of the same sign, the matrix $I + J$ over $\mathbb{F}_2$ reverses all four signs, where every entry of J is 1. Thus dimension and the Brown invariant $a - b$ modulo 8 determine the needed diagonal representative.

Put $B = 2b$. In diagonal coordinates it is the standard dot product. The following are actual isometries: permutations of coordinates with equal coefficient; $x \mapsto x + B(x, w)w$ for $q(w) = 1$; and, on two positive and two negative coordinates, $(I + J)P$, where P interchanges the positive and negative pairs. For the last transformation, the norm numerators of the four coordinates sum to 8, and each column is the complement of a basis vector of the opposite sign. Its columns therefore have the original norms and are pairwise B-orthogonal.

Modulo equal-coefficient permutations, a vector is represented by its two support sizes (i, j). A transvection with support (r, s) and overlap (u, v) acts, when $u + v$ is odd, by

$$(i, j) \mapsto (i + r - 2u, j + s - 2v), \quad r + 3s \equiv 2 \pmod 4.$$

All feasible overlaps are included. For the four-coordinate isometry the targets are $(i - u + v, j - v + u)$ for even overlap and $(i - u + 2 - v, j - v + 2 - u)$ for odd overlap. The finite graph on these support pairs is connected for each fixed norm after separating zero and the characteristic vector (a, b), for every $1 \le a + b \le 20$ and every a, b [C18]. Each support pair represents $\binom{a}{i}\binom{b}{j}$ vectors. Consequently each such class is a single orbit, and those sizes justify the compression used by the finite transition. The zero-dimensional case is immediate. The bound 20 covers the present root inputs and their residual generator numbers; no assertion for arbitrary dimension is needed.

The 3-adic outer sum. With a single order-9 factor $A = \langle v \rangle \perp W$, $q(v) = 2a/9$ with $3 \nmid a$, every isotropic subgroup is killed by 3: an element of order 9 has a unit norm numerator modulo 3, which the elementary part cannot cancel. Put $K = \langle 3v \rangle$. If an isotropic subgroup contains K, it is K plus an isotropic subspace U of W, and its reduction removes the order-9 factor. If it avoids K, it is the graph of a map $U \to K$, with $3^{\dim U}$ choices, and the reduction retains an order-9 factor together with the Witt reduction of W. With more than one order-9 factor the code enumerates actual finite subgroups and their orthogonal quotients. This outer sum over primitive closures is distinct from the primitive-gluing count below.

The sums at $p = 3$ with $\mathbb{Z}/9$: the reduction. Let $A = D_3(T')$, $B = D_3(Y)$, and let $C \subset A \perp B$ be the isotropic graph of a primitive gluing. Primitivity says $C \cap A = C \cap B = 0$, and the ambient discriminant is 3-elementary precisely when $3C^\perp \subset C$. More generally use an odd prime p below.

The projection $C^\perp \to A$ is onto: the character on the gluing subgroup in B prescribed by any $a \in A$ extends to B, and non-degeneracy represents it by pairing with an element of B. The same holds for the other projection. The elementary condition then forces pA and pB into the gluing subgroups and

identifies them. This argument does not assume an exponent bound on B: if A has exponent at most p^2, the resulting isomorphism $pA \cong pB$ proves that B has exponent at most p^2 as well.

On pA define a separate quadratic form $\tilde{Q}_A(px) = (p^2/2)q_A(x)$ modulo p. It is well defined since changing x by an element of $A[p]$ changes this value by a multiple of p. It is non-degenerate on the p^2 constituents. This is not the restriction of q_A, and the original q_A need not descend to A/pA. Define $\tilde{Q}_B$ in the same way.

The restriction $\varphi_0 : pA \to pB$ of the gluing is an anti-isometry for these new forms. To see this, choose $(a, b) \in C^\perp$ for each a; then $pb = \varphi_0(pa)$, and orthogonality to $(px, \varphi_0(px)) \in C$, multiplied by p, gives the anti-isometry identity on the polar forms. Since p is odd this also determines the quadratic forms. Put $I = \mathrm{graph}(\varphi_0)$. The correct reduction is $E = I^\perp/I$. It is p-elementary: orthogonality to I and non-degeneracy force $pb = \varphi_0(pa)$ for every $(a, b) \in I^\perp$.

Further gluings correspond to isotropic subspaces $J = C/I$ of E. Primitivity is preserved by requiring J to meet each of the images of $A[p]$ and $B[p]$ trivially. These images inject into E but may intersect each other.

Two constituents. In rank 20 with ambient discriminant 3^6, two surviving order-9 constituents require A_8^2 in the original root lattice. If $A = (\mathbb{Z}/9)^2 \perp \mathbb{F}_3^w$, its rank n is at least $16 + 2w$: extra 3-elementary constituents cost at least two root dimensions each, and saturation preserving the two order-9 constituents cannot increase w. Put $N = 20 - n$ and $B = (\mathbb{Z}/9)^2 \perp \mathbb{F}_3^b$. If $h = \dim J$, orders give

$$6 = (4 + w) + (4 + b) - 2(2 + h), \quad b = 2 - w + 2h.$$

The minimal number $2 + b$ of generators of B is at most $N \le 4 - 2w$. Therefore $w + 2h \le 0$, forcing $w = h = 0$, $b = 2$, $N = 4$. No nonzero J is possible. For A_8, $q(x) = 8/9$, so $\tilde{Q}_A(3x) = 1$ modulo 3. The two constituents give the anisotropic binary form $x_1^2 + x_2^2$ over $\mathbb{F}_3$. Its orthogonal group has order $2(3 + 1) = 8$, giving the eight anti-isometry graphs when the opposite form exists. Thus the extra branches excluded by the implementation cannot occur in this application.

One constituent. Write $A = \langle v \rangle \perp W$, $B = \langle y \rangle \perp V$ with $q(v) = 2a/9$ and $q(y) = -2a/9$, and put $w = \dim W$, $b = \dim V$. Let $\mathrm{Anti}(W, V; h)$ count anti-isometry graphs from h-dimensional subspaces of W into V, including degenerate subspaces.

If C has no element of order 9, the graph on $3A$ has two choices. For either one, E is a hyperbolic plane $\langle e, f \rangle$ plus $W \perp V$, with $e = [(3v, 0)]$, $f = [(v, \pm y)]$. The subspace J has no f component and is the graph of a linear map from an anti-isometry graph in $W \perp V$ to $\langle e \rangle$. For $h = \dim(C/I)$ there are 3^h such maps. This branch has multiplicity $2 \cdot 3^h$ $\mathrm{Anti}(W, V; h)$ and $|C| = 3^{1+h}$, whence $b = s - 2 - w + 2h$.

If C has an order-9 element, let G count all pairs (v', y') of such elements with $q(v') + q(y') = 0$. Their cyclic graphs C_1 have six generators each, giving $G/6$ choices. The non-degenerate cyclic factors split off; their complements are isometric to W, V. Now put $h = \dim(C/C_1)$, which is one less than $\dim(C/I)$. The remaining gluing contributes $\mathrm{Anti}(W, V; h)$. Each final $C \cong \mathbb{Z}/9 \times (\mathbb{Z}/3)^h$ contains 3^h cyclic subgroups C_1 of order 9, so the multiplicity is $(G/6)\mathrm{Anti}(W, V; h)/3^h$. Here $|C| = 3^{2+h}$ and $b = s - w + 2h$.

In the elementary-form convention $3q$, let δ_W, δ_V be the determinant characters and φ that of the ambient discriminant. Witt reduction gives $\delta_V = \varphi\delta_W \cdot (-1)^{h+1}$ in the first branch and $\delta_V = \varphi\delta_W \cdot (-1)^h$ in the second. These are the two local branches and multiplicities used in [C17]. The coefficients

arise from finite graph choices and the two stated overcounts, not from a form on $A/3A$. The outer sum over primitive closures and the local mass factors remain the separate operations specified above.

Theorem D.11. $\mu_F^{\emptyset}(20,6) = 3{,}275{,}143\ /\ 7{,}464{,}960 \approx 0.43874$ for $p_0 = 3$, and $\mu_F^{\emptyset}(16,4) = 1/\ 288{,}000$ for $p_0 = 5$.

In the first case 811 of the 10,632 root types have non-zero mass and none is negative; in the second 329 [C17]. The 24 classes F of the genus $(20,6)$ listed in Appendix B.4 are pairwise non-isometric, rootless, and the sum of $1/\ |\mathrm{Aut}\ F|$ over them equals Theorem D.11; so the list is complete. For $(16,4)$, $p_0 = 5$, the value is attained by a single lattice with $|\mathrm{Aut}\ F| = 288{,}000$ (the fixed lattice of an element of frame shape $5^4 1^{12}$, that is of type $5-(4,16)-4$, of Λ_{32}', Table A1), which is therefore the only rootless lattice of the genus.

Table D5. Checks of (D.10)–(D.12) [C17]. They are consistency checks; the values rest on the proofs above.

check	result
$2m_p$ against [CS88, Table 2] ($p = 3,5,7$, $n \le 5$) and the standard mass against [CS88, Tables 3, 6]	equal
(D.12) against Sage's implementation of the local factors of [CS88], for the 40 parabolic sublattices of E_8^3 (including type I and bound components at 2)	equal
$p_0 = 3$: masses by root system against the complete lists of the genera $(12,6)$, $(12,8)$, $(12,10)$, $(14,7)$, $(14,9)$, $(16,8)$, all classes with roots included	equal for every root system (e.g. $\mu_F^{\emptyset}(16,8) = 59/\ 1{,}036{,}800$, 79 root systems)
$p_0 = 3$: $(8,0)$ and $(16,0)$	$\mu_F^{\emptyset} = 0$ (E_8; E_8^2, D_{16}^+)
$p_0 = 5$: masses by root system against the complete lists of the genera of rank below 24, as for $p_0 = 3$; in rank 24, the rootless mass against a value computed directly	equal

Remark (vectors of norm 2/3 in $F^{\#}$). For the type $3-(6,20)-6$ the extremal gluings need $F^{\#}$ free of vectors of norm $2/3$ (Lemma 4.3). For a rootless F and $v \in F^{\#}$ of norm $2/3$, $r = 3v \in F$ is primitive of norm 6 with $(r,F) = 3\mathbb{Z}$, and $(F, \pm v) \leftrightarrow Y = r^{\perp}$ is a bijection onto the rootless lattices of the genus $\mathcal{K}_{19}$ of even lattices of rank 19 and determinant $2 \cdot 3^5$ with discriminant form $\langle 1/2 \rangle \perp q'$, where q' is the restriction of q_F to the orthogonal complement of a vector of norm $2/3$ in $D(F) \cong \mathbb{F}_3^6$ (well defined by Witt's theorem). Indeed $[F : \mathbb{Z}r \perp Y] = 2$, glued along $r/2$ with $q = 3/2$, and at 3 the summand $\mathbb{Z}_3 r$ splits off; the genus is determined by the discriminant form [Nik79, Cor. 1.9.4]. Moreover $|\mathrm{Aut}(F, \pm v)| = 2\ |\mathrm{Aut}\ Y|$, the factor 2 being the reflection in r, which lies in $\mathrm{Aut}\ F$ since $(r,F) = 3\mathbb{Z}$ and $(r,r) = 6$; the same method gives $\mu^{\emptyset}(\mathcal{K}_{19}) = 1/648$ [C17], so the lattices F with such vectors carry total weight $\sum n(F)/\ |\mathrm{Aut}\ F| = 1/\ 1{,}296$, where $n(F)$ is the number of pairs $\pm v$ of vectors of norm $2/3$ in $F^{\#}$. In the list exactly two classes have such vectors, each with one pair $\pm v$ and $|\mathrm{Aut}\ F| = 2{,}592$, which accounts for $1/\ 1{,}296$. This is a check; the list is complete by Theorem D.11.

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